Nordic Model analysis: TALIS 2024

Blinded

2026-10-26

Setup

Load data and create training and test data

Create training and test data

```
set.seed(1312)

CountrySizes <- BTGINTT4 %>%
  count(CNTRY) %>%
  mutate(Nover=n-4000) %>%
  mutate( # select all over 4000 or 10%, whichever is bigger to be test data. this limits tra
ining data size for large countries, which was suggested by a reviewer
    Ntrain=ifelse(Nover/n <= 0.1, as.integer(n*0.1) , Nover)) %>%
  select(-Nover,-n)

BTGINTT4 %>%
  count(CNTRY,IDSCHOOL) %>%
  merge(CountrySizes) %>%
  mutate(train=0) %>%
  group_by(CNTRY) %>%
  group_modify(~{
    while ( sum(.x$train) < .x$Ntrain[1] ) {
      a <- sample(1:nrow(.x),1)
      .x[a, "train"] <- .x[a,"n"]
    }
    .x
  } ) %>%
  as.data.frame %>%
  filter(train>0) %>%
  with( paste.(CNTRY,IDSCHOOL) ) ->
  TestSchools

Test <- BTGINTT4 %>% filter(paste.(CNTRY,IDSCHOOL) %in% TestSchools )
BTGINTT4 <- BTGINTT4 %>% filter(!( paste.(CNTRY,IDSCHOOL) %in% TestSchools ) )

cat("\nNrows of test data: ", nrow(Test))
```

```
##
## Nrows of test data:  30515
```

```
cat("\nNrows of training data: ", nrow(BTGINTT4))
```

```
##
## Nrows of training data:  158831
```

```
cat("\nPercentage of data in test data: ", nrow(Test)/(nrow(Test)+nrow(BTGINTT4)) )
```

```
##
## Percentage of data in test data:  0.16116
```

```
print("\nRelative size of teh Nordic and non-nordic teachers training data: ")
```

```
## [1] "\nRelative size of teh Nordic and non-nordic teachers training data: "
```

```
BTGINTT4$Nordic  |> table()
```

```
##
##      0      1
## 147340  11491
```

```
BTGINTT4$Nordic  |> table() |> prop.table()
```

```
##
##          0          1
## 0.92765266 0.07234734
```

```
print("\nRelative size of teh Nordic and non-nordic teachers test data: ")
```

```
## [1] "\nRelative size of teh Nordic and non-nordic teachers test data: "
```

```
Test$Nordic  |> table()
```

```
##
##     0     1
## 29198  1317
```

```
Test$Nordic  |> table() |> prop.table()
```

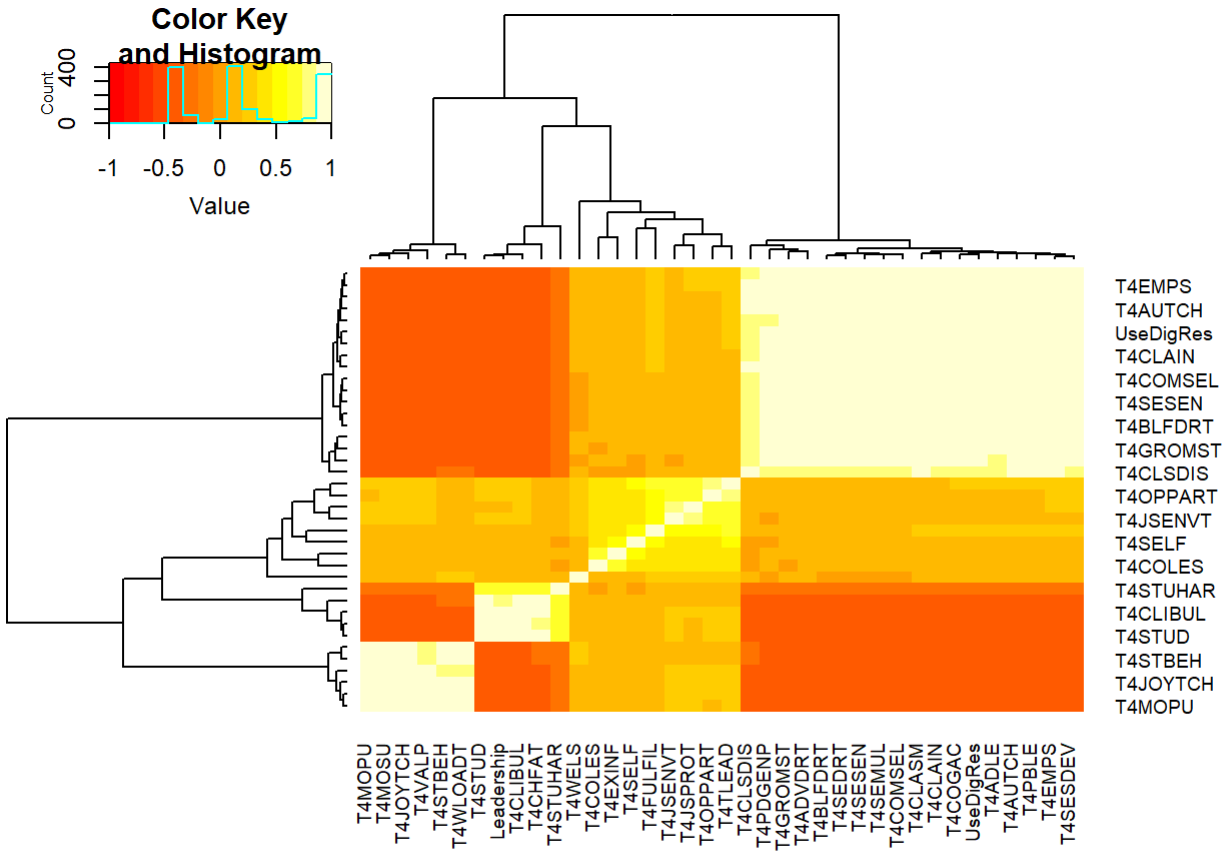
```
##
##         0         1
## 0.9568409 0.0431591
```

Test for high correlations and remove those to reduce multicollinearity issues

There are some scales that are correlated over 0.79. We combine these scales to ensure that high collinearity does not impact our ability to explore variable importance.

Scale1	Scale2	Correlation
T4DRTIIA	T4DRTWCI	0.8067
T4INSTLE	T4RELEAD	0.7988

```
BTGINTT4 %>% select(T4EXINF,T4COLES,T4SELF,T4SEMUL,T4GROMST,T4SESEN,T4COMSEL,T4SEDRT,T4BLFDR
T,T4ADVDRT,T4CLAIN,T4COGAC,T4CLASM,T4PDGENP,T4EMPS,T4CLSDIS,T4ADLE,T4PBLE,T4AUTCH,T4FULFIL,T4
OPPART,T4SESDEV,T4TLEAD,T4STUD,T4CLIBUL,T4STUHAR,T4CHFAT,T4MOPU,T4MOSU,T4WELS,T4WLOADT,T4STBE
H,T4JSENVT,T4JSPROT,T4VALP,T4JOYTCH,UseDigRes,Leadership) %>% cor(use="pair") %>% gplots::hea
tmap.2(trace="none",cexCol=.7,cexRow=.7,margins=c(8,8), dendrogram="both" )
```



Test for ICCs at the school and country level

```
require(lmer)
```

```
## Loading required package: lmer
```

```
## Warning in library(package, lib.loc = lib.loc, character.only = TRUE, logical.return = TRU
E, : there is no package called 'lmer'
```

```
VarComp <- function(.) {
  v <- VarCorr(.)
  s <- sigma(.)^2
  nms <- names(v)
  for (i in nms) {
    if (length(v[[i]])==1) {
      v[[i]] <- c(v[[i]])
    } else {
      # browser()
      c <- attr(v[[i]], "correlation")
      n <- names(diag(v[[i]]))
      nl <- length(n)
      v[[i]] <- c(diag(v[[i]]), attr(v[[i]], "correlation")[lower.tri(diag(nl))])
      ncor <- combn(nl,2)
      names(v[[i]]) <- c(n, apply(ncor,2,function(x)
        paste0("C",paste(n[ x[1] ],n[ x[2] ],sep=":"))) )
    }
  }
  return(c(unlist(v),Residual=s))
}
VarPer <- function(.) {
  v <- VarComp(.)
  v<- v[grep("\\.C",names(v),invert=T,value=T)]
  round(100*v/sum(v),1)
}

iccs <- lapply(c("T4EXINF","T4COLES","T4SELF","T4SEMUL","T4GROMST","T4SESEN","T4COMSEL","T4SE
DRT","T4BLFDRT","T4ADVDRT","T4CLAIN","T4COGAC","T4CLASM","T4PDGENP","T4EMPS","T4CLSDIS","T4AD
LE","T4PBLE","T4AUTCH","T4FULFIL","T4OPPART","T4SESDEV","T4TLEAD","T4STUD","T4CLIBUL","T4STUH
AR","T4CHFAT","T4MOPU","T4MOSU","T4WELS","T4WLOADT","T4STBEH","T4JSENVT","T4JSPROT","T4VAL
P","T4JOYTCH","UseDigRes","Leadership"),
               function(x) {
                 lme4::lmer(as.formula(paste0(x,"~1+(1|CNTRY)+(1|IDSCHOOL)")), BTGINTT4, cont
rol=lmerControl("bobyqa")) |> VarPer()
                 })
```

```
## boundary (singular) fit: see help('isSingular')
```

```
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
## boundary (singular) fit: see help('isSingular')
```

```
names(iccs) <- c("T4EXINF","T4COLES","T4SELF","T4SEMUL","T4GROMST","T4SESEN","T4COMSEL","T4SE
DRT","T4BLFDRT","T4ADVDRT","T4CLAIN","T4COGAC","T4CLASM","T4PDGENP","T4EMPS","T4CLSDIS","T4AD
LE","T4PBLE","T4AUTCH","T4FULFIL","T4OPPART","T4SESDEV","T4TLEAD","T4STUD","T4CLIBUL","T4STUH
AR","T4CHFAT","T4MOPU","T4MOSU","T4WELS","T4WLOADT","T4STBEH","T4JSENVT","T4JSPROT","T4VAL
P","T4JOYTCH","UseDigRes","Leadership")
bind_rows(iccs,.id="Variable") ->iccs

iccs %>% knitr::kable(caption="Table of percentages of variation in variables that is between
schools",digits=4)
```

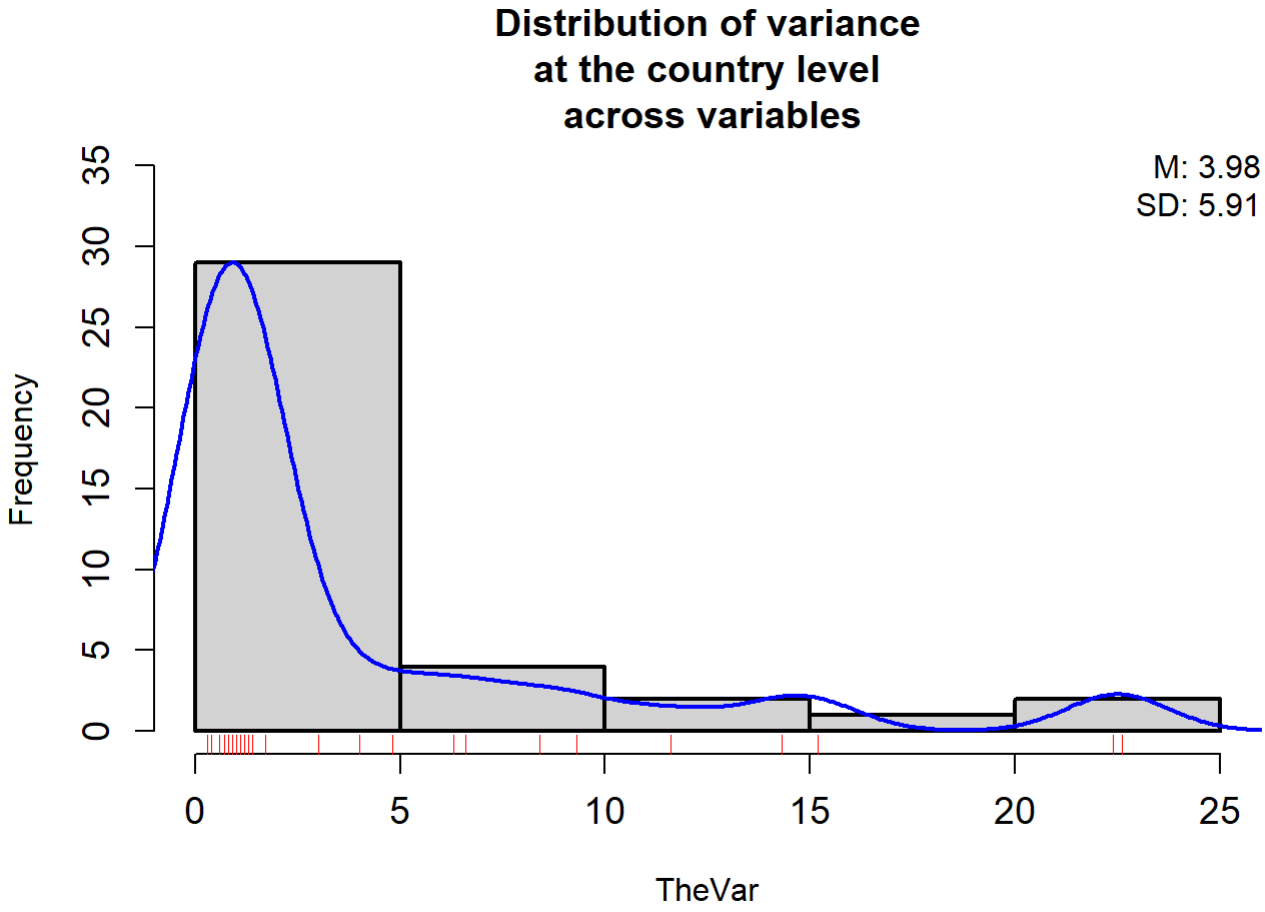
Table of percentages of variation in variables that is between schools

Variable	IDSCHOOL	CNTRY	Residual
T4EXINF	0.1	6.3	93.6
T4COLES	0.2	15.2	84.5
T4SELF	0.1	14.3	85.6
T4SEMUL	0.0	1.1	98.9
T4GROMST	0.0	0.6	99.4
T4SESEN	0.0	0.9	99.1
T4COMSEL	0.0	0.7	99.3

Variable	IDSCHOOL	CNTRY	Residual
T4SEDRT	0.0	0.7	99.3
T4BLFDRT	0.0	1.4	98.6
T4ADVDRT	0.0	0.3	99.7
T4CLAIN	0.0	1.0	99.0
T4COGAC	0.0	1.4	98.6
T4CLASM	0.0	1.0	99.0
T4PDGENP	0.0	1.4	98.6
T4EMPS	0.0	0.7	99.3
T4CLSDIS	0.0	1.2	98.8
T4ADLE	0.0	4.0	96.0
T4PBLE	0.0	1.0	99.0
T4AUTCH	0.0	0.7	99.3
T4FULFIL	0.1	11.6	88.2
T4OPPART	0.2	9.3	90.5
T4SESDEV	0.0	1.3	98.7
T4TLEAD	0.2	8.4	91.4
T4STUD	0.0	0.4	99.6
T4CLIBUL	0.0	0.6	99.4
T4STUHAR	0.0	22.4	77.6
T4CHFAT	0.0	0.4	99.6
T4MOPU	0.0	0.6	99.4
T4MOSU	0.0	0.7	99.3
T4WELS	0.2	22.6	77.2
T4WLOADT	0.0	0.9	99.1
T4STBEH	0.0	0.8	99.2
T4JSENVT	0.3	4.8	94.9
T4JSPROT	0.2	6.6	93.3
T4VALP	0.0	3.0	97.0
T4JOYTCH	0.0	0.6	99.4
UseDigRes	0.0	1.7	98.3
Leadership	0.0	0.8	99.2

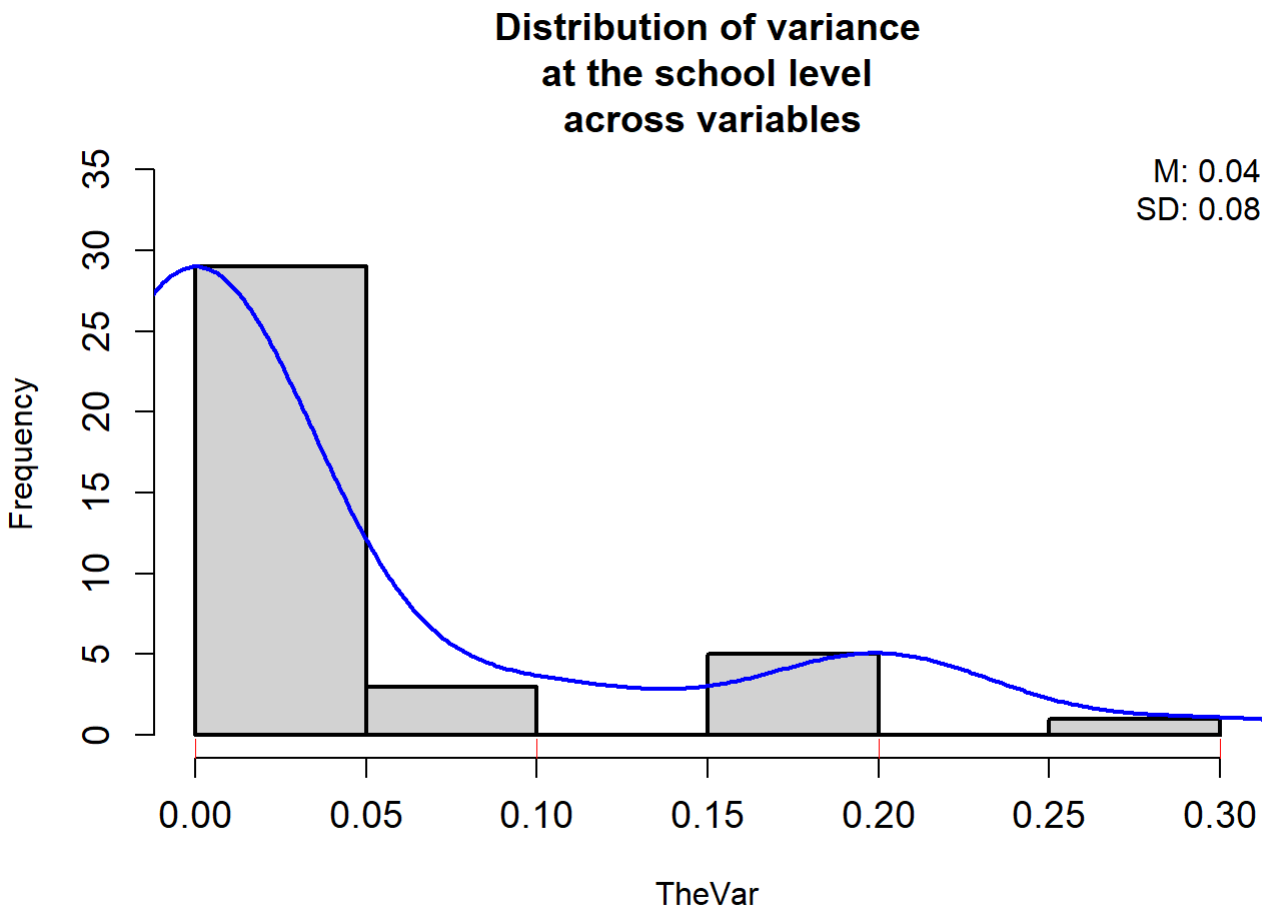
```
hist.rug(iccs[["CNTRY"   ]],main="Distribution of variance\nat the country level\n across var
iables")
```

```
##    vars  n mean   sd median trimmed  mad min  max range skew kurtosis   se
## X1    1 38 3.98 5.91   1.05    2.82 0.67 0.3 22.6  22.3 1.93     2.82 0.96
```




```
hist.rug(iccs[["IDSCHOOL"]],main="Distribution of variance\nat the school level\n across vari
ables")
```

```
##    vars  n mean   sd median trimmed mad min max range skew kurtosis   se
## X1    1 38 0.04 0.08      0    0.03   0   0 0.3   0.3 1.64     1.32 0.01
```



BART Model

run model

The CV model to verify ideal parameters is not run due to the amount of time it takes to run.

```
model_cv <- xbart(Nordic~ T4EXINF+T4COLES+T4SELF+T4SEMUL+T4GROMST+T4SESEN+T4COMSEL+T4SEDRT+T4
BLFDRT+T4ADVDRT+T4CLAIN+T4COGAC+T4CLASM+T4PDGENP+T4EMPS+T4CLSDIS+T4ADLE+T4PBLE+T4AUTCH+T4FULF
IL+T4OPPART+T4SESDEV+T4TLEAD+T4STUD+T4CLIBUL+T4STUHAR+T4CHFAT+T4MOPU+T4MOSU+T4WELS+T4WLOADT+T
4STBEH+T4JSENVT+T4JSPROT+T4VALP+T4JOYTCH+UseDigRes+Leadership,
  data = BTGINTT4,n.burn = c(20000,1000,1000), n.samples=2000,
  method="k-fold",n.test=5,n.reps=10,
  n.trees = c(50,100,200),
  k=c(1.5, 2.0),
  power=c(1.5,2),
  base=c(0.95,0.99),
  loss = "log",
  n.threads = 20,
  seed=1121
)

apply(model_cv,c(2,3,4),mean)
apply(model_cv,c(2,3,4,5),mean) %>% as.data.frame %>% rownames_to_column(var="Ntree") %>% gat
her(var,val,-Ntree) %>% arrange(-val) %>% head(5)
```

```
X = BTGINTT4 %>% select(T4EXINF,T4COLES,T4SELF,T4SEMUL,T4GROMST,T4SESEN,T4COMSEL,T4SEDRT,T4BL
FDRT,T4ADVDRT,T4CLAIN,T4COGAC,T4CLASM,T4PDGENP,T4EMPS,T4CLSDIS,T4ADLE,T4PBLE,T4AUTCH,T4FULFI
L,T4OPPART,T4SESDEV,T4TLEAD,T4STUD,T4CLIBUL,T4STUHAR,T4CHFAT,T4MOPU,T4MOSU,T4WELS,T4WLOADT,T4
STBEH,T4JSENVT,T4JSPROT,T4VALP,T4JOYTCH,UseDigRes,Leadership) %>% as.matrix
Xtest = Test %>% select(T4EXINF,T4COLES,T4SELF,T4SEMUL,T4GROMST,T4SESEN,T4COMSEL,T4SEDRT,T4BL
FDRT,T4ADVDRT,T4CLAIN,T4COGAC,T4CLASM,T4PDGENP,T4EMPS,T4CLSDIS,T4ADLE,T4PBLE,T4AUTCH,T4FULFI
L,T4OPPART,T4SESDEV,T4TLEAD,T4STUD,T4CLIBUL,T4STUHAR,T4CHFAT,T4MOPU,T4MOSU,T4WELS,T4WLOADT,T4
STBEH,T4JSENVT,T4JSPROT,T4VALP,T4JOYTCH,UseDigRes,Leadership) %>% as.matrix

model <- bart(
  x.train = as.matrix(X),
  y.train = factor(BTGINTT4$Nordic),
  x.test = Xtest,
  ntree = 50, power=2, base=0.99, k=1.5,
  nskip = 70000, ndpost = 40000, keepevery = 100,
  nchain=8, nthread=8,
  seed=561,
  keeptrees = TRUE
)
```

```
##
## Running BART with binary y
##
## number of trees: 50
## number of chains: 8, default number of threads 8
## tree thinning rate: 100
## Prior:
##  k prior fixed to 1.500000
##  power and base for tree prior: 2.000000 0.990000
##  use quantiles for rule cut points: false
##  proposal probabilities: birth/death 0.50, swap 0.10, change 0.40; birth 0.50
## data:
##  number of training observations: 158831
##  number of test observations: 30515
##  number of explanatory variables: 38
##
## Cutoff rules c in x<=c vs x>c
## Number of cutoffs: (var: number of possible c):
## (1: 100) (2: 100) (3: 100) (4: 100) (5: 100)
## (6: 100) (7: 100) (8: 100) (9: 100) (10: 100)
## (11: 100) (12: 100) (13: 100) (14: 100) (15: 100)
## (16: 100) (17: 100) (18: 100) (19: 100) (20: 100)
## (21: 100) (22: 100) (23: 100) (24: 100) (25: 100)
## (26: 100) (27: 100) (28: 100) (29: 100) (30: 100)
## (31: 100) (32: 100) (33: 100) (34: 100) (35: 100)
## (36: 100) (37: 100) (38: 100)
## offsets:
##  reg : 0.00 0.00 0.00 0.00 0.00
##  test: 0.00 0.00 0.00 0.00 0.00
## Running mcmc loop:
## [7] iteration: 100 (of 40000)
## [5] iteration: 100 (of 40000)
## [4] iteration: 100 (of 40000)
## [1] iteration: 100 (of 40000)
## [6] iteration: 100 (of 40000)
## [3] iteration: 100 (of 40000)
## [8] iteration: 100 (of 40000)
## [2] iteration: 100 (of 40000)
## [7] iteration: 200 (of 40000)
## [1] iteration: 200 (of 40000)
## [4] iteration: 200 (of 40000)
## [5] iteration: 200 (of 40000)
## [6] iteration: 200 (of 40000)
## [3] iteration: 200 (of 40000)
## [8] iteration: 200 (of 40000)
## [2] iteration: 200 (of 40000)
## [7] iteration: 300 (of 40000)
## [5] iteration: 300 (of 40000)
## [4] iteration: 300 (of 40000)
## [6] iteration: 300 (of 40000)
## [1] iteration: 300 (of 40000)
## [3] iteration: 300 (of 40000)
## [8] iteration: 300 (of 40000)
## [2] iteration: 300 (of 40000)
## [7] iteration: 400 (of 40000)
```

```
## [5] iteration: 400 (of 40000)
## [6] iteration: 400 (of 40000)
## [4] iteration: 400 (of 40000)
## [1] iteration: 400 (of 40000)
## [8] iteration: 400 (of 40000)
## [3] iteration: 400 (of 40000)
## [2] iteration: 400 (of 40000)
## [7] iteration: 500 (of 40000)
## [5] iteration: 500 (of 40000)
## [6] iteration: 500 (of 40000)
## [4] iteration: 500 (of 40000)
## [1] iteration: 500 (of 40000)
## [3] iteration: 500 (of 40000)
## [8] iteration: 500 (of 40000)
## [2] iteration: 500 (of 40000)
## [7] iteration: 600 (of 40000)
## [5] iteration: 600 (of 40000)
## [6] iteration: 600 (of 40000)
## [4] iteration: 600 (of 40000)
## [1] iteration: 600 (of 40000)
## [3] iteration: 600 (of 40000)
## [8] iteration: 600 (of 40000)
## [2] iteration: 600 (of 40000)
## [7] iteration: 700 (of 40000)
## [5] iteration: 700 (of 40000)
## [6] iteration: 700 (of 40000)
## [4] iteration: 700 (of 40000)
## [1] iteration: 700 (of 40000)
## [3] iteration: 700 (of 40000)
## [8] iteration: 700 (of 40000)
## [2] iteration: 700 (of 40000)
## [7] iteration: 800 (of 40000)
## [5] iteration: 800 (of 40000)
## [6] iteration: 800 (of 40000)
## [4] iteration: 800 (of 40000)
## [1] iteration: 800 (of 40000)
## [8] iteration: 800 (of 40000)
## [3] iteration: 800 (of 40000)
## [2] iteration: 800 (of 40000)
## [7] iteration: 900 (of 40000)
## [5] iteration: 900 (of 40000)
## [6] iteration: 900 (of 40000)
## [4] iteration: 900 (of 40000)
## [1] iteration: 900 (of 40000)
## [8] iteration: 900 (of 40000)
## [3] iteration: 900 (of 40000)
## [2] iteration: 900 (of 40000)
## [7] iteration: 1000 (of 40000)
## [5] iteration: 1000 (of 40000)
## [6] iteration: 1000 (of 40000)
## [4] iteration: 1000 (of 40000)
## [1] iteration: 1000 (of 40000)
## [8] iteration: 1000 (of 40000)
## [3] iteration: 1000 (of 40000)
## [2] iteration: 1000 (of 40000)
## [7] iteration: 1100 (of 40000)
```

```
## [5] iteration: 1100 (of 40000)
## [6] iteration: 1100 (of 40000)
## [4] iteration: 1100 (of 40000)
## [1] iteration: 1100 (of 40000)
## [8] iteration: 1100 (of 40000)
## [3] iteration: 1100 (of 40000)
## [2] iteration: 1100 (of 40000)
## [7] iteration: 1200 (of 40000)
## [5] iteration: 1200 (of 40000)
## [6] iteration: 1200 (of 40000)
## [4] iteration: 1200 (of 40000)
## [1] iteration: 1200 (of 40000)
## [8] iteration: 1200 (of 40000)
## [3] iteration: 1200 (of 40000)
## [2] iteration: 1200 (of 40000)
## [7] iteration: 1300 (of 40000)
## [5] iteration: 1300 (of 40000)
## [6] iteration: 1300 (of 40000)
## [4] iteration: 1300 (of 40000)
## [1] iteration: 1300 (of 40000)
## [8] iteration: 1300 (of 40000)
## [3] iteration: 1300 (of 40000)
## [2] iteration: 1300 (of 40000)
## [7] iteration: 1400 (of 40000)
## [5] iteration: 1400 (of 40000)
## [6] iteration: 1400 (of 40000)
## [4] iteration: 1400 (of 40000)
## [1] iteration: 1400 (of 40000)
## [3] iteration: 1400 (of 40000)
## [8] iteration: 1400 (of 40000)
## [2] iteration: 1400 (of 40000)
## [7] iteration: 1500 (of 40000)
## [5] iteration: 1500 (of 40000)
## [6] iteration: 1500 (of 40000)
## [4] iteration: 1500 (of 40000)
## [1] iteration: 1500 (of 40000)
## [3] iteration: 1500 (of 40000)
## [8] iteration: 1500 (of 40000)
## [2] iteration: 1500 (of 40000)
## [7] iteration: 1600 (of 40000)
## [5] iteration: 1600 (of 40000)
## [6] iteration: 1600 (of 40000)
## [4] iteration: 1600 (of 40000)
## [1] iteration: 1600 (of 40000)
## [3] iteration: 1600 (of 40000)
## [8] iteration: 1600 (of 40000)
## [2] iteration: 1600 (of 40000)
## [7] iteration: 1700 (of 40000)
## [6] iteration: 1700 (of 40000)
## [5] iteration: 1700 (of 40000)
## [4] iteration: 1700 (of 40000)
## [1] iteration: 1700 (of 40000)
## [8] iteration: 1700 (of 40000)
## [3] iteration: 1700 (of 40000)
## [2] iteration: 1700 (of 40000)
## [7] iteration: 1800 (of 40000)
```

```
## [6] iteration: 1800 (of 40000)
## [5] iteration: 1800 (of 40000)
## [1] iteration: 1800 (of 40000)
## [4] iteration: 1800 (of 40000)
## [8] iteration: 1800 (of 40000)
## [3] iteration: 1800 (of 40000)
## [2] iteration: 1800 (of 40000)
## [7] iteration: 1900 (of 40000)
## [6] iteration: 1900 (of 40000)
## [5] iteration: 1900 (of 40000)
## [4] iteration: 1900 (of 40000)
## [8] iteration: 1900 (of 40000)
## [1] iteration: 1900 (of 40000)
## [3] iteration: 1900 (of 40000)
## [2] iteration: 1900 (of 40000)
## [7] iteration: 2000 (of 40000)
## [6] iteration: 2000 (of 40000)
## [5] iteration: 2000 (of 40000)
## [4] iteration: 2000 (of 40000)
## [8] iteration: 2000 (of 40000)
## [1] iteration: 2000 (of 40000)
## [3] iteration: 2000 (of 40000)
## [2] iteration: 2000 (of 40000)
## [7] iteration: 2100 (of 40000)
## [6] iteration: 2100 (of 40000)
## [5] iteration: 2100 (of 40000)
## [4] iteration: 2100 (of 40000)
## [8] iteration: 2100 (of 40000)
## [1] iteration: 2100 (of 40000)
## [3] iteration: 2100 (of 40000)
## [2] iteration: 2100 (of 40000)
## [7] iteration: 2200 (of 40000)
## [6] iteration: 2200 (of 40000)
## [5] iteration: 2200 (of 40000)
## [4] iteration: 2200 (of 40000)
## [8] iteration: 2200 (of 40000)
## [1] iteration: 2200 (of 40000)
## [3] iteration: 2200 (of 40000)
## [2] iteration: 2200 (of 40000)
## [7] iteration: 2300 (of 40000)
## [6] iteration: 2300 (of 40000)
## [5] iteration: 2300 (of 40000)
## [4] iteration: 2300 (of 40000)
## [8] iteration: 2300 (of 40000)
## [1] iteration: 2300 (of 40000)
## [3] iteration: 2300 (of 40000)
## [2] iteration: 2300 (of 40000)
## [7] iteration: 2400 (of 40000)
## [6] iteration: 2400 (of 40000)
## [5] iteration: 2400 (of 40000)
## [4] iteration: 2400 (of 40000)
## [8] iteration: 2400 (of 40000)
## [1] iteration: 2400 (of 40000)
## [3] iteration: 2400 (of 40000)
## [2] iteration: 2400 (of 40000)
## [7] iteration: 2500 (of 40000)
```

```
## [6] iteration: 2500 (of 40000)
## [5] iteration: 2500 (of 40000)
## [4] iteration: 2500 (of 40000)
## [8] iteration: 2500 (of 40000)
## [1] iteration: 2500 (of 40000)
## [3] iteration: 2500 (of 40000)
## [2] iteration: 2500 (of 40000)
## [7] iteration: 2600 (of 40000)
## [6] iteration: 2600 (of 40000)
## [5] iteration: 2600 (of 40000)
## [4] iteration: 2600 (of 40000)
## [8] iteration: 2600 (of 40000)
## [1] iteration: 2600 (of 40000)
## [3] iteration: 2600 (of 40000)
## [2] iteration: 2600 (of 40000)
## [7] iteration: 2700 (of 40000)
## [6] iteration: 2700 (of 40000)
## [5] iteration: 2700 (of 40000)
## [4] iteration: 2700 (of 40000)
## [8] iteration: 2700 (of 40000)
## [1] iteration: 2700 (of 40000)
## [3] iteration: 2700 (of 40000)
## [7] iteration: 2800 (of 40000)
## [2] iteration: 2700 (of 40000)
## [6] iteration: 2800 (of 40000)
## [5] iteration: 2800 (of 40000)
## [4] iteration: 2800 (of 40000)
## [8] iteration: 2800 (of 40000)
## [1] iteration: 2800 (of 40000)
## [3] iteration: 2800 (of 40000)
## [7] iteration: 2900 (of 40000)
## [2] iteration: 2800 (of 40000)
## [6] iteration: 2900 (of 40000)
## [5] iteration: 2900 (of 40000)
## [4] iteration: 2900 (of 40000)
## [8] iteration: 2900 (of 40000)
## [1] iteration: 2900 (of 40000)
## [3] iteration: 2900 (of 40000)
## [7] iteration: 3000 (of 40000)
## [2] iteration: 2900 (of 40000)
## [6] iteration: 3000 (of 40000)
## [5] iteration: 3000 (of 40000)
## [4] iteration: 3000 (of 40000)
## [8] iteration: 3000 (of 40000)
## [1] iteration: 3000 (of 40000)
## [3] iteration: 3000 (of 40000)
## [7] iteration: 3100 (of 40000)
## [2] iteration: 3000 (of 40000)
## [6] iteration: 3100 (of 40000)
## [5] iteration: 3100 (of 40000)
## [4] iteration: 3100 (of 40000)
## [8] iteration: 3100 (of 40000)
## [1] iteration: 3100 (of 40000)
## [3] iteration: 3100 (of 40000)
## [7] iteration: 3200 (of 40000)
## [2] iteration: 3100 (of 40000)
```

```
## [6] iteration: 3200 (of 40000)
## [5] iteration: 3200 (of 40000)
## [4] iteration: 3200 (of 40000)
## [8] iteration: 3200 (of 40000)
## [1] iteration: 3200 (of 40000)
## [7] iteration: 3300 (of 40000)
## [3] iteration: 3200 (of 40000)
## [2] iteration: 3200 (of 40000)
## [6] iteration: 3300 (of 40000)
## [5] iteration: 3300 (of 40000)
## [4] iteration: 3300 (of 40000)
## [8] iteration: 3300 (of 40000)
## [1] iteration: 3300 (of 40000)
## [7] iteration: 3400 (of 40000)
## [3] iteration: 3300 (of 40000)
## [2] iteration: 3300 (of 40000)
## [6] iteration: 3400 (of 40000)
## [5] iteration: 3400 (of 40000)
## [4] iteration: 3400 (of 40000)
## [8] iteration: 3400 (of 40000)
## [1] iteration: 3400 (of 40000)
## [7] iteration: 3500 (of 40000)
## [3] iteration: 3400 (of 40000)
## [2] iteration: 3400 (of 40000)
## [6] iteration: 3500 (of 40000)
## [5] iteration: 3500 (of 40000)
## [4] iteration: 3500 (of 40000)
## [8] iteration: 3500 (of 40000)
## [1] iteration: 3500 (of 40000)
## [7] iteration: 3600 (of 40000)
## [3] iteration: 3500 (of 40000)
## [2] iteration: 3500 (of 40000)
## [6] iteration: 3600 (of 40000)
## [5] iteration: 3600 (of 40000)
## [4] iteration: 3600 (of 40000)
## [8] iteration: 3600 (of 40000)
## [7] iteration: 3700 (of 40000)
## [1] iteration: 3600 (of 40000)
## [3] iteration: 3600 (of 40000)
## [2] iteration: 3600 (of 40000)
## [6] iteration: 3700 (of 40000)
## [5] iteration: 3700 (of 40000)
## [4] iteration: 3700 (of 40000)
## [8] iteration: 3700 (of 40000)
## [7] iteration: 3800 (of 40000)
## [1] iteration: 3700 (of 40000)
## [3] iteration: 3700 (of 40000)
## [2] iteration: 3700 (of 40000)
## [6] iteration: 3800 (of 40000)
## [5] iteration: 3800 (of 40000)
## [4] iteration: 3800 (of 40000)
## [7] iteration: 3900 (of 40000)
## [8] iteration: 3800 (of 40000)
## [1] iteration: 3800 (of 40000)
## [3] iteration: 3800 (of 40000)
## [2] iteration: 3800 (of 40000)
```

```
## [6] iteration: 3900 (of 40000)
## [5] iteration: 3900 (of 40000)
## [4] iteration: 3900 (of 40000)
## [7] iteration: 4000 (of 40000)
## [8] iteration: 3900 (of 40000)
## [1] iteration: 3900 (of 40000)
## [3] iteration: 3900 (of 40000)
## [2] iteration: 3900 (of 40000)
## [6] iteration: 4000 (of 40000)
## [5] iteration: 4000 (of 40000)
## [4] iteration: 4000 (of 40000)
## [7] iteration: 4100 (of 40000)
## [8] iteration: 4000 (of 40000)
## [1] iteration: 4000 (of 40000)
## [3] iteration: 4000 (of 40000)
## [2] iteration: 4000 (of 40000)
## [6] iteration: 4100 (of 40000)
## [5] iteration: 4100 (of 40000)
## [7] iteration: 4200 (of 40000)
## [4] iteration: 4100 (of 40000)
## [8] iteration: 4100 (of 40000)
## [1] iteration: 4100 (of 40000)
## [3] iteration: 4100 (of 40000)
## [2] iteration: 4100 (of 40000)
## [6] iteration: 4200 (of 40000)
## [5] iteration: 4200 (of 40000)
## [7] iteration: 4300 (of 40000)
## [4] iteration: 4200 (of 40000)
## [8] iteration: 4200 (of 40000)
## [1] iteration: 4200 (of 40000)
## [3] iteration: 4200 (of 40000)
## [2] iteration: 4200 (of 40000)
## [6] iteration: 4300 (of 40000)
## [5] iteration: 4300 (of 40000)
## [7] iteration: 4400 (of 40000)
## [4] iteration: 4300 (of 40000)
## [8] iteration: 4300 (of 40000)
## [1] iteration: 4300 (of 40000)
## [3] iteration: 4300 (of 40000)
## [2] iteration: 4300 (of 40000)
## [6] iteration: 4400 (of 40000)
## [5] iteration: 4400 (of 40000)
## [7] iteration: 4500 (of 40000)
## [4] iteration: 4400 (of 40000)
## [8] iteration: 4400 (of 40000)
## [1] iteration: 4400 (of 40000)
## [3] iteration: 4400 (of 40000)
## [2] iteration: 4400 (of 40000)
## [6] iteration: 4500 (of 40000)
## [5] iteration: 4500 (of 40000)
## [7] iteration: 4600 (of 40000)
## [4] iteration: 4500 (of 40000)
## [8] iteration: 4500 (of 40000)
## [1] iteration: 4500 (of 40000)
## [3] iteration: 4500 (of 40000)
## [2] iteration: 4500 (of 40000)
```

```
## [6] iteration: 4600 (of 40000)
## [5] iteration: 4600 (of 40000)
## [7] iteration: 4700 (of 40000)
## [4] iteration: 4600 (of 40000)
## [8] iteration: 4600 (of 40000)
## [1] iteration: 4600 (of 40000)
## [3] iteration: 4600 (of 40000)
## [6] iteration: 4700 (of 40000)
## [2] iteration: 4600 (of 40000)
## [5] iteration: 4700 (of 40000)
## [7] iteration: 4800 (of 40000)
## [4] iteration: 4700 (of 40000)
## [8] iteration: 4700 (of 40000)
## [1] iteration: 4700 (of 40000)
## [3] iteration: 4700 (of 40000)
## [6] iteration: 4800 (of 40000)
## [2] iteration: 4700 (of 40000)
## [5] iteration: 4800 (of 40000)
## [7] iteration: 4900 (of 40000)
## [4] iteration: 4800 (of 40000)
## [8] iteration: 4800 (of 40000)
## [1] iteration: 4800 (of 40000)
## [3] iteration: 4800 (of 40000)
## [6] iteration: 4900 (of 40000)
## [2] iteration: 4800 (of 40000)
## [5] iteration: 4900 (of 40000)
## [7] iteration: 5000 (of 40000)
## [4] iteration: 4900 (of 40000)
## [8] iteration: 4900 (of 40000)
## [1] iteration: 4900 (of 40000)
## [3] iteration: 4900 (of 40000)
## [6] iteration: 5000 (of 40000)
## [2] iteration: 4900 (of 40000)
## [5] iteration: 5000 (of 40000)
## [7] iteration: 5100 (of 40000)
## [4] iteration: 5000 (of 40000)
## [1] iteration: 5000 (of 40000)
## [8] iteration: 5000 (of 40000)
## [6] iteration: 5100 (of 40000)
## [3] iteration: 5000 (of 40000)
## [2] iteration: 5000 (of 40000)
## [5] iteration: 5100 (of 40000)
## [7] iteration: 5200 (of 40000)
## [4] iteration: 5100 (of 40000)
## [1] iteration: 5100 (of 40000)
## [8] iteration: 5100 (of 40000)
## [6] iteration: 5200 (of 40000)
## [3] iteration: 5100 (of 40000)
## [5] iteration: 5200 (of 40000)
## [2] iteration: 5100 (of 40000)
## [7] iteration: 5300 (of 40000)
## [4] iteration: 5200 (of 40000)
## [1] iteration: 5200 (of 40000)
## [8] iteration: 5200 (of 40000)
## [6] iteration: 5300 (of 40000)
## [3] iteration: 5200 (of 40000)
```

```
## [5] iteration: 5300 (of 40000)
## [2] iteration: 5200 (of 40000)
## [7] iteration: 5400 (of 40000)
## [4] iteration: 5300 (of 40000)
## [1] iteration: 5300 (of 40000)
## [8] iteration: 5300 (of 40000)
## [6] iteration: 5400 (of 40000)
## [3] iteration: 5300 (of 40000)
## [5] iteration: 5400 (of 40000)
## [2] iteration: 5300 (of 40000)
## [7] iteration: 5500 (of 40000)
## [4] iteration: 5400 (of 40000)
## [1] iteration: 5400 (of 40000)
## [8] iteration: 5400 (of 40000)
## [6] iteration: 5500 (of 40000)
## [3] iteration: 5400 (of 40000)
## [5] iteration: 5500 (of 40000)
## [2] iteration: 5400 (of 40000)
## [7] iteration: 5600 (of 40000)
## [4] iteration: 5500 (of 40000)
## [1] iteration: 5500 (of 40000)
## [8] iteration: 5500 (of 40000)
## [6] iteration: 5600 (of 40000)
## [3] iteration: 5500 (of 40000)
## [5] iteration: 5600 (of 40000)
## [2] iteration: 5500 (of 40000)
## [7] iteration: 5700 (of 40000)
## [4] iteration: 5600 (of 40000)
## [1] iteration: 5600 (of 40000)
## [8] iteration: 5600 (of 40000)
## [6] iteration: 5700 (of 40000)
## [3] iteration: 5600 (of 40000)
## [5] iteration: 5700 (of 40000)
## [2] iteration: 5600 (of 40000)
## [7] iteration: 5800 (of 40000)
## [4] iteration: 5700 (of 40000)
## [1] iteration: 5700 (of 40000)
## [8] iteration: 5700 (of 40000)
## [6] iteration: 5800 (of 40000)
## [3] iteration: 5700 (of 40000)
## [5] iteration: 5800 (of 40000)
## [2] iteration: 5700 (of 40000)
## [7] iteration: 5900 (of 40000)
## [4] iteration: 5800 (of 40000)
## [1] iteration: 5800 (of 40000)
## [8] iteration: 5800 (of 40000)
## [6] iteration: 5900 (of 40000)
## [5] iteration: 5900 (of 40000)
## [3] iteration: 5800 (of 40000)
## [2] iteration: 5800 (of 40000)
## [7] iteration: 6000 (of 40000)
## [4] iteration: 5900 (of 40000)
## [1] iteration: 5900 (of 40000)
## [8] iteration: 5900 (of 40000)
## [6] iteration: 6000 (of 40000)
## [5] iteration: 6000 (of 40000)
```

```
## [3] iteration: 5900 (of 40000)
## [2] iteration: 5900 (of 40000)
## [7] iteration: 6100 (of 40000)
## [4] iteration: 6000 (of 40000)
## [1] iteration: 6000 (of 40000)
## [8] iteration: 6000 (of 40000)
## [6] iteration: 6100 (of 40000)
## [5] iteration: 6100 (of 40000)
## [3] iteration: 6000 (of 40000)
## [2] iteration: 6000 (of 40000)
## [7] iteration: 6200 (of 40000)
## [4] iteration: 6100 (of 40000)
## [1] iteration: 6100 (of 40000)
## [8] iteration: 6100 (of 40000)
## [6] iteration: 6200 (of 40000)
## [5] iteration: 6200 (of 40000)
## [3] iteration: 6100 (of 40000)
## [7] iteration: 6300 (of 40000)
## [2] iteration: 6100 (of 40000)
## [4] iteration: 6200 (of 40000)
## [1] iteration: 6200 (of 40000)
## [8] iteration: 6200 (of 40000)
## [6] iteration: 6300 (of 40000)
## [5] iteration: 6300 (of 40000)
## [7] iteration: 6400 (of 40000)
## [3] iteration: 6200 (of 40000)
## [2] iteration: 6200 (of 40000)
## [1] iteration: 6300 (of 40000)
## [4] iteration: 6300 (of 40000)
## [8] iteration: 6300 (of 40000)
## [6] iteration: 6400 (of 40000)
## [5] iteration: 6400 (of 40000)
## [7] iteration: 6500 (of 40000)
## [3] iteration: 6300 (of 40000)
## [2] iteration: 6300 (of 40000)
## [1] iteration: 6400 (of 40000)
## [4] iteration: 6400 (of 40000)
## [8] iteration: 6400 (of 40000)
## [6] iteration: 6500 (of 40000)
## [5] iteration: 6500 (of 40000)
## [7] iteration: 6600 (of 40000)
## [3] iteration: 6400 (of 40000)
## [2] iteration: 6400 (of 40000)
## [1] iteration: 6500 (of 40000)
## [4] iteration: 6500 (of 40000)
## [8] iteration: 6500 (of 40000)
## [6] iteration: 6600 (of 40000)
## [5] iteration: 6600 (of 40000)
## [7] iteration: 6700 (of 40000)
## [3] iteration: 6500 (of 40000)
## [2] iteration: 6500 (of 40000)
## [1] iteration: 6600 (of 40000)
## [4] iteration: 6600 (of 40000)
## [8] iteration: 6600 (of 40000)
## [6] iteration: 6700 (of 40000)
## [5] iteration: 6700 (of 40000)
```

```
## [7] iteration: 6800 (of 40000)
## [3] iteration: 6600 (of 40000)
## [2] iteration: 6600 (of 40000)
## [1] iteration: 6700 (of 40000)
## [4] iteration: 6700 (of 40000)
## [8] iteration: 6700 (of 40000)
## [6] iteration: 6800 (of 40000)
## [5] iteration: 6800 (of 40000)
## [7] iteration: 6900 (of 40000)
## [3] iteration: 6700 (of 40000)
## [2] iteration: 6700 (of 40000)
## [1] iteration: 6800 (of 40000)
## [4] iteration: 6800 (of 40000)
## [6] iteration: 6900 (of 40000)
## [8] iteration: 6800 (of 40000)
## [5] iteration: 6900 (of 40000)
## [7] iteration: 7000 (of 40000)
## [3] iteration: 6800 (of 40000)
## [2] iteration: 6800 (of 40000)
## [1] iteration: 6900 (of 40000)
## [4] iteration: 6900 (of 40000)
## [6] iteration: 7000 (of 40000)
## [8] iteration: 6900 (of 40000)
## [5] iteration: 7000 (of 40000)
## [7] iteration: 7100 (of 40000)
## [3] iteration: 6900 (of 40000)
## [2] iteration: 6900 (of 40000)
## [1] iteration: 7000 (of 40000)
## [4] iteration: 7000 (of 40000)
## [6] iteration: 7100 (of 40000)
## [8] iteration: 7000 (of 40000)
## [5] iteration: 7100 (of 40000)
## [7] iteration: 7200 (of 40000)
## [2] iteration: 7000 (of 40000)
## [3] iteration: 7000 (of 40000)
## [1] iteration: 7100 (of 40000)
## [4] iteration: 7100 (of 40000)
## [6] iteration: 7200 (of 40000)
## [8] iteration: 7100 (of 40000)
## [5] iteration: 7200 (of 40000)
## [7] iteration: 7300 (of 40000)
## [3] iteration: 7100 (of 40000)
## [2] iteration: 7100 (of 40000)
## [1] iteration: 7200 (of 40000)
## [4] iteration: 7200 (of 40000)
## [6] iteration: 7300 (of 40000)
## [8] iteration: 7200 (of 40000)
## [5] iteration: 7300 (of 40000)
## [7] iteration: 7400 (of 40000)
## [3] iteration: 7200 (of 40000)
## [2] iteration: 7200 (of 40000)
## [1] iteration: 7300 (of 40000)
## [4] iteration: 7300 (of 40000)
## [6] iteration: 7400 (of 40000)
## [8] iteration: 7300 (of 40000)
## [5] iteration: 7400 (of 40000)
```

```
## [7] iteration: 7500 (of 40000)
## [2] iteration: 7300 (of 40000)
## [3] iteration: 7300 (of 40000)
## [1] iteration: 7400 (of 40000)
## [4] iteration: 7400 (of 40000)
## [6] iteration: 7500 (of 40000)
## [8] iteration: 7400 (of 40000)
## [5] iteration: 7500 (of 40000)
## [7] iteration: 7600 (of 40000)
## [3] iteration: 7400 (of 40000)
## [2] iteration: 7400 (of 40000)
## [1] iteration: 7500 (of 40000)
## [4] iteration: 7500 (of 40000)
## [6] iteration: 7600 (of 40000)
## [8] iteration: 7500 (of 40000)
## [7] iteration: 7700 (of 40000)
## [5] iteration: 7600 (of 40000)
## [3] iteration: 7500 (of 40000)
## [2] iteration: 7500 (of 40000)
## [1] iteration: 7600 (of 40000)
## [4] iteration: 7600 (of 40000)
## [6] iteration: 7700 (of 40000)
## [8] iteration: 7600 (of 40000)
## [7] iteration: 7800 (of 40000)
## [5] iteration: 7700 (of 40000)
## [3] iteration: 7600 (of 40000)
## [2] iteration: 7600 (of 40000)
## [1] iteration: 7700 (of 40000)
## [4] iteration: 7700 (of 40000)
## [6] iteration: 7800 (of 40000)
## [8] iteration: 7700 (of 40000)
## [7] iteration: 7900 (of 40000)
## [5] iteration: 7800 (of 40000)
## [3] iteration: 7700 (of 40000)
## [2] iteration: 7700 (of 40000)
## [1] iteration: 7800 (of 40000)
## [4] iteration: 7800 (of 40000)
## [6] iteration: 7900 (of 40000)
## [8] iteration: 7800 (of 40000)
## [7] iteration: 8000 (of 40000)
## [5] iteration: 7900 (of 40000)
## [3] iteration: 7800 (of 40000)
## [2] iteration: 7800 (of 40000)
## [1] iteration: 7900 (of 40000)
## [4] iteration: 7900 (of 40000)
## [6] iteration: 8000 (of 40000)
## [8] iteration: 7900 (of 40000)
## [7] iteration: 8100 (of 40000)
## [5] iteration: 8000 (of 40000)
## [3] iteration: 7900 (of 40000)
## [2] iteration: 7900 (of 40000)
## [1] iteration: 8000 (of 40000)
## [4] iteration: 8000 (of 40000)
## [6] iteration: 8100 (of 40000)
## [7] iteration: 8200 (of 40000)
## [8] iteration: 8000 (of 40000)
```

```
## [5] iteration: 8100 (of 40000)
## [3] iteration: 8000 (of 40000)
## [2] iteration: 8000 (of 40000)
## [1] iteration: 8100 (of 40000)
## [4] iteration: 8100 (of 40000)
## [6] iteration: 8200 (of 40000)
## [7] iteration: 8300 (of 40000)
## [8] iteration: 8100 (of 40000)
## [5] iteration: 8200 (of 40000)
## [3] iteration: 8100 (of 40000)
## [1] iteration: 8200 (of 40000)
## [2] iteration: 8100 (of 40000)
## [4] iteration: 8200 (of 40000)
## [6] iteration: 8300 (of 40000)
## [7] iteration: 8400 (of 40000)
## [8] iteration: 8200 (of 40000)
## [5] iteration: 8300 (of 40000)
## [3] iteration: 8200 (of 40000)
## [1] iteration: 8300 (of 40000)
## [2] iteration: 8200 (of 40000)
## [4] iteration: 8300 (of 40000)
## [6] iteration: 8400 (of 40000)
## [7] iteration: 8500 (of 40000)
## [8] iteration: 8300 (of 40000)
## [5] iteration: 8400 (of 40000)
## [3] iteration: 8300 (of 40000)
## [1] iteration: 8400 (of 40000)
## [2] iteration: 8300 (of 40000)
## [6] iteration: 8500 (of 40000)
## [4] iteration: 8400 (of 40000)
## [7] iteration: 8600 (of 40000)
## [8] iteration: 8400 (of 40000)
## [5] iteration: 8500 (of 40000)
## [3] iteration: 8400 (of 40000)
## [1] iteration: 8500 (of 40000)
## [2] iteration: 8400 (of 40000)
## [6] iteration: 8600 (of 40000)
## [7] iteration: 8700 (of 40000)
## [4] iteration: 8500 (of 40000)
## [8] iteration: 8500 (of 40000)
## [5] iteration: 8600 (of 40000)
## [3] iteration: 8500 (of 40000)
## [1] iteration: 8600 (of 40000)
## [2] iteration: 8500 (of 40000)
## [7] iteration: 8800 (of 40000)
## [6] iteration: 8700 (of 40000)
## [4] iteration: 8600 (of 40000)
## [8] iteration: 8600 (of 40000)
## [5] iteration: 8700 (of 40000)
## [1] iteration: 8700 (of 40000)
## [3] iteration: 8600 (of 40000)
## [2] iteration: 8600 (of 40000)
## [7] iteration: 8900 (of 40000)
## [6] iteration: 8800 (of 40000)
## [4] iteration: 8700 (of 40000)
## [8] iteration: 8700 (of 40000)
```

```
## [5] iteration: 8800 (of 40000)
## [1] iteration: 8800 (of 40000)
## [3] iteration: 8700 (of 40000)
## [2] iteration: 8700 (of 40000)
## [7] iteration: 9000 (of 40000)
## [6] iteration: 8900 (of 40000)
## [4] iteration: 8800 (of 40000)
## [8] iteration: 8800 (of 40000)
## [5] iteration: 8900 (of 40000)
## [1] iteration: 8900 (of 40000)
## [3] iteration: 8800 (of 40000)
## [2] iteration: 8800 (of 40000)
## [7] iteration: 9100 (of 40000)
## [6] iteration: 9000 (of 40000)
## [8] iteration: 8900 (of 40000)
## [4] iteration: 8900 (of 40000)
## [5] iteration: 9000 (of 40000)
## [1] iteration: 9000 (of 40000)
## [3] iteration: 8900 (of 40000)
## [7] iteration: 9200 (of 40000)
## [2] iteration: 8900 (of 40000)
## [6] iteration: 9100 (of 40000)
## [8] iteration: 9000 (of 40000)
## [4] iteration: 9000 (of 40000)
## [5] iteration: 9100 (of 40000)
## [1] iteration: 9100 (of 40000)
## [3] iteration: 9000 (of 40000)
## [7] iteration: 9300 (of 40000)
## [2] iteration: 9000 (of 40000)
## [6] iteration: 9200 (of 40000)
## [8] iteration: 9100 (of 40000)
## [4] iteration: 9100 (of 40000)
## [5] iteration: 9200 (of 40000)
## [1] iteration: 9200 (of 40000)
## [3] iteration: 9100 (of 40000)
## [7] iteration: 9400 (of 40000)
## [2] iteration: 9100 (of 40000)
## [6] iteration: 9300 (of 40000)
## [8] iteration: 9200 (of 40000)
## [4] iteration: 9200 (of 40000)
## [5] iteration: 9300 (of 40000)
## [1] iteration: 9300 (of 40000)
## [3] iteration: 9200 (of 40000)
## [7] iteration: 9500 (of 40000)
## [2] iteration: 9200 (of 40000)
## [6] iteration: 9400 (of 40000)
## [8] iteration: 9300 (of 40000)
## [4] iteration: 9300 (of 40000)
## [5] iteration: 9400 (of 40000)
## [1] iteration: 9400 (of 40000)
## [3] iteration: 9300 (of 40000)
## [7] iteration: 9600 (of 40000)
## [2] iteration: 9300 (of 40000)
## [6] iteration: 9500 (of 40000)
## [8] iteration: 9400 (of 40000)
## [5] iteration: 9500 (of 40000)
```

```
## [4] iteration: 9400 (of 40000)
## [1] iteration: 9500 (of 40000)
## [3] iteration: 9400 (of 40000)
## [7] iteration: 9700 (of 40000)
## [2] iteration: 9400 (of 40000)
## [6] iteration: 9600 (of 40000)
## [8] iteration: 9500 (of 40000)
## [5] iteration: 9600 (of 40000)
## [4] iteration: 9500 (of 40000)
## [1] iteration: 9600 (of 40000)
## [3] iteration: 9500 (of 40000)
## [7] iteration: 9800 (of 40000)
## [6] iteration: 9700 (of 40000)
## [2] iteration: 9500 (of 40000)
## [8] iteration: 9600 (of 40000)
## [5] iteration: 9700 (of 40000)
## [4] iteration: 9600 (of 40000)
## [1] iteration: 9700 (of 40000)
## [7] iteration: 9900 (of 40000)
## [3] iteration: 9600 (of 40000)
## [6] iteration: 9800 (of 40000)
## [2] iteration: 9600 (of 40000)
## [8] iteration: 9700 (of 40000)
## [5] iteration: 9800 (of 40000)
## [4] iteration: 9700 (of 40000)
## [1] iteration: 9800 (of 40000)
## [7] iteration: 10000 (of 40000)
## [3] iteration: 9700 (of 40000)
## [6] iteration: 9900 (of 40000)
## [8] iteration: 9800 (of 40000)
## [2] iteration: 9700 (of 40000)
## [5] iteration: 9900 (of 40000)
## [4] iteration: 9800 (of 40000)
## [1] iteration: 9900 (of 40000)
## [7] iteration: 10100 (of 40000)
## [3] iteration: 9800 (of 40000)
## [6] iteration: 10000 (of 40000)
## [8] iteration: 9900 (of 40000)
## [5] iteration: 10000 (of 40000)
## [2] iteration: 9800 (of 40000)
## [4] iteration: 9900 (of 40000)
## [1] iteration: 10000 (of 40000)
## [7] iteration: 10200 (of 40000)
## [3] iteration: 9900 (of 40000)
## [6] iteration: 10100 (of 40000)
## [5] iteration: 10100 (of 40000)
## [8] iteration: 10000 (of 40000)
## [2] iteration: 9900 (of 40000)
## [4] iteration: 10000 (of 40000)
## [7] iteration: 10300 (of 40000)
## [1] iteration: 10100 (of 40000)
## [3] iteration: 10000 (of 40000)
## [6] iteration: 10200 (of 40000)
## [5] iteration: 10200 (of 40000)
## [8] iteration: 10100 (of 40000)
## [2] iteration: 10000 (of 40000)
```

```
## [4] iteration: 10100 (of 40000)
## [7] iteration: 10400 (of 40000)
## [1] iteration: 10200 (of 40000)
## [3] iteration: 10100 (of 40000)
## [6] iteration: 10300 (of 40000)
## [5] iteration: 10300 (of 40000)
## [8] iteration: 10200 (of 40000)
## [4] iteration: 10200 (of 40000)
## [2] iteration: 10100 (of 40000)
## [7] iteration: 10500 (of 40000)
## [1] iteration: 10300 (of 40000)
## [3] iteration: 10200 (of 40000)
## [6] iteration: 10400 (of 40000)
## [5] iteration: 10400 (of 40000)
## [8] iteration: 10300 (of 40000)
## [4] iteration: 10300 (of 40000)
## [2] iteration: 10200 (of 40000)
## [7] iteration: 10600 (of 40000)
## [1] iteration: 10400 (of 40000)
## [3] iteration: 10300 (of 40000)
## [6] iteration: 10500 (of 40000)
## [5] iteration: 10500 (of 40000)
## [8] iteration: 10400 (of 40000)
## [4] iteration: 10400 (of 40000)
## [2] iteration: 10300 (of 40000)
## [7] iteration: 10700 (of 40000)
## [1] iteration: 10500 (of 40000)
## [6] iteration: 10600 (of 40000)
## [3] iteration: 10400 (of 40000)
## [5] iteration: 10600 (of 40000)
## [8] iteration: 10500 (of 40000)
## [4] iteration: 10500 (of 40000)
## [2] iteration: 10400 (of 40000)
## [7] iteration: 10800 (of 40000)
## [1] iteration: 10600 (of 40000)
## [6] iteration: 10700 (of 40000)
## [3] iteration: 10500 (of 40000)
## [5] iteration: 10700 (of 40000)
## [8] iteration: 10600 (of 40000)
## [4] iteration: 10600 (of 40000)
## [2] iteration: 10500 (of 40000)
## [7] iteration: 10900 (of 40000)
## [1] iteration: 10700 (of 40000)
## [6] iteration: 10800 (of 40000)
## [3] iteration: 10600 (of 40000)
## [5] iteration: 10800 (of 40000)
## [8] iteration: 10700 (of 40000)
## [4] iteration: 10700 (of 40000)
## [2] iteration: 10600 (of 40000)
## [7] iteration: 11000 (of 40000)
## [1] iteration: 10800 (of 40000)
## [6] iteration: 10900 (of 40000)
## [3] iteration: 10700 (of 40000)
## [5] iteration: 10900 (of 40000)
## [8] iteration: 10800 (of 40000)
## [4] iteration: 10800 (of 40000)
```

```
## [2] iteration: 10700 (of 40000)
## [7] iteration: 11100 (of 40000)
## [1] iteration: 10900 (of 40000)
## [6] iteration: 11000 (of 40000)
## [3] iteration: 10800 (of 40000)
## [5] iteration: 11000 (of 40000)
## [8] iteration: 10900 (of 40000)
## [4] iteration: 10900 (of 40000)
## [2] iteration: 10800 (of 40000)
## [7] iteration: 11200 (of 40000)
## [1] iteration: 11000 (of 40000)
## [6] iteration: 11100 (of 40000)
## [5] iteration: 11100 (of 40000)
## [3] iteration: 10900 (of 40000)
## [8] iteration: 11000 (of 40000)
## [4] iteration: 11000 (of 40000)
## [2] iteration: 10900 (of 40000)
## [7] iteration: 11300 (of 40000)
## [1] iteration: 11100 (of 40000)
## [5] iteration: 11200 (of 40000)
## [6] iteration: 11200 (of 40000)
## [3] iteration: 11000 (of 40000)
## [8] iteration: 11100 (of 40000)
## [4] iteration: 11100 (of 40000)
## [2] iteration: 11000 (of 40000)
## [7] iteration: 11400 (of 40000)
## [1] iteration: 11200 (of 40000)
## [5] iteration: 11300 (of 40000)
## [6] iteration: 11300 (of 40000)
## [3] iteration: 11100 (of 40000)
## [8] iteration: 11200 (of 40000)
## [4] iteration: 11200 (of 40000)
## [2] iteration: 11100 (of 40000)
## [7] iteration: 11500 (of 40000)
## [1] iteration: 11300 (of 40000)
## [5] iteration: 11400 (of 40000)
## [6] iteration: 11400 (of 40000)
## [3] iteration: 11200 (of 40000)
## [8] iteration: 11300 (of 40000)
## [4] iteration: 11300 (of 40000)
## [2] iteration: 11200 (of 40000)
## [7] iteration: 11600 (of 40000)
## [1] iteration: 11400 (of 40000)
## [5] iteration: 11500 (of 40000)
## [6] iteration: 11500 (of 40000)
## [3] iteration: 11300 (of 40000)
## [8] iteration: 11400 (of 40000)
## [4] iteration: 11400 (of 40000)
## [2] iteration: 11300 (of 40000)
## [7] iteration: 11700 (of 40000)
## [1] iteration: 11500 (of 40000)
## [5] iteration: 11600 (of 40000)
## [6] iteration: 11600 (of 40000)
## [3] iteration: 11400 (of 40000)
## [8] iteration: 11500 (of 40000)
## [4] iteration: 11500 (of 40000)
```

```
## [2] iteration: 11400 (of 40000)
## [7] iteration: 11800 (of 40000)
## [1] iteration: 11600 (of 40000)
## [5] iteration: 11700 (of 40000)
## [6] iteration: 11700 (of 40000)
## [3] iteration: 11500 (of 40000)
## [8] iteration: 11600 (of 40000)
## [4] iteration: 11600 (of 40000)
## [2] iteration: 11500 (of 40000)
## [7] iteration: 11900 (of 40000)
## [1] iteration: 11700 (of 40000)
## [5] iteration: 11800 (of 40000)
## [6] iteration: 11800 (of 40000)
## [3] iteration: 11600 (of 40000)
## [8] iteration: 11700 (of 40000)
## [4] iteration: 11700 (of 40000)
## [2] iteration: 11600 (of 40000)
## [7] iteration: 12000 (of 40000)
## [1] iteration: 11800 (of 40000)
## [5] iteration: 11900 (of 40000)
## [6] iteration: 11900 (of 40000)
## [3] iteration: 11700 (of 40000)
## [8] iteration: 11800 (of 40000)
## [4] iteration: 11800 (of 40000)
## [7] iteration: 12100 (of 40000)
## [2] iteration: 11700 (of 40000)
## [1] iteration: 11900 (of 40000)
## [5] iteration: 12000 (of 40000)
## [6] iteration: 12000 (of 40000)
## [3] iteration: 11800 (of 40000)
## [8] iteration: 11900 (of 40000)
## [4] iteration: 11900 (of 40000)
## [7] iteration: 12200 (of 40000)
## [2] iteration: 11800 (of 40000)
## [1] iteration: 12000 (of 40000)
## [5] iteration: 12100 (of 40000)
## [6] iteration: 12100 (of 40000)
## [3] iteration: 11900 (of 40000)
## [8] iteration: 12000 (of 40000)
## [4] iteration: 12000 (of 40000)
## [7] iteration: 12300 (of 40000)
## [2] iteration: 11900 (of 40000)
## [5] iteration: 12200 (of 40000)
## [1] iteration: 12100 (of 40000)
## [6] iteration: 12200 (of 40000)
## [8] iteration: 12100 (of 40000)
## [3] iteration: 12000 (of 40000)
## [4] iteration: 12100 (of 40000)
## [7] iteration: 12400 (of 40000)
## [2] iteration: 12000 (of 40000)
## [5] iteration: 12300 (of 40000)
## [1] iteration: 12200 (of 40000)
## [6] iteration: 12300 (of 40000)
## [8] iteration: 12200 (of 40000)
## [3] iteration: 12100 (of 40000)
## [7] iteration: 12500 (of 40000)
```

```
## [4] iteration: 12200 (of 40000)
## [2] iteration: 12100 (of 40000)
## [5] iteration: 12400 (of 40000)
## [1] iteration: 12300 (of 40000)
## [6] iteration: 12400 (of 40000)
## [8] iteration: 12300 (of 40000)
## [3] iteration: 12200 (of 40000)
## [7] iteration: 12600 (of 40000)
## [4] iteration: 12300 (of 40000)
## [2] iteration: 12200 (of 40000)
## [5] iteration: 12500 (of 40000)
## [1] iteration: 12400 (of 40000)
## [6] iteration: 12500 (of 40000)
## [8] iteration: 12400 (of 40000)
## [3] iteration: 12300 (of 40000)
## [7] iteration: 12700 (of 40000)
## [4] iteration: 12400 (of 40000)
## [2] iteration: 12300 (of 40000)
## [5] iteration: 12600 (of 40000)
## [6] iteration: 12600 (of 40000)
## [1] iteration: 12500 (of 40000)
## [8] iteration: 12500 (of 40000)
## [3] iteration: 12400 (of 40000)
## [7] iteration: 12800 (of 40000)
## [4] iteration: 12500 (of 40000)
## [2] iteration: 12400 (of 40000)
## [5] iteration: 12700 (of 40000)
## [6] iteration: 12700 (of 40000)
## [1] iteration: 12600 (of 40000)
## [8] iteration: 12600 (of 40000)
## [3] iteration: 12500 (of 40000)
## [7] iteration: 12900 (of 40000)
## [4] iteration: 12600 (of 40000)
## [2] iteration: 12500 (of 40000)
## [5] iteration: 12800 (of 40000)
## [6] iteration: 12800 (of 40000)
## [1] iteration: 12700 (of 40000)
## [8] iteration: 12700 (of 40000)
## [3] iteration: 12600 (of 40000)
## [7] iteration: 13000 (of 40000)
## [4] iteration: 12700 (of 40000)
## [2] iteration: 12600 (of 40000)
## [5] iteration: 12900 (of 40000)
## [6] iteration: 12900 (of 40000)
## [1] iteration: 12800 (of 40000)
## [8] iteration: 12800 (of 40000)
## [7] iteration: 13100 (of 40000)
## [3] iteration: 12700 (of 40000)
## [4] iteration: 12800 (of 40000)
## [2] iteration: 12700 (of 40000)
## [5] iteration: 13000 (of 40000)
## [6] iteration: 13000 (of 40000)
## [1] iteration: 12900 (of 40000)
## [8] iteration: 12900 (of 40000)
## [7] iteration: 13200 (of 40000)
## [3] iteration: 12800 (of 40000)
```

```
## [4] iteration: 12900 (of 40000)
## [2] iteration: 12800 (of 40000)
## [5] iteration: 13100 (of 40000)
## [6] iteration: 13100 (of 40000)
## [1] iteration: 13000 (of 40000)
## [7] iteration: 13300 (of 40000)
## [8] iteration: 13000 (of 40000)
## [3] iteration: 12900 (of 40000)
## [4] iteration: 13000 (of 40000)
## [2] iteration: 12900 (of 40000)
## [5] iteration: 13200 (of 40000)
## [6] iteration: 13200 (of 40000)
## [1] iteration: 13100 (of 40000)
## [7] iteration: 13400 (of 40000)
## [8] iteration: 13100 (of 40000)
## [3] iteration: 13000 (of 40000)
## [4] iteration: 13100 (of 40000)
## [2] iteration: 13000 (of 40000)
## [6] iteration: 13300 (of 40000)
## [5] iteration: 13300 (of 40000)
## [1] iteration: 13200 (of 40000)
## [7] iteration: 13500 (of 40000)
## [8] iteration: 13200 (of 40000)
## [3] iteration: 13100 (of 40000)
## [4] iteration: 13200 (of 40000)
## [2] iteration: 13100 (of 40000)
## [6] iteration: 13400 (of 40000)
## [5] iteration: 13400 (of 40000)
## [1] iteration: 13300 (of 40000)
## [7] iteration: 13600 (of 40000)
## [8] iteration: 13300 (of 40000)
## [3] iteration: 13200 (of 40000)
## [4] iteration: 13300 (of 40000)
## [6] iteration: 13500 (of 40000)
## [2] iteration: 13200 (of 40000)
## [5] iteration: 13500 (of 40000)
## [1] iteration: 13400 (of 40000)
## [7] iteration: 13700 (of 40000)
## [8] iteration: 13400 (of 40000)
## [3] iteration: 13300 (of 40000)
## [4] iteration: 13400 (of 40000)
## [6] iteration: 13600 (of 40000)
## [5] iteration: 13600 (of 40000)
## [2] iteration: 13300 (of 40000)
## [1] iteration: 13500 (of 40000)
## [7] iteration: 13800 (of 40000)
## [8] iteration: 13500 (of 40000)
## [3] iteration: 13400 (of 40000)
## [4] iteration: 13500 (of 40000)
## [6] iteration: 13700 (of 40000)
## [5] iteration: 13700 (of 40000)
## [2] iteration: 13400 (of 40000)
## [1] iteration: 13600 (of 40000)
## [7] iteration: 13900 (of 40000)
## [8] iteration: 13600 (of 40000)
## [3] iteration: 13500 (of 40000)
```

```
## [4] iteration: 13600 (of 40000)
## [6] iteration: 13800 (of 40000)
## [5] iteration: 13800 (of 40000)
## [2] iteration: 13500 (of 40000)
## [1] iteration: 13700 (of 40000)
## [7] iteration: 14000 (of 40000)
## [8] iteration: 13700 (of 40000)
## [3] iteration: 13600 (of 40000)
## [4] iteration: 13700 (of 40000)
## [6] iteration: 13900 (of 40000)
## [5] iteration: 13900 (of 40000)
## [1] iteration: 13800 (of 40000)
## [2] iteration: 13600 (of 40000)
## [7] iteration: 14100 (of 40000)
## [8] iteration: 13800 (of 40000)
## [3] iteration: 13700 (of 40000)
## [4] iteration: 13800 (of 40000)
## [6] iteration: 14000 (of 40000)
## [5] iteration: 14000 (of 40000)
## [1] iteration: 13900 (of 40000)
## [2] iteration: 13700 (of 40000)
## [7] iteration: 14200 (of 40000)
## [8] iteration: 13900 (of 40000)
## [3] iteration: 13800 (of 40000)
## [4] iteration: 13900 (of 40000)
## [6] iteration: 14100 (of 40000)
## [5] iteration: 14100 (of 40000)
## [1] iteration: 14000 (of 40000)
## [2] iteration: 13800 (of 40000)
## [7] iteration: 14300 (of 40000)
## [8] iteration: 14000 (of 40000)
## [3] iteration: 13900 (of 40000)
## [4] iteration: 14000 (of 40000)
## [6] iteration: 14200 (of 40000)
## [5] iteration: 14200 (of 40000)
## [1] iteration: 14100 (of 40000)
## [2] iteration: 13900 (of 40000)
## [7] iteration: 14400 (of 40000)
## [8] iteration: 14100 (of 40000)
## [3] iteration: 14000 (of 40000)
## [4] iteration: 14100 (of 40000)
## [6] iteration: 14300 (of 40000)
## [5] iteration: 14300 (of 40000)
## [1] iteration: 14200 (of 40000)
## [7] iteration: 14500 (of 40000)
## [2] iteration: 14000 (of 40000)
## [8] iteration: 14200 (of 40000)
## [3] iteration: 14100 (of 40000)
## [4] iteration: 14200 (of 40000)
## [6] iteration: 14400 (of 40000)
## [5] iteration: 14400 (of 40000)
## [1] iteration: 14300 (of 40000)
## [7] iteration: 14600 (of 40000)
## [2] iteration: 14100 (of 40000)
## [8] iteration: 14300 (of 40000)
## [3] iteration: 14200 (of 40000)
```

```
## [4] iteration: 14300 (of 40000)
## [6] iteration: 14500 (of 40000)
## [5] iteration: 14500 (of 40000)
## [1] iteration: 14400 (of 40000)
## [7] iteration: 14700 (of 40000)
## [2] iteration: 14200 (of 40000)
## [8] iteration: 14400 (of 40000)
## [3] iteration: 14300 (of 40000)
## [4] iteration: 14400 (of 40000)
## [6] iteration: 14600 (of 40000)
## [5] iteration: 14600 (of 40000)
## [1] iteration: 14500 (of 40000)
## [7] iteration: 14800 (of 40000)
## [2] iteration: 14300 (of 40000)
## [8] iteration: 14500 (of 40000)
## [3] iteration: 14400 (of 40000)
## [4] iteration: 14500 (of 40000)
## [6] iteration: 14700 (of 40000)
## [5] iteration: 14700 (of 40000)
## [1] iteration: 14600 (of 40000)
## [7] iteration: 14900 (of 40000)
## [2] iteration: 14400 (of 40000)
## [8] iteration: 14600 (of 40000)
## [3] iteration: 14500 (of 40000)
## [4] iteration: 14600 (of 40000)
## [6] iteration: 14800 (of 40000)
## [5] iteration: 14800 (of 40000)
## [7] iteration: 15000 (of 40000)
## [1] iteration: 14700 (of 40000)
## [2] iteration: 14500 (of 40000)
## [8] iteration: 14700 (of 40000)
## [3] iteration: 14600 (of 40000)
## [4] iteration: 14700 (of 40000)
## [6] iteration: 14900 (of 40000)
## [5] iteration: 14900 (of 40000)
## [7] iteration: 15100 (of 40000)
## [1] iteration: 14800 (of 40000)
## [2] iteration: 14600 (of 40000)
## [8] iteration: 14800 (of 40000)
## [3] iteration: 14700 (of 40000)
## [4] iteration: 14800 (of 40000)
## [6] iteration: 15000 (of 40000)
## [7] iteration: 15200 (of 40000)
## [5] iteration: 15000 (of 40000)
## [1] iteration: 14900 (of 40000)
## [2] iteration: 14700 (of 40000)
## [8] iteration: 14900 (of 40000)
## [3] iteration: 14800 (of 40000)
## [4] iteration: 14900 (of 40000)
## [6] iteration: 15100 (of 40000)
## [7] iteration: 15300 (of 40000)
## [5] iteration: 15100 (of 40000)
## [1] iteration: 15000 (of 40000)
## [2] iteration: 14800 (of 40000)
## [8] iteration: 15000 (of 40000)
## [3] iteration: 14900 (of 40000)
```

```
## [4] iteration: 15000 (of 40000)
## [6] iteration: 15200 (of 40000)
## [7] iteration: 15400 (of 40000)
## [5] iteration: 15200 (of 40000)
## [1] iteration: 15100 (of 40000)
## [8] iteration: 15100 (of 40000)
## [2] iteration: 14900 (of 40000)
## [3] iteration: 15000 (of 40000)
## [4] iteration: 15100 (of 40000)
## [6] iteration: 15300 (of 40000)
## [7] iteration: 15500 (of 40000)
## [5] iteration: 15300 (of 40000)
## [1] iteration: 15200 (of 40000)
## [8] iteration: 15200 (of 40000)
## [2] iteration: 15000 (of 40000)
## [3] iteration: 15100 (of 40000)
## [6] iteration: 15400 (of 40000)
## [4] iteration: 15200 (of 40000)
## [7] iteration: 15600 (of 40000)
## [5] iteration: 15400 (of 40000)
## [1] iteration: 15300 (of 40000)
## [8] iteration: 15300 (of 40000)
## [2] iteration: 15100 (of 40000)
## [3] iteration: 15200 (of 40000)
## [6] iteration: 15500 (of 40000)
## [4] iteration: 15300 (of 40000)
## [7] iteration: 15700 (of 40000)
## [5] iteration: 15500 (of 40000)
## [1] iteration: 15400 (of 40000)
## [8] iteration: 15400 (of 40000)
## [2] iteration: 15200 (of 40000)
## [3] iteration: 15300 (of 40000)
## [6] iteration: 15600 (of 40000)
## [4] iteration: 15400 (of 40000)
## [7] iteration: 15800 (of 40000)
## [5] iteration: 15600 (of 40000)
## [1] iteration: 15500 (of 40000)
## [8] iteration: 15500 (of 40000)
## [2] iteration: 15300 (of 40000)
## [3] iteration: 15400 (of 40000)
## [6] iteration: 15700 (of 40000)
## [4] iteration: 15500 (of 40000)
## [7] iteration: 15900 (of 40000)
## [5] iteration: 15700 (of 40000)
## [1] iteration: 15600 (of 40000)
## [8] iteration: 15600 (of 40000)
## [2] iteration: 15400 (of 40000)
## [6] iteration: 15800 (of 40000)
## [3] iteration: 15500 (of 40000)
## [4] iteration: 15600 (of 40000)
## [7] iteration: 16000 (of 40000)
## [5] iteration: 15800 (of 40000)
## [1] iteration: 15700 (of 40000)
## [8] iteration: 15700 (of 40000)
## [2] iteration: 15500 (of 40000)
## [6] iteration: 15900 (of 40000)
```

```
## [3] iteration: 15600 (of 40000)
## [4] iteration: 15700 (of 40000)
## [7] iteration: 16100 (of 40000)
## [5] iteration: 15900 (of 40000)
## [1] iteration: 15800 (of 40000)
## [8] iteration: 15800 (of 40000)
## [2] iteration: 15600 (of 40000)
## [6] iteration: 16000 (of 40000)
## [3] iteration: 15700 (of 40000)
## [4] iteration: 15800 (of 40000)
## [7] iteration: 16200 (of 40000)
## [5] iteration: 16000 (of 40000)
## [1] iteration: 15900 (of 40000)
## [8] iteration: 15900 (of 40000)
## [6] iteration: 16100 (of 40000)
## [2] iteration: 15700 (of 40000)
## [3] iteration: 15800 (of 40000)
## [4] iteration: 15900 (of 40000)
## [7] iteration: 16300 (of 40000)
## [5] iteration: 16100 (of 40000)
## [1] iteration: 16000 (of 40000)
## [8] iteration: 16000 (of 40000)
## [6] iteration: 16200 (of 40000)
## [2] iteration: 15800 (of 40000)
## [3] iteration: 15900 (of 40000)
## [4] iteration: 16000 (of 40000)
## [7] iteration: 16400 (of 40000)
## [1] iteration: 16100 (of 40000)
## [5] iteration: 16200 (of 40000)
## [8] iteration: 16100 (of 40000)
## [6] iteration: 16300 (of 40000)
## [2] iteration: 15900 (of 40000)
## [3] iteration: 16000 (of 40000)
## [4] iteration: 16100 (of 40000)
## [7] iteration: 16500 (of 40000)
## [1] iteration: 16200 (of 40000)
## [5] iteration: 16300 (of 40000)
## [8] iteration: 16200 (of 40000)
## [6] iteration: 16400 (of 40000)
## [2] iteration: 16000 (of 40000)
## [3] iteration: 16100 (of 40000)
## [4] iteration: 16200 (of 40000)
## [7] iteration: 16600 (of 40000)
## [1] iteration: 16300 (of 40000)
## [5] iteration: 16400 (of 40000)
## [8] iteration: 16300 (of 40000)
## [6] iteration: 16500 (of 40000)
## [2] iteration: 16100 (of 40000)
## [3] iteration: 16200 (of 40000)
## [4] iteration: 16300 (of 40000)
## [7] iteration: 16700 (of 40000)
## [1] iteration: 16400 (of 40000)
## [5] iteration: 16500 (of 40000)
## [8] iteration: 16400 (of 40000)
## [6] iteration: 16600 (of 40000)
## [2] iteration: 16200 (of 40000)
```

```
## [3] iteration: 16300 (of 40000)
## [4] iteration: 16400 (of 40000)
## [7] iteration: 16800 (of 40000)
## [5] iteration: 16600 (of 40000)
## [1] iteration: 16500 (of 40000)
## [8] iteration: 16500 (of 40000)
## [6] iteration: 16700 (of 40000)
## [2] iteration: 16300 (of 40000)
## [3] iteration: 16400 (of 40000)
## [4] iteration: 16500 (of 40000)
## [7] iteration: 16900 (of 40000)
## [5] iteration: 16700 (of 40000)
## [1] iteration: 16600 (of 40000)
## [8] iteration: 16600 (of 40000)
## [6] iteration: 16800 (of 40000)
## [2] iteration: 16400 (of 40000)
## [3] iteration: 16500 (of 40000)
## [4] iteration: 16600 (of 40000)
## [7] iteration: 17000 (of 40000)
## [5] iteration: 16800 (of 40000)
## [1] iteration: 16700 (of 40000)
## [8] iteration: 16700 (of 40000)
## [6] iteration: 16900 (of 40000)
## [3] iteration: 16600 (of 40000)
## [2] iteration: 16500 (of 40000)
## [7] iteration: 17100 (of 40000)
## [4] iteration: 16700 (of 40000)
## [5] iteration: 16900 (of 40000)
## [1] iteration: 16800 (of 40000)
## [8] iteration: 16800 (of 40000)
## [6] iteration: 17000 (of 40000)
## [3] iteration: 16700 (of 40000)
## [2] iteration: 16600 (of 40000)
## [7] iteration: 17200 (of 40000)
## [4] iteration: 16800 (of 40000)
## [5] iteration: 17000 (of 40000)
## [1] iteration: 16900 (of 40000)
## [8] iteration: 16900 (of 40000)
## [6] iteration: 17100 (of 40000)
## [7] iteration: 17300 (of 40000)
## [3] iteration: 16800 (of 40000)
## [2] iteration: 16700 (of 40000)
## [4] iteration: 16900 (of 40000)
## [5] iteration: 17100 (of 40000)
## [1] iteration: 17000 (of 40000)
## [8] iteration: 17000 (of 40000)
## [6] iteration: 17200 (of 40000)
## [7] iteration: 17400 (of 40000)
## [2] iteration: 16800 (of 40000)
## [3] iteration: 16900 (of 40000)
## [4] iteration: 17000 (of 40000)
## [5] iteration: 17200 (of 40000)
## [1] iteration: 17100 (of 40000)
## [8] iteration: 17100 (of 40000)
## [6] iteration: 17300 (of 40000)
## [7] iteration: 17500 (of 40000)
```

```
## [2] iteration: 16900 (of 40000)
## [3] iteration: 17000 (of 40000)
## [4] iteration: 17100 (of 40000)
## [5] iteration: 17300 (of 40000)
## [1] iteration: 17200 (of 40000)
## [8] iteration: 17200 (of 40000)
## [6] iteration: 17400 (of 40000)
## [7] iteration: 17600 (of 40000)
## [2] iteration: 17000 (of 40000)
## [3] iteration: 17100 (of 40000)
## [4] iteration: 17200 (of 40000)
## [5] iteration: 17400 (of 40000)
## [1] iteration: 17300 (of 40000)
## [8] iteration: 17300 (of 40000)
## [6] iteration: 17500 (of 40000)
## [7] iteration: 17700 (of 40000)
## [2] iteration: 17100 (of 40000)
## [3] iteration: 17200 (of 40000)
## [4] iteration: 17300 (of 40000)
## [5] iteration: 17500 (of 40000)
## [1] iteration: 17400 (of 40000)
## [8] iteration: 17400 (of 40000)
## [6] iteration: 17600 (of 40000)
## [7] iteration: 17800 (of 40000)
## [3] iteration: 17300 (of 40000)
## [2] iteration: 17200 (of 40000)
## [4] iteration: 17400 (of 40000)
## [5] iteration: 17600 (of 40000)
## [1] iteration: 17500 (of 40000)
## [8] iteration: 17500 (of 40000)
## [6] iteration: 17700 (of 40000)
## [7] iteration: 17900 (of 40000)
## [3] iteration: 17400 (of 40000)
## [2] iteration: 17300 (of 40000)
## [4] iteration: 17500 (of 40000)
## [5] iteration: 17700 (of 40000)
## [1] iteration: 17600 (of 40000)
## [8] iteration: 17600 (of 40000)
## [6] iteration: 17800 (of 40000)
## [7] iteration: 18000 (of 40000)
## [3] iteration: 17500 (of 40000)
## [2] iteration: 17400 (of 40000)
## [4] iteration: 17600 (of 40000)
## [5] iteration: 17800 (of 40000)
## [1] iteration: 17700 (of 40000)
## [8] iteration: 17700 (of 40000)
## [6] iteration: 17900 (of 40000)
## [7] iteration: 18100 (of 40000)
## [3] iteration: 17600 (of 40000)
## [4] iteration: 17700 (of 40000)
## [2] iteration: 17500 (of 40000)
## [5] iteration: 17900 (of 40000)
## [1] iteration: 17800 (of 40000)
## [8] iteration: 17800 (of 40000)
## [6] iteration: 18000 (of 40000)
## [7] iteration: 18200 (of 40000)
```

```
## [3] iteration: 17700 (of 40000)
## [4] iteration: 17800 (of 40000)
## [2] iteration: 17600 (of 40000)
## [5] iteration: 18000 (of 40000)
## [1] iteration: 17900 (of 40000)
## [8] iteration: 17900 (of 40000)
## [6] iteration: 18100 (of 40000)
## [7] iteration: 18300 (of 40000)
## [3] iteration: 17800 (of 40000)
## [2] iteration: 17700 (of 40000)
## [4] iteration: 17900 (of 40000)
## [5] iteration: 18100 (of 40000)
## [1] iteration: 18000 (of 40000)
## [8] iteration: 18000 (of 40000)
## [6] iteration: 18200 (of 40000)
## [7] iteration: 18400 (of 40000)
## [3] iteration: 17900 (of 40000)
## [2] iteration: 17800 (of 40000)
## [4] iteration: 18000 (of 40000)
## [5] iteration: 18200 (of 40000)
## [1] iteration: 18100 (of 40000)
## [8] iteration: 18100 (of 40000)
## [6] iteration: 18300 (of 40000)
## [7] iteration: 18500 (of 40000)
## [3] iteration: 18000 (of 40000)
## [2] iteration: 17900 (of 40000)
## [4] iteration: 18100 (of 40000)
## [5] iteration: 18300 (of 40000)
## [1] iteration: 18200 (of 40000)
## [8] iteration: 18200 (of 40000)
## [6] iteration: 18400 (of 40000)
## [7] iteration: 18600 (of 40000)
## [3] iteration: 18100 (of 40000)
## [2] iteration: 18000 (of 40000)
## [4] iteration: 18200 (of 40000)
## [5] iteration: 18400 (of 40000)
## [1] iteration: 18300 (of 40000)
## [6] iteration: 18500 (of 40000)
## [8] iteration: 18300 (of 40000)
## [7] iteration: 18700 (of 40000)
## [3] iteration: 18200 (of 40000)
## [4] iteration: 18300 (of 40000)
## [2] iteration: 18100 (of 40000)
## [5] iteration: 18500 (of 40000)
## [1] iteration: 18400 (of 40000)
## [6] iteration: 18600 (of 40000)
## [8] iteration: 18400 (of 40000)
## [7] iteration: 18800 (of 40000)
## [3] iteration: 18300 (of 40000)
## [4] iteration: 18400 (of 40000)
## [2] iteration: 18200 (of 40000)
## [5] iteration: 18600 (of 40000)
## [1] iteration: 18500 (of 40000)
## [6] iteration: 18700 (of 40000)
## [8] iteration: 18500 (of 40000)
## [7] iteration: 18900 (of 40000)
```

```
## [3] iteration: 18400 (of 40000)
## [4] iteration: 18500 (of 40000)
## [5] iteration: 18700 (of 40000)
## [2] iteration: 18300 (of 40000)
## [1] iteration: 18600 (of 40000)
## [6] iteration: 18800 (of 40000)
## [8] iteration: 18600 (of 40000)
## [7] iteration: 19000 (of 40000)
## [3] iteration: 18500 (of 40000)
## [4] iteration: 18600 (of 40000)
## [5] iteration: 18800 (of 40000)
## [2] iteration: 18400 (of 40000)
## [1] iteration: 18700 (of 40000)
## [6] iteration: 18900 (of 40000)
## [8] iteration: 18700 (of 40000)
## [7] iteration: 19100 (of 40000)
## [3] iteration: 18600 (of 40000)
## [4] iteration: 18700 (of 40000)
## [5] iteration: 18900 (of 40000)
## [2] iteration: 18500 (of 40000)
## [1] iteration: 18800 (of 40000)
## [6] iteration: 19000 (of 40000)
## [8] iteration: 18800 (of 40000)
## [7] iteration: 19200 (of 40000)
## [3] iteration: 18700 (of 40000)
## [4] iteration: 18800 (of 40000)
## [5] iteration: 19000 (of 40000)
## [2] iteration: 18600 (of 40000)
## [6] iteration: 19100 (of 40000)
## [1] iteration: 18900 (of 40000)
## [8] iteration: 18900 (of 40000)
## [7] iteration: 19300 (of 40000)
## [3] iteration: 18800 (of 40000)
## [4] iteration: 18900 (of 40000)
## [5] iteration: 19100 (of 40000)
## [2] iteration: 18700 (of 40000)
## [6] iteration: 19200 (of 40000)
## [1] iteration: 19000 (of 40000)
## [8] iteration: 19000 (of 40000)
## [7] iteration: 19400 (of 40000)
## [3] iteration: 18900 (of 40000)
## [4] iteration: 19000 (of 40000)
## [5] iteration: 19200 (of 40000)
## [2] iteration: 18800 (of 40000)
## [6] iteration: 19300 (of 40000)
## [1] iteration: 19100 (of 40000)
## [8] iteration: 19100 (of 40000)
## [7] iteration: 19500 (of 40000)
## [3] iteration: 19000 (of 40000)
## [4] iteration: 19100 (of 40000)
## [5] iteration: 19300 (of 40000)
## [2] iteration: 18900 (of 40000)
## [6] iteration: 19400 (of 40000)
## [1] iteration: 19200 (of 40000)
## [8] iteration: 19200 (of 40000)
## [7] iteration: 19600 (of 40000)
```

```
## [3] iteration: 19100 (of 40000)
## [4] iteration: 19200 (of 40000)
## [5] iteration: 19400 (of 40000)
## [6] iteration: 19500 (of 40000)
## [2] iteration: 19000 (of 40000)
## [1] iteration: 19300 (of 40000)
## [8] iteration: 19300 (of 40000)
## [7] iteration: 19700 (of 40000)
## [3] iteration: 19200 (of 40000)
## [4] iteration: 19300 (of 40000)
## [5] iteration: 19500 (of 40000)
## [6] iteration: 19600 (of 40000)
## [2] iteration: 19100 (of 40000)
## [1] iteration: 19400 (of 40000)
## [8] iteration: 19400 (of 40000)
## [7] iteration: 19800 (of 40000)
## [4] iteration: 19400 (of 40000)
## [3] iteration: 19300 (of 40000)
## [5] iteration: 19600 (of 40000)
## [6] iteration: 19700 (of 40000)
## [2] iteration: 19200 (of 40000)
## [1] iteration: 19500 (of 40000)
## [8] iteration: 19500 (of 40000)
## [7] iteration: 19900 (of 40000)
## [4] iteration: 19500 (of 40000)
## [3] iteration: 19400 (of 40000)
## [6] iteration: 19800 (of 40000)
## [5] iteration: 19700 (of 40000)
## [2] iteration: 19300 (of 40000)
## [1] iteration: 19600 (of 40000)
## [8] iteration: 19600 (of 40000)
## [7] iteration: 20000 (of 40000)
## [4] iteration: 19600 (of 40000)
## [3] iteration: 19500 (of 40000)
## [6] iteration: 19900 (of 40000)
## [5] iteration: 19800 (of 40000)
## [2] iteration: 19400 (of 40000)
## [1] iteration: 19700 (of 40000)
## [8] iteration: 19700 (of 40000)
## [7] iteration: 20100 (of 40000)
## [4] iteration: 19700 (of 40000)
## [6] iteration: 20000 (of 40000)
## [3] iteration: 19600 (of 40000)
## [5] iteration: 19900 (of 40000)
## [2] iteration: 19500 (of 40000)
## [1] iteration: 19800 (of 40000)
## [8] iteration: 19800 (of 40000)
## [7] iteration: 20200 (of 40000)
## [6] iteration: 20100 (of 40000)
## [4] iteration: 19800 (of 40000)
## [3] iteration: 19700 (of 40000)
## [5] iteration: 20000 (of 40000)
## [2] iteration: 19600 (of 40000)
## [8] iteration: 19900 (of 40000)
## [1] iteration: 19900 (of 40000)
## [7] iteration: 20300 (of 40000)
```

```
## [6] iteration: 20200 (of 40000)
## [4] iteration: 19900 (of 40000)
## [3] iteration: 19800 (of 40000)
## [5] iteration: 20100 (of 40000)
## [8] iteration: 20000 (of 40000)
## [2] iteration: 19700 (of 40000)
## [1] iteration: 20000 (of 40000)
## [7] iteration: 20400 (of 40000)
## [6] iteration: 20300 (of 40000)
## [4] iteration: 20000 (of 40000)
## [5] iteration: 20200 (of 40000)
## [3] iteration: 19900 (of 40000)
## [8] iteration: 20100 (of 40000)
## [1] iteration: 20100 (of 40000)
## [2] iteration: 19800 (of 40000)
## [7] iteration: 20500 (of 40000)
## [6] iteration: 20400 (of 40000)
## [4] iteration: 20100 (of 40000)
## [5] iteration: 20300 (of 40000)
## [3] iteration: 20000 (of 40000)
## [8] iteration: 20200 (of 40000)
## [1] iteration: 20200 (of 40000)
## [2] iteration: 19900 (of 40000)
## [7] iteration: 20600 (of 40000)
## [6] iteration: 20500 (of 40000)
## [4] iteration: 20200 (of 40000)
## [5] iteration: 20400 (of 40000)
## [3] iteration: 20100 (of 40000)
## [8] iteration: 20300 (of 40000)
## [1] iteration: 20300 (of 40000)
## [7] iteration: 20700 (of 40000)
## [2] iteration: 20000 (of 40000)
## [6] iteration: 20600 (of 40000)
## [4] iteration: 20300 (of 40000)
## [5] iteration: 20500 (of 40000)
## [3] iteration: 20200 (of 40000)
## [8] iteration: 20400 (of 40000)
## [1] iteration: 20400 (of 40000)
## [7] iteration: 20800 (of 40000)
## [2] iteration: 20100 (of 40000)
## [6] iteration: 20700 (of 40000)
## [4] iteration: 20400 (of 40000)
## [5] iteration: 20600 (of 40000)
## [3] iteration: 20300 (of 40000)
## [1] iteration: 20500 (of 40000)
## [8] iteration: 20500 (of 40000)
## [7] iteration: 20900 (of 40000)
## [2] iteration: 20200 (of 40000)
## [6] iteration: 20800 (of 40000)
## [4] iteration: 20500 (of 40000)
## [5] iteration: 20700 (of 40000)
## [3] iteration: 20400 (of 40000)
## [7] iteration: 21000 (of 40000)
## [1] iteration: 20600 (of 40000)
## [8] iteration: 20600 (of 40000)
## [2] iteration: 20300 (of 40000)
```

```
## [6] iteration: 20900 (of 40000)
## [4] iteration: 20600 (of 40000)
## [5] iteration: 20800 (of 40000)
## [3] iteration: 20500 (of 40000)
## [7] iteration: 21100 (of 40000)
## [1] iteration: 20700 (of 40000)
## [8] iteration: 20700 (of 40000)
## [2] iteration: 20400 (of 40000)
## [6] iteration: 21000 (of 40000)
## [4] iteration: 20700 (of 40000)
## [5] iteration: 20900 (of 40000)
## [3] iteration: 20600 (of 40000)
## [7] iteration: 21200 (of 40000)
## [1] iteration: 20800 (of 40000)
## [8] iteration: 20800 (of 40000)
## [2] iteration: 20500 (of 40000)
## [6] iteration: 21100 (of 40000)
## [4] iteration: 20800 (of 40000)
## [5] iteration: 21000 (of 40000)
## [3] iteration: 20700 (of 40000)
## [7] iteration: 21300 (of 40000)
## [1] iteration: 20900 (of 40000)
## [8] iteration: 20900 (of 40000)
## [2] iteration: 20600 (of 40000)
## [6] iteration: 21200 (of 40000)
## [4] iteration: 20900 (of 40000)
## [5] iteration: 21100 (of 40000)
## [7] iteration: 21400 (of 40000)
## [3] iteration: 20800 (of 40000)
## [1] iteration: 21000 (of 40000)
## [8] iteration: 21000 (of 40000)
## [2] iteration: 20700 (of 40000)
## [6] iteration: 21300 (of 40000)
## [4] iteration: 21000 (of 40000)
## [5] iteration: 21200 (of 40000)
## [7] iteration: 21500 (of 40000)
## [3] iteration: 20900 (of 40000)
## [1] iteration: 21100 (of 40000)
## [8] iteration: 21100 (of 40000)
## [2] iteration: 20800 (of 40000)
## [6] iteration: 21400 (of 40000)
## [4] iteration: 21100 (of 40000)
## [7] iteration: 21600 (of 40000)
## [5] iteration: 21300 (of 40000)
## [3] iteration: 21000 (of 40000)
## [1] iteration: 21200 (of 40000)
## [8] iteration: 21200 (of 40000)
## [2] iteration: 20900 (of 40000)
## [6] iteration: 21500 (of 40000)
## [4] iteration: 21200 (of 40000)
## [7] iteration: 21700 (of 40000)
## [5] iteration: 21400 (of 40000)
## [3] iteration: 21100 (of 40000)
## [1] iteration: 21300 (of 40000)
## [8] iteration: 21300 (of 40000)
## [2] iteration: 21000 (of 40000)
```

```
## [6] iteration: 21600 (of 40000)
## [4] iteration: 21300 (of 40000)
## [7] iteration: 21800 (of 40000)
## [5] iteration: 21500 (of 40000)
## [1] iteration: 21400 (of 40000)
## [3] iteration: 21200 (of 40000)
## [8] iteration: 21400 (of 40000)
## [2] iteration: 21100 (of 40000)
## [6] iteration: 21700 (of 40000)
## [4] iteration: 21400 (of 40000)
## [7] iteration: 21900 (of 40000)
## [5] iteration: 21600 (of 40000)
## [1] iteration: 21500 (of 40000)
## [3] iteration: 21300 (of 40000)
## [8] iteration: 21500 (of 40000)
## [2] iteration: 21200 (of 40000)
## [6] iteration: 21800 (of 40000)
## [4] iteration: 21500 (of 40000)
## [7] iteration: 22000 (of 40000)
## [5] iteration: 21700 (of 40000)
## [1] iteration: 21600 (of 40000)
## [3] iteration: 21400 (of 40000)
## [8] iteration: 21600 (of 40000)
## [2] iteration: 21300 (of 40000)
## [6] iteration: 21900 (of 40000)
## [7] iteration: 22100 (of 40000)
## [4] iteration: 21600 (of 40000)
## [5] iteration: 21800 (of 40000)
## [1] iteration: 21700 (of 40000)
## [3] iteration: 21500 (of 40000)
## [8] iteration: 21700 (of 40000)
## [2] iteration: 21400 (of 40000)
## [6] iteration: 22000 (of 40000)
## [7] iteration: 22200 (of 40000)
## [4] iteration: 21700 (of 40000)
## [5] iteration: 21900 (of 40000)
## [1] iteration: 21800 (of 40000)
## [3] iteration: 21600 (of 40000)
## [8] iteration: 21800 (of 40000)
## [2] iteration: 21500 (of 40000)
## [6] iteration: 22100 (of 40000)
## [7] iteration: 22300 (of 40000)
## [4] iteration: 21800 (of 40000)
## [5] iteration: 22000 (of 40000)
## [1] iteration: 21900 (of 40000)
## [3] iteration: 21700 (of 40000)
## [8] iteration: 21900 (of 40000)
## [2] iteration: 21600 (of 40000)
## [6] iteration: 22200 (of 40000)
## [7] iteration: 22400 (of 40000)
## [4] iteration: 21900 (of 40000)
## [5] iteration: 22100 (of 40000)
## [1] iteration: 22000 (of 40000)
## [3] iteration: 21800 (of 40000)
## [8] iteration: 22000 (of 40000)
## [6] iteration: 22300 (of 40000)
```

```
## [2] iteration: 21700 (of 40000)
## [7] iteration: 22500 (of 40000)
## [4] iteration: 22000 (of 40000)
## [5] iteration: 22200 (of 40000)
## [1] iteration: 22100 (of 40000)
## [8] iteration: 22100 (of 40000)
## [3] iteration: 21900 (of 40000)
## [6] iteration: 22400 (of 40000)
## [2] iteration: 21800 (of 40000)
## [7] iteration: 22600 (of 40000)
## [4] iteration: 22100 (of 40000)
## [5] iteration: 22300 (of 40000)
## [1] iteration: 22200 (of 40000)
## [8] iteration: 22200 (of 40000)
## [3] iteration: 22000 (of 40000)
## [6] iteration: 22500 (of 40000)
## [2] iteration: 21900 (of 40000)
## [7] iteration: 22700 (of 40000)
## [4] iteration: 22200 (of 40000)
## [5] iteration: 22400 (of 40000)
## [1] iteration: 22300 (of 40000)
## [8] iteration: 22300 (of 40000)
## [3] iteration: 22100 (of 40000)
## [6] iteration: 22600 (of 40000)
## [2] iteration: 22000 (of 40000)
## [7] iteration: 22800 (of 40000)
## [4] iteration: 22300 (of 40000)
## [5] iteration: 22500 (of 40000)
## [1] iteration: 22400 (of 40000)
## [8] iteration: 22400 (of 40000)
## [3] iteration: 22200 (of 40000)
## [6] iteration: 22700 (of 40000)
## [2] iteration: 22100 (of 40000)
## [7] iteration: 22900 (of 40000)
## [4] iteration: 22400 (of 40000)
## [5] iteration: 22600 (of 40000)
## [1] iteration: 22500 (of 40000)
## [8] iteration: 22500 (of 40000)
## [3] iteration: 22300 (of 40000)
## [6] iteration: 22800 (of 40000)
## [2] iteration: 22200 (of 40000)
## [7] iteration: 23000 (of 40000)
## [5] iteration: 22700 (of 40000)
## [4] iteration: 22500 (of 40000)
## [1] iteration: 22600 (of 40000)
## [8] iteration: 22600 (of 40000)
## [3] iteration: 22400 (of 40000)
## [6] iteration: 22900 (of 40000)
## [2] iteration: 22300 (of 40000)
## [7] iteration: 23100 (of 40000)
## [5] iteration: 22800 (of 40000)
## [4] iteration: 22600 (of 40000)
## [1] iteration: 22700 (of 40000)
## [3] iteration: 22500 (of 40000)
## [8] iteration: 22700 (of 40000)
## [6] iteration: 23000 (of 40000)
```

```
## [7] iteration: 23200 (of 40000)
## [2] iteration: 22400 (of 40000)
## [5] iteration: 22900 (of 40000)
## [4] iteration: 22700 (of 40000)
## [1] iteration: 22800 (of 40000)
## [3] iteration: 22600 (of 40000)
## [8] iteration: 22800 (of 40000)
## [6] iteration: 23100 (of 40000)
## [7] iteration: 23300 (of 40000)
## [2] iteration: 22500 (of 40000)
## [5] iteration: 23000 (of 40000)
## [4] iteration: 22800 (of 40000)
## [1] iteration: 22900 (of 40000)
## [8] iteration: 22900 (of 40000)
## [3] iteration: 22700 (of 40000)
## [6] iteration: 23200 (of 40000)
## [7] iteration: 23400 (of 40000)
## [2] iteration: 22600 (of 40000)
## [5] iteration: 23100 (of 40000)
## [4] iteration: 22900 (of 40000)
## [1] iteration: 23000 (of 40000)
## [8] iteration: 23000 (of 40000)
## [3] iteration: 22800 (of 40000)
## [6] iteration: 23300 (of 40000)
## [7] iteration: 23500 (of 40000)
## [2] iteration: 22700 (of 40000)
## [5] iteration: 23200 (of 40000)
## [4] iteration: 23000 (of 40000)
## [1] iteration: 23100 (of 40000)
## [8] iteration: 23100 (of 40000)
## [3] iteration: 22900 (of 40000)
## [6] iteration: 23400 (of 40000)
## [7] iteration: 23600 (of 40000)
## [2] iteration: 22800 (of 40000)
## [5] iteration: 23300 (of 40000)
## [4] iteration: 23100 (of 40000)
## [1] iteration: 23200 (of 40000)
## [8] iteration: 23200 (of 40000)
## [6] iteration: 23500 (of 40000)
## [3] iteration: 23000 (of 40000)
## [7] iteration: 23700 (of 40000)
## [2] iteration: 22900 (of 40000)
## [5] iteration: 23400 (of 40000)
## [4] iteration: 23200 (of 40000)
## [1] iteration: 23300 (of 40000)
## [8] iteration: 23300 (of 40000)
## [6] iteration: 23600 (of 40000)
## [3] iteration: 23100 (of 40000)
## [7] iteration: 23800 (of 40000)
## [2] iteration: 23000 (of 40000)
## [5] iteration: 23500 (of 40000)
## [4] iteration: 23300 (of 40000)
## [1] iteration: 23400 (of 40000)
## [8] iteration: 23400 (of 40000)
## [6] iteration: 23700 (of 40000)
## [3] iteration: 23200 (of 40000)
```

```
## [7] iteration: 23900 (of 40000)
## [5] iteration: 23600 (of 40000)
## [2] iteration: 23100 (of 40000)
## [4] iteration: 23400 (of 40000)
## [1] iteration: 23500 (of 40000)
## [6] iteration: 23800 (of 40000)
## [8] iteration: 23500 (of 40000)
## [3] iteration: 23300 (of 40000)
## [7] iteration: 24000 (of 40000)
## [5] iteration: 23700 (of 40000)
## [2] iteration: 23200 (of 40000)
## [4] iteration: 23500 (of 40000)
## [1] iteration: 23600 (of 40000)
## [8] iteration: 23600 (of 40000)
## [6] iteration: 23900 (of 40000)
## [3] iteration: 23400 (of 40000)
## [7] iteration: 24100 (of 40000)
## [5] iteration: 23800 (of 40000)
## [2] iteration: 23300 (of 40000)
## [4] iteration: 23600 (of 40000)
## [1] iteration: 23700 (of 40000)
## [6] iteration: 24000 (of 40000)
## [8] iteration: 23700 (of 40000)
## [7] iteration: 24200 (of 40000)
## [3] iteration: 23500 (of 40000)
## [5] iteration: 23900 (of 40000)
## [2] iteration: 23400 (of 40000)
## [4] iteration: 23700 (of 40000)
## [1] iteration: 23800 (of 40000)
## [6] iteration: 24100 (of 40000)
## [8] iteration: 23800 (of 40000)
## [7] iteration: 24300 (of 40000)
## [3] iteration: 23600 (of 40000)
## [5] iteration: 24000 (of 40000)
## [2] iteration: 23500 (of 40000)
## [4] iteration: 23800 (of 40000)
## [1] iteration: 23900 (of 40000)
## [6] iteration: 24200 (of 40000)
## [8] iteration: 23900 (of 40000)
## [7] iteration: 24400 (of 40000)
## [3] iteration: 23700 (of 40000)
## [5] iteration: 24100 (of 40000)
## [2] iteration: 23600 (of 40000)
## [4] iteration: 23900 (of 40000)
## [1] iteration: 24000 (of 40000)
## [6] iteration: 24300 (of 40000)
## [8] iteration: 24000 (of 40000)
## [7] iteration: 24500 (of 40000)
## [3] iteration: 23800 (of 40000)
## [5] iteration: 24200 (of 40000)
## [2] iteration: 23700 (of 40000)
## [4] iteration: 24000 (of 40000)
## [1] iteration: 24100 (of 40000)
## [6] iteration: 24400 (of 40000)
## [8] iteration: 24100 (of 40000)
## [7] iteration: 24600 (of 40000)
```

```
## [3] iteration: 23900 (of 40000)
## [5] iteration: 24300 (of 40000)
## [4] iteration: 24100 (of 40000)
## [2] iteration: 23800 (of 40000)
## [1] iteration: 24200 (of 40000)
## [6] iteration: 24500 (of 40000)
## [8] iteration: 24200 (of 40000)
## [7] iteration: 24700 (of 40000)
## [3] iteration: 24000 (of 40000)
## [5] iteration: 24400 (of 40000)
## [4] iteration: 24200 (of 40000)
## [2] iteration: 23900 (of 40000)
## [1] iteration: 24300 (of 40000)
## [6] iteration: 24600 (of 40000)
## [7] iteration: 24800 (of 40000)
## [8] iteration: 24300 (of 40000)
## [3] iteration: 24100 (of 40000)
## [5] iteration: 24500 (of 40000)
## [4] iteration: 24300 (of 40000)
## [2] iteration: 24000 (of 40000)
## [1] iteration: 24400 (of 40000)
## [6] iteration: 24700 (of 40000)
## [7] iteration: 24900 (of 40000)
## [8] iteration: 24400 (of 40000)
## [3] iteration: 24200 (of 40000)
## [5] iteration: 24600 (of 40000)
## [4] iteration: 24400 (of 40000)
## [2] iteration: 24100 (of 40000)
## [1] iteration: 24500 (of 40000)
## [6] iteration: 24800 (of 40000)
## [7] iteration: 25000 (of 40000)
## [8] iteration: 24500 (of 40000)
## [3] iteration: 24300 (of 40000)
## [5] iteration: 24700 (of 40000)
## [4] iteration: 24500 (of 40000)
## [1] iteration: 24600 (of 40000)
## [2] iteration: 24200 (of 40000)
## [6] iteration: 24900 (of 40000)
## [7] iteration: 25100 (of 40000)
## [8] iteration: 24600 (of 40000)
## [5] iteration: 24800 (of 40000)
## [3] iteration: 24400 (of 40000)
## [4] iteration: 24600 (of 40000)
## [1] iteration: 24700 (of 40000)
## [2] iteration: 24300 (of 40000)
## [6] iteration: 25000 (of 40000)
## [7] iteration: 25200 (of 40000)
## [8] iteration: 24700 (of 40000)
## [5] iteration: 24900 (of 40000)
## [3] iteration: 24500 (of 40000)
## [4] iteration: 24700 (of 40000)
## [1] iteration: 24800 (of 40000)
## [2] iteration: 24400 (of 40000)
## [6] iteration: 25100 (of 40000)
## [7] iteration: 25300 (of 40000)
## [8] iteration: 24800 (of 40000)
```

```
## [5] iteration: 25000 (of 40000)
## [3] iteration: 24600 (of 40000)
## [4] iteration: 24800 (of 40000)
## [1] iteration: 24900 (of 40000)
## [2] iteration: 24500 (of 40000)
## [6] iteration: 25200 (of 40000)
## [7] iteration: 25400 (of 40000)
## [8] iteration: 24900 (of 40000)
## [5] iteration: 25100 (of 40000)
## [3] iteration: 24700 (of 40000)
## [1] iteration: 25000 (of 40000)
## [4] iteration: 24900 (of 40000)
## [2] iteration: 24600 (of 40000)
## [6] iteration: 25300 (of 40000)
## [7] iteration: 25500 (of 40000)
## [8] iteration: 25000 (of 40000)
## [5] iteration: 25200 (of 40000)
## [3] iteration: 24800 (of 40000)
## [1] iteration: 25100 (of 40000)
## [4] iteration: 25000 (of 40000)
## [2] iteration: 24700 (of 40000)
## [6] iteration: 25400 (of 40000)
## [7] iteration: 25600 (of 40000)
## [8] iteration: 25100 (of 40000)
## [5] iteration: 25300 (of 40000)
## [3] iteration: 24900 (of 40000)
## [1] iteration: 25200 (of 40000)
## [4] iteration: 25100 (of 40000)
## [6] iteration: 25500 (of 40000)
## [2] iteration: 24800 (of 40000)
## [7] iteration: 25700 (of 40000)
## [8] iteration: 25200 (of 40000)
## [5] iteration: 25400 (of 40000)
## [3] iteration: 25000 (of 40000)
## [1] iteration: 25300 (of 40000)
## [4] iteration: 25200 (of 40000)
## [6] iteration: 25600 (of 40000)
## [2] iteration: 24900 (of 40000)
## [7] iteration: 25800 (of 40000)
## [8] iteration: 25300 (of 40000)
## [5] iteration: 25500 (of 40000)
## [3] iteration: 25100 (of 40000)
## [1] iteration: 25400 (of 40000)
## [4] iteration: 25300 (of 40000)
## [6] iteration: 25700 (of 40000)
## [7] iteration: 25900 (of 40000)
## [2] iteration: 25000 (of 40000)
## [8] iteration: 25400 (of 40000)
## [5] iteration: 25600 (of 40000)
## [3] iteration: 25200 (of 40000)
## [1] iteration: 25500 (of 40000)
## [4] iteration: 25400 (of 40000)
## [6] iteration: 25800 (of 40000)
## [7] iteration: 26000 (of 40000)
## [2] iteration: 25100 (of 40000)
## [8] iteration: 25500 (of 40000)
```

```
## [5] iteration: 25700 (of 40000)
## [1] iteration: 25600 (of 40000)
## [3] iteration: 25300 (of 40000)
## [6] iteration: 25900 (of 40000)
## [4] iteration: 25500 (of 40000)
## [7] iteration: 26100 (of 40000)
## [2] iteration: 25200 (of 40000)
## [8] iteration: 25600 (of 40000)
## [5] iteration: 25800 (of 40000)
## [1] iteration: 25700 (of 40000)
## [6] iteration: 26000 (of 40000)
## [3] iteration: 25400 (of 40000)
## [4] iteration: 25600 (of 40000)
## [7] iteration: 26200 (of 40000)
## [2] iteration: 25300 (of 40000)
## [8] iteration: 25700 (of 40000)
## [5] iteration: 25900 (of 40000)
## [1] iteration: 25800 (of 40000)
## [6] iteration: 26100 (of 40000)
## [3] iteration: 25500 (of 40000)
## [4] iteration: 25700 (of 40000)
## [7] iteration: 26300 (of 40000)
## [2] iteration: 25400 (of 40000)
## [8] iteration: 25800 (of 40000)
## [5] iteration: 26000 (of 40000)
## [1] iteration: 25900 (of 40000)
## [6] iteration: 26200 (of 40000)
## [3] iteration: 25600 (of 40000)
## [7] iteration: 26400 (of 40000)
## [4] iteration: 25800 (of 40000)
## [2] iteration: 25500 (of 40000)
## [8] iteration: 25900 (of 40000)
## [5] iteration: 26100 (of 40000)
## [1] iteration: 26000 (of 40000)
## [6] iteration: 26300 (of 40000)
## [7] iteration: 26500 (of 40000)
## [3] iteration: 25700 (of 40000)
## [4] iteration: 25900 (of 40000)
## [2] iteration: 25600 (of 40000)
## [8] iteration: 26000 (of 40000)
## [5] iteration: 26200 (of 40000)
## [1] iteration: 26100 (of 40000)
## [6] iteration: 26400 (of 40000)
## [7] iteration: 26600 (of 40000)
## [3] iteration: 25800 (of 40000)
## [4] iteration: 26000 (of 40000)
## [2] iteration: 25700 (of 40000)
## [8] iteration: 26100 (of 40000)
## [1] iteration: 26200 (of 40000)
## [5] iteration: 26300 (of 40000)
## [6] iteration: 26500 (of 40000)
## [7] iteration: 26700 (of 40000)
## [4] iteration: 26100 (of 40000)
## [3] iteration: 25900 (of 40000)
## [2] iteration: 25800 (of 40000)
## [8] iteration: 26200 (of 40000)
```

```
## [1] iteration: 26300 (of 40000)
## [5] iteration: 26400 (of 40000)
## [6] iteration: 26600 (of 40000)
## [7] iteration: 26800 (of 40000)
## [4] iteration: 26200 (of 40000)
## [3] iteration: 26000 (of 40000)
## [2] iteration: 25900 (of 40000)
## [8] iteration: 26300 (of 40000)
## [1] iteration: 26400 (of 40000)
## [6] iteration: 26700 (of 40000)
## [5] iteration: 26500 (of 40000)
## [7] iteration: 26900 (of 40000)
## [4] iteration: 26300 (of 40000)
## [3] iteration: 26100 (of 40000)
## [2] iteration: 26000 (of 40000)
## [8] iteration: 26400 (of 40000)
## [1] iteration: 26500 (of 40000)
## [6] iteration: 26800 (of 40000)
## [5] iteration: 26600 (of 40000)
## [7] iteration: 27000 (of 40000)
## [4] iteration: 26400 (of 40000)
## [3] iteration: 26200 (of 40000)
## [2] iteration: 26100 (of 40000)
## [8] iteration: 26500 (of 40000)
## [6] iteration: 26900 (of 40000)
## [1] iteration: 26600 (of 40000)
## [5] iteration: 26700 (of 40000)
## [7] iteration: 27100 (of 40000)
## [4] iteration: 26500 (of 40000)
## [3] iteration: 26300 (of 40000)
## [2] iteration: 26200 (of 40000)
## [8] iteration: 26600 (of 40000)
## [6] iteration: 27000 (of 40000)
## [1] iteration: 26700 (of 40000)
## [7] iteration: 27200 (of 40000)
## [5] iteration: 26800 (of 40000)
## [4] iteration: 26600 (of 40000)
## [3] iteration: 26400 (of 40000)
## [2] iteration: 26300 (of 40000)
## [8] iteration: 26700 (of 40000)
## [6] iteration: 27100 (of 40000)
## [1] iteration: 26800 (of 40000)
## [7] iteration: 27300 (of 40000)
## [5] iteration: 26900 (of 40000)
## [4] iteration: 26700 (of 40000)
## [3] iteration: 26500 (of 40000)
## [2] iteration: 26400 (of 40000)
## [8] iteration: 26800 (of 40000)
## [6] iteration: 27200 (of 40000)
## [1] iteration: 26900 (of 40000)
## [7] iteration: 27400 (of 40000)
## [5] iteration: 27000 (of 40000)
## [3] iteration: 26600 (of 40000)
## [4] iteration: 26800 (of 40000)
## [2] iteration: 26500 (of 40000)
## [8] iteration: 26900 (of 40000)
```

```
## [6] iteration: 27300 (of 40000)
## [1] iteration: 27000 (of 40000)
## [7] iteration: 27500 (of 40000)
## [5] iteration: 27100 (of 40000)
## [3] iteration: 26700 (of 40000)
## [4] iteration: 26900 (of 40000)
## [2] iteration: 26600 (of 40000)
## [8] iteration: 27000 (of 40000)
## [6] iteration: 27400 (of 40000)
## [7] iteration: 27600 (of 40000)
## [1] iteration: 27100 (of 40000)
## [5] iteration: 27200 (of 40000)
## [3] iteration: 26800 (of 40000)
## [4] iteration: 27000 (of 40000)
## [2] iteration: 26700 (of 40000)
## [8] iteration: 27100 (of 40000)
## [6] iteration: 27500 (of 40000)
## [7] iteration: 27700 (of 40000)
## [1] iteration: 27200 (of 40000)
## [5] iteration: 27300 (of 40000)
## [3] iteration: 26900 (of 40000)
## [4] iteration: 27100 (of 40000)
## [2] iteration: 26800 (of 40000)
## [8] iteration: 27200 (of 40000)
## [6] iteration: 27600 (of 40000)
## [7] iteration: 27800 (of 40000)
## [1] iteration: 27300 (of 40000)
## [5] iteration: 27400 (of 40000)
## [3] iteration: 27000 (of 40000)
## [4] iteration: 27200 (of 40000)
## [2] iteration: 26900 (of 40000)
## [8] iteration: 27300 (of 40000)
## [6] iteration: 27700 (of 40000)
## [7] iteration: 27900 (of 40000)
## [1] iteration: 27400 (of 40000)
## [5] iteration: 27500 (of 40000)
## [3] iteration: 27100 (of 40000)
## [4] iteration: 27300 (of 40000)
## [2] iteration: 27000 (of 40000)
## [8] iteration: 27400 (of 40000)
## [6] iteration: 27800 (of 40000)
## [7] iteration: 28000 (of 40000)
## [1] iteration: 27500 (of 40000)
## [5] iteration: 27600 (of 40000)
## [3] iteration: 27200 (of 40000)
## [4] iteration: 27400 (of 40000)
## [2] iteration: 27100 (of 40000)
## [6] iteration: 27900 (of 40000)
## [7] iteration: 28100 (of 40000)
## [8] iteration: 27500 (of 40000)
## [1] iteration: 27600 (of 40000)
## [5] iteration: 27700 (of 40000)
## [3] iteration: 27300 (of 40000)
## [4] iteration: 27500 (of 40000)
## [2] iteration: 27200 (of 40000)
## [6] iteration: 28000 (of 40000)
```

```
## [7] iteration: 28200 (of 40000)
## [8] iteration: 27600 (of 40000)
## [1] iteration: 27700 (of 40000)
## [5] iteration: 27800 (of 40000)
## [3] iteration: 27400 (of 40000)
## [4] iteration: 27600 (of 40000)
## [2] iteration: 27300 (of 40000)
## [6] iteration: 28100 (of 40000)
## [7] iteration: 28300 (of 40000)
## [8] iteration: 27700 (of 40000)
## [1] iteration: 27800 (of 40000)
## [5] iteration: 27900 (of 40000)
## [4] iteration: 27700 (of 40000)
## [3] iteration: 27500 (of 40000)
## [2] iteration: 27400 (of 40000)
## [6] iteration: 28200 (of 40000)
## [7] iteration: 28400 (of 40000)
## [8] iteration: 27800 (of 40000)
## [1] iteration: 27900 (of 40000)
## [5] iteration: 28000 (of 40000)
## [4] iteration: 27800 (of 40000)
## [3] iteration: 27600 (of 40000)
## [2] iteration: 27500 (of 40000)
## [6] iteration: 28300 (of 40000)
## [7] iteration: 28500 (of 40000)
## [8] iteration: 27900 (of 40000)
## [1] iteration: 28000 (of 40000)
## [5] iteration: 28100 (of 40000)
## [4] iteration: 27900 (of 40000)
## [3] iteration: 27700 (of 40000)
## [6] iteration: 28400 (of 40000)
## [2] iteration: 27600 (of 40000)
## [7] iteration: 28600 (of 40000)
## [8] iteration: 28000 (of 40000)
## [1] iteration: 28100 (of 40000)
## [5] iteration: 28200 (of 40000)
## [4] iteration: 28000 (of 40000)
## [3] iteration: 27800 (of 40000)
## [6] iteration: 28500 (of 40000)
## [2] iteration: 27700 (of 40000)
## [7] iteration: 28700 (of 40000)
## [8] iteration: 28100 (of 40000)
## [1] iteration: 28200 (of 40000)
## [5] iteration: 28300 (of 40000)
## [4] iteration: 28100 (of 40000)
## [3] iteration: 27900 (of 40000)
## [6] iteration: 28600 (of 40000)
## [2] iteration: 27800 (of 40000)
## [7] iteration: 28800 (of 40000)
## [8] iteration: 28200 (of 40000)
## [1] iteration: 28300 (of 40000)
## [5] iteration: 28400 (of 40000)
## [4] iteration: 28200 (of 40000)
## [3] iteration: 28000 (of 40000)
## [6] iteration: 28700 (of 40000)
## [7] iteration: 28900 (of 40000)
```

```
## [2] iteration: 27900 (of 40000)
## [8] iteration: 28300 (of 40000)
## [1] iteration: 28400 (of 40000)
## [5] iteration: 28500 (of 40000)
## [4] iteration: 28300 (of 40000)
## [3] iteration: 28100 (of 40000)
## [6] iteration: 28800 (of 40000)
## [7] iteration: 29000 (of 40000)
## [2] iteration: 28000 (of 40000)
## [8] iteration: 28400 (of 40000)
## [1] iteration: 28500 (of 40000)
## [5] iteration: 28600 (of 40000)
## [4] iteration: 28400 (of 40000)
## [3] iteration: 28200 (of 40000)
## [6] iteration: 28900 (of 40000)
## [7] iteration: 29100 (of 40000)
## [2] iteration: 28100 (of 40000)
## [8] iteration: 28500 (of 40000)
## [1] iteration: 28600 (of 40000)
## [5] iteration: 28700 (of 40000)
## [4] iteration: 28500 (of 40000)
## [3] iteration: 28300 (of 40000)
## [6] iteration: 29000 (of 40000)
## [7] iteration: 29200 (of 40000)
## [2] iteration: 28200 (of 40000)
## [8] iteration: 28600 (of 40000)
## [1] iteration: 28700 (of 40000)
## [5] iteration: 28800 (of 40000)
## [4] iteration: 28600 (of 40000)
## [3] iteration: 28400 (of 40000)
## [6] iteration: 29100 (of 40000)
## [7] iteration: 29300 (of 40000)
## [2] iteration: 28300 (of 40000)
## [8] iteration: 28700 (of 40000)
## [1] iteration: 28800 (of 40000)
## [5] iteration: 28900 (of 40000)
## [4] iteration: 28700 (of 40000)
## [6] iteration: 29200 (of 40000)
## [3] iteration: 28500 (of 40000)
## [7] iteration: 29400 (of 40000)
## [2] iteration: 28400 (of 40000)
## [8] iteration: 28800 (of 40000)
## [1] iteration: 28900 (of 40000)
## [5] iteration: 29000 (of 40000)
## [4] iteration: 28800 (of 40000)
## [6] iteration: 29300 (of 40000)
## [3] iteration: 28600 (of 40000)
## [7] iteration: 29500 (of 40000)
## [2] iteration: 28500 (of 40000)
## [8] iteration: 28900 (of 40000)
## [1] iteration: 29000 (of 40000)
## [5] iteration: 29100 (of 40000)
## [4] iteration: 28900 (of 40000)
## [6] iteration: 29400 (of 40000)
## [7] iteration: 29600 (of 40000)
## [3] iteration: 28700 (of 40000)
```

```
## [2] iteration: 28600 (of 40000)
## [8] iteration: 29000 (of 40000)
## [1] iteration: 29100 (of 40000)
## [5] iteration: 29200 (of 40000)
## [4] iteration: 29000 (of 40000)
## [6] iteration: 29500 (of 40000)
## [7] iteration: 29700 (of 40000)
## [3] iteration: 28800 (of 40000)
## [2] iteration: 28700 (of 40000)
## [8] iteration: 29100 (of 40000)
## [1] iteration: 29200 (of 40000)
## [5] iteration: 29300 (of 40000)
## [4] iteration: 29100 (of 40000)
## [6] iteration: 29600 (of 40000)
## [7] iteration: 29800 (of 40000)
## [3] iteration: 28900 (of 40000)
## [2] iteration: 28800 (of 40000)
## [8] iteration: 29200 (of 40000)
## [1] iteration: 29300 (of 40000)
## [4] iteration: 29200 (of 40000)
## [5] iteration: 29400 (of 40000)
## [6] iteration: 29700 (of 40000)
## [7] iteration: 29900 (of 40000)
## [3] iteration: 29000 (of 40000)
## [2] iteration: 28900 (of 40000)
## [8] iteration: 29300 (of 40000)
## [1] iteration: 29400 (of 40000)
## [4] iteration: 29300 (of 40000)
## [6] iteration: 29800 (of 40000)
## [5] iteration: 29500 (of 40000)
## [7] iteration: 30000 (of 40000)
## [3] iteration: 29100 (of 40000)
## [2] iteration: 29000 (of 40000)
## [8] iteration: 29400 (of 40000)
## [1] iteration: 29500 (of 40000)
## [6] iteration: 29900 (of 40000)
## [4] iteration: 29400 (of 40000)
## [5] iteration: 29600 (of 40000)
## [7] iteration: 30100 (of 40000)
## [3] iteration: 29200 (of 40000)
## [8] iteration: 29500 (of 40000)
## [1] iteration: 29600 (of 40000)
## [2] iteration: 29100 (of 40000)
## [6] iteration: 30000 (of 40000)
## [4] iteration: 29500 (of 40000)
## [5] iteration: 29700 (of 40000)
## [7] iteration: 30200 (of 40000)
## [3] iteration: 29300 (of 40000)
## [8] iteration: 29600 (of 40000)
## [1] iteration: 29700 (of 40000)
## [2] iteration: 29200 (of 40000)
## [6] iteration: 30100 (of 40000)
## [4] iteration: 29600 (of 40000)
## [7] iteration: 30300 (of 40000)
## [5] iteration: 29800 (of 40000)
## [3] iteration: 29400 (of 40000)
```

```
## [8] iteration: 29700 (of 40000)
## [1] iteration: 29800 (of 40000)
## [2] iteration: 29300 (of 40000)
## [6] iteration: 30200 (of 40000)
## [4] iteration: 29700 (of 40000)
## [7] iteration: 30400 (of 40000)
## [5] iteration: 29900 (of 40000)
## [3] iteration: 29500 (of 40000)
## [8] iteration: 29800 (of 40000)
## [1] iteration: 29900 (of 40000)
## [2] iteration: 29400 (of 40000)
## [6] iteration: 30300 (of 40000)
## [7] iteration: 30500 (of 40000)
## [4] iteration: 29800 (of 40000)
## [5] iteration: 30000 (of 40000)
## [3] iteration: 29600 (of 40000)
## [8] iteration: 29900 (of 40000)
## [1] iteration: 30000 (of 40000)
## [2] iteration: 29500 (of 40000)
## [6] iteration: 30400 (of 40000)
## [7] iteration: 30600 (of 40000)
## [4] iteration: 29900 (of 40000)
## [5] iteration: 30100 (of 40000)
## [3] iteration: 29700 (of 40000)
## [8] iteration: 30000 (of 40000)
## [1] iteration: 30100 (of 40000)
## [2] iteration: 29600 (of 40000)
## [6] iteration: 30500 (of 40000)
## [7] iteration: 30700 (of 40000)
## [4] iteration: 30000 (of 40000)
## [5] iteration: 30200 (of 40000)
## [3] iteration: 29800 (of 40000)
## [8] iteration: 30100 (of 40000)
## [1] iteration: 30200 (of 40000)
## [2] iteration: 29700 (of 40000)
## [6] iteration: 30600 (of 40000)
## [7] iteration: 30800 (of 40000)
## [4] iteration: 30100 (of 40000)
## [5] iteration: 30300 (of 40000)
## [3] iteration: 29900 (of 40000)
## [8] iteration: 30200 (of 40000)
## [1] iteration: 30300 (of 40000)
## [2] iteration: 29800 (of 40000)
## [6] iteration: 30700 (of 40000)
## [7] iteration: 30900 (of 40000)
## [4] iteration: 30200 (of 40000)
## [5] iteration: 30400 (of 40000)
## [1] iteration: 30400 (of 40000)
## [8] iteration: 30300 (of 40000)
## [3] iteration: 30000 (of 40000)
## [2] iteration: 29900 (of 40000)
## [6] iteration: 30800 (of 40000)
## [7] iteration: 31000 (of 40000)
## [4] iteration: 30300 (of 40000)
## [5] iteration: 30500 (of 40000)
## [1] iteration: 30500 (of 40000)
```

```
## [8] iteration: 30400 (of 40000)
## [3] iteration: 30100 (of 40000)
## [2] iteration: 30000 (of 40000)
## [6] iteration: 30900 (of 40000)
## [7] iteration: 31100 (of 40000)
## [4] iteration: 30400 (of 40000)
## [5] iteration: 30600 (of 40000)
## [1] iteration: 30600 (of 40000)
## [8] iteration: 30500 (of 40000)
## [3] iteration: 30200 (of 40000)
## [2] iteration: 30100 (of 40000)
## [6] iteration: 31000 (of 40000)
## [7] iteration: 31200 (of 40000)
## [4] iteration: 30500 (of 40000)
## [5] iteration: 30700 (of 40000)
## [1] iteration: 30700 (of 40000)
## [8] iteration: 30600 (of 40000)
## [3] iteration: 30300 (of 40000)
## [2] iteration: 30200 (of 40000)
## [6] iteration: 31100 (of 40000)
## [7] iteration: 31300 (of 40000)
## [5] iteration: 30800 (of 40000)
## [4] iteration: 30600 (of 40000)
## [1] iteration: 30800 (of 40000)
## [8] iteration: 30700 (of 40000)
## [3] iteration: 30400 (of 40000)
## [6] iteration: 31200 (of 40000)
## [2] iteration: 30300 (of 40000)
## [7] iteration: 31400 (of 40000)
## [5] iteration: 30900 (of 40000)
## [4] iteration: 30700 (of 40000)
## [1] iteration: 30900 (of 40000)
## [8] iteration: 30800 (of 40000)
## [3] iteration: 30500 (of 40000)
## [6] iteration: 31300 (of 40000)
## [2] iteration: 30400 (of 40000)
## [7] iteration: 31500 (of 40000)
## [5] iteration: 31000 (of 40000)
## [4] iteration: 30800 (of 40000)
## [1] iteration: 31000 (of 40000)
## [8] iteration: 30900 (of 40000)
## [3] iteration: 30600 (of 40000)
## [6] iteration: 31400 (of 40000)
## [2] iteration: 30500 (of 40000)
## [7] iteration: 31600 (of 40000)
## [5] iteration: 31100 (of 40000)
## [4] iteration: 30900 (of 40000)
## [1] iteration: 31100 (of 40000)
## [8] iteration: 31000 (of 40000)
## [3] iteration: 30700 (of 40000)
## [6] iteration: 31500 (of 40000)
## [2] iteration: 30600 (of 40000)
## [7] iteration: 31700 (of 40000)
## [5] iteration: 31200 (of 40000)
## [4] iteration: 31000 (of 40000)
## [1] iteration: 31200 (of 40000)
```

```
## [8] iteration: 31100 (of 40000)
## [3] iteration: 30800 (of 40000)
## [6] iteration: 31600 (of 40000)
## [2] iteration: 30700 (of 40000)
## [7] iteration: 31800 (of 40000)
## [5] iteration: 31300 (of 40000)
## [4] iteration: 31100 (of 40000)
## [1] iteration: 31300 (of 40000)
## [8] iteration: 31200 (of 40000)
## [6] iteration: 31700 (of 40000)
## [3] iteration: 30900 (of 40000)
## [2] iteration: 30800 (of 40000)
## [7] iteration: 31900 (of 40000)
## [5] iteration: 31400 (of 40000)
## [4] iteration: 31200 (of 40000)
## [1] iteration: 31400 (of 40000)
## [8] iteration: 31300 (of 40000)
## [6] iteration: 31800 (of 40000)
## [3] iteration: 31000 (of 40000)
## [2] iteration: 30900 (of 40000)
## [7] iteration: 32000 (of 40000)
## [5] iteration: 31500 (of 40000)
## [4] iteration: 31300 (of 40000)
## [1] iteration: 31500 (of 40000)
## [8] iteration: 31400 (of 40000)
## [6] iteration: 31900 (of 40000)
## [3] iteration: 31100 (of 40000)
## [2] iteration: 31000 (of 40000)
## [7] iteration: 32100 (of 40000)
## [5] iteration: 31600 (of 40000)
## [4] iteration: 31400 (of 40000)
## [1] iteration: 31600 (of 40000)
## [8] iteration: 31500 (of 40000)
## [6] iteration: 32000 (of 40000)
## [3] iteration: 31200 (of 40000)
## [7] iteration: 32200 (of 40000)
## [2] iteration: 31100 (of 40000)
## [5] iteration: 31700 (of 40000)
## [4] iteration: 31500 (of 40000)
## [1] iteration: 31700 (of 40000)
## [8] iteration: 31600 (of 40000)
## [6] iteration: 32100 (of 40000)
## [3] iteration: 31300 (of 40000)
## [7] iteration: 32300 (of 40000)
## [2] iteration: 31200 (of 40000)
## [5] iteration: 31800 (of 40000)
## [1] iteration: 31800 (of 40000)
## [4] iteration: 31600 (of 40000)
## [8] iteration: 31700 (of 40000)
## [6] iteration: 32200 (of 40000)
## [3] iteration: 31400 (of 40000)
## [7] iteration: 32400 (of 40000)
## [2] iteration: 31300 (of 40000)
## [5] iteration: 31900 (of 40000)
## [1] iteration: 31900 (of 40000)
## [4] iteration: 31700 (of 40000)
```

```
## [8] iteration: 31800 (of 40000)
## [6] iteration: 32300 (of 40000)
## [3] iteration: 31500 (of 40000)
## [7] iteration: 32500 (of 40000)
## [2] iteration: 31400 (of 40000)
## [5] iteration: 32000 (of 40000)
## [1] iteration: 32000 (of 40000)
## [4] iteration: 31800 (of 40000)
## [8] iteration: 31900 (of 40000)
## [6] iteration: 32400 (of 40000)
## [7] iteration: 32600 (of 40000)
## [3] iteration: 31600 (of 40000)
## [2] iteration: 31500 (of 40000)
## [5] iteration: 32100 (of 40000)
## [1] iteration: 32100 (of 40000)
## [4] iteration: 31900 (of 40000)
## [6] iteration: 32500 (of 40000)
## [8] iteration: 32000 (of 40000)
## [7] iteration: 32700 (of 40000)
## [3] iteration: 31700 (of 40000)
## [2] iteration: 31600 (of 40000)
## [5] iteration: 32200 (of 40000)
## [1] iteration: 32200 (of 40000)
## [6] iteration: 32600 (of 40000)
## [4] iteration: 32000 (of 40000)
## [8] iteration: 32100 (of 40000)
## [7] iteration: 32800 (of 40000)
## [3] iteration: 31800 (of 40000)
## [2] iteration: 31700 (of 40000)
## [5] iteration: 32300 (of 40000)
## [1] iteration: 32300 (of 40000)
## [6] iteration: 32700 (of 40000)
## [4] iteration: 32100 (of 40000)
## [8] iteration: 32200 (of 40000)
## [7] iteration: 32900 (of 40000)
## [3] iteration: 31900 (of 40000)
## [2] iteration: 31800 (of 40000)
## [5] iteration: 32400 (of 40000)
## [1] iteration: 32400 (of 40000)
## [6] iteration: 32800 (of 40000)
## [4] iteration: 32200 (of 40000)
## [8] iteration: 32300 (of 40000)
## [7] iteration: 33000 (of 40000)
## [3] iteration: 32000 (of 40000)
## [2] iteration: 31900 (of 40000)
## [5] iteration: 32500 (of 40000)
## [1] iteration: 32500 (of 40000)
## [6] iteration: 32900 (of 40000)
## [4] iteration: 32300 (of 40000)
## [8] iteration: 32400 (of 40000)
## [7] iteration: 33100 (of 40000)
## [3] iteration: 32100 (of 40000)
## [2] iteration: 32000 (of 40000)
## [5] iteration: 32600 (of 40000)
## [1] iteration: 32600 (of 40000)
## [6] iteration: 33000 (of 40000)
```

```
## [8] iteration: 32500 (of 40000)
## [4] iteration: 32400 (of 40000)
## [7] iteration: 33200 (of 40000)
## [3] iteration: 32200 (of 40000)
## [2] iteration: 32100 (of 40000)
## [5] iteration: 32700 (of 40000)
## [1] iteration: 32700 (of 40000)
## [6] iteration: 33100 (of 40000)
## [4] iteration: 32500 (of 40000)
## [8] iteration: 32600 (of 40000)
## [7] iteration: 33300 (of 40000)
## [3] iteration: 32300 (of 40000)
## [2] iteration: 32200 (of 40000)
## [1] iteration: 32800 (of 40000)
## [5] iteration: 32800 (of 40000)
## [6] iteration: 33200 (of 40000)
## [8] iteration: 32700 (of 40000)
## [4] iteration: 32600 (of 40000)
## [7] iteration: 33400 (of 40000)
## [3] iteration: 32400 (of 40000)
## [2] iteration: 32300 (of 40000)
## [1] iteration: 32900 (of 40000)
## [5] iteration: 32900 (of 40000)
## [6] iteration: 33300 (of 40000)
## [4] iteration: 32700 (of 40000)
## [8] iteration: 32800 (of 40000)
## [7] iteration: 33500 (of 40000)
## [3] iteration: 32500 (of 40000)
## [2] iteration: 32400 (of 40000)
## [1] iteration: 33000 (of 40000)
## [5] iteration: 33000 (of 40000)
## [6] iteration: 33400 (of 40000)
## [4] iteration: 32800 (of 40000)
## [8] iteration: 32900 (of 40000)
## [7] iteration: 33600 (of 40000)
## [3] iteration: 32600 (of 40000)
## [2] iteration: 32500 (of 40000)
## [1] iteration: 33100 (of 40000)
## [6] iteration: 33500 (of 40000)
## [5] iteration: 33100 (of 40000)
## [8] iteration: 33000 (of 40000)
## [4] iteration: 32900 (of 40000)
## [7] iteration: 33700 (of 40000)
## [3] iteration: 32700 (of 40000)
## [2] iteration: 32600 (of 40000)
## [1] iteration: 33200 (of 40000)
## [6] iteration: 33600 (of 40000)
## [5] iteration: 33200 (of 40000)
## [8] iteration: 33100 (of 40000)
## [4] iteration: 33000 (of 40000)
## [7] iteration: 33800 (of 40000)
## [3] iteration: 32800 (of 40000)
## [2] iteration: 32700 (of 40000)
## [1] iteration: 33300 (of 40000)
## [6] iteration: 33700 (of 40000)
## [5] iteration: 33300 (of 40000)
```

```
## [8] iteration: 33200 (of 40000)
## [4] iteration: 33100 (of 40000)
## [7] iteration: 33900 (of 40000)
## [3] iteration: 32900 (of 40000)
## [2] iteration: 32800 (of 40000)
## [6] iteration: 33800 (of 40000)
## [1] iteration: 33400 (of 40000)
## [5] iteration: 33400 (of 40000)
## [8] iteration: 33300 (of 40000)
## [4] iteration: 33200 (of 40000)
## [7] iteration: 34000 (of 40000)
## [3] iteration: 33000 (of 40000)
## [2] iteration: 32900 (of 40000)
## [6] iteration: 33900 (of 40000)
## [1] iteration: 33500 (of 40000)
## [5] iteration: 33500 (of 40000)
## [8] iteration: 33400 (of 40000)
## [4] iteration: 33300 (of 40000)
## [7] iteration: 34100 (of 40000)
## [3] iteration: 33100 (of 40000)
## [2] iteration: 33000 (of 40000)
## [6] iteration: 34000 (of 40000)
## [1] iteration: 33600 (of 40000)
## [5] iteration: 33600 (of 40000)
## [8] iteration: 33500 (of 40000)
## [4] iteration: 33400 (of 40000)
## [7] iteration: 34200 (of 40000)
## [3] iteration: 33200 (of 40000)
## [6] iteration: 34100 (of 40000)
## [2] iteration: 33100 (of 40000)
## [1] iteration: 33700 (of 40000)
## [5] iteration: 33700 (of 40000)
## [8] iteration: 33600 (of 40000)
## [4] iteration: 33500 (of 40000)
## [7] iteration: 34300 (of 40000)
## [3] iteration: 33300 (of 40000)
## [6] iteration: 34200 (of 40000)
## [2] iteration: 33200 (of 40000)
## [1] iteration: 33800 (of 40000)
## [5] iteration: 33800 (of 40000)
## [8] iteration: 33700 (of 40000)
## [4] iteration: 33600 (of 40000)
## [7] iteration: 34400 (of 40000)
## [3] iteration: 33400 (of 40000)
## [6] iteration: 34300 (of 40000)
## [2] iteration: 33300 (of 40000)
## [1] iteration: 33900 (of 40000)
## [5] iteration: 33900 (of 40000)
## [8] iteration: 33800 (of 40000)
## [4] iteration: 33700 (of 40000)
## [7] iteration: 34500 (of 40000)
## [3] iteration: 33500 (of 40000)
## [6] iteration: 34400 (of 40000)
## [2] iteration: 33400 (of 40000)
## [1] iteration: 34000 (of 40000)
## [5] iteration: 34000 (of 40000)
```

```
## [8] iteration: 33900 (of 40000)
## [7] iteration: 34600 (of 40000)
## [4] iteration: 33800 (of 40000)
## [3] iteration: 33600 (of 40000)
## [6] iteration: 34500 (of 40000)
## [2] iteration: 33500 (of 40000)
## [1] iteration: 34100 (of 40000)
## [5] iteration: 34100 (of 40000)
## [7] iteration: 34700 (of 40000)
## [8] iteration: 34000 (of 40000)
## [4] iteration: 33900 (of 40000)
## [6] iteration: 34600 (of 40000)
## [3] iteration: 33700 (of 40000)
## [2] iteration: 33600 (of 40000)
## [1] iteration: 34200 (of 40000)
## [5] iteration: 34200 (of 40000)
## [7] iteration: 34800 (of 40000)
## [8] iteration: 34100 (of 40000)
## [4] iteration: 34000 (of 40000)
## [6] iteration: 34700 (of 40000)
## [3] iteration: 33800 (of 40000)
## [1] iteration: 34300 (of 40000)
## [2] iteration: 33700 (of 40000)
## [5] iteration: 34300 (of 40000)
## [7] iteration: 34900 (of 40000)
## [4] iteration: 34100 (of 40000)
## [8] iteration: 34200 (of 40000)
## [6] iteration: 34800 (of 40000)
## [3] iteration: 33900 (of 40000)
## [1] iteration: 34400 (of 40000)
## [2] iteration: 33800 (of 40000)
## [5] iteration: 34400 (of 40000)
## [7] iteration: 35000 (of 40000)
## [4] iteration: 34200 (of 40000)
## [8] iteration: 34300 (of 40000)
## [6] iteration: 34900 (of 40000)
## [3] iteration: 34000 (of 40000)
## [1] iteration: 34500 (of 40000)
## [2] iteration: 33900 (of 40000)
## [5] iteration: 34500 (of 40000)
## [7] iteration: 35100 (of 40000)
## [8] iteration: 34400 (of 40000)
## [4] iteration: 34300 (of 40000)
## [6] iteration: 35000 (of 40000)
## [3] iteration: 34100 (of 40000)
## [1] iteration: 34600 (of 40000)
## [2] iteration: 34000 (of 40000)
## [5] iteration: 34600 (of 40000)
## [7] iteration: 35200 (of 40000)
## [8] iteration: 34500 (of 40000)
## [4] iteration: 34400 (of 40000)
## [6] iteration: 35100 (of 40000)
## [3] iteration: 34200 (of 40000)
## [1] iteration: 34700 (of 40000)
## [2] iteration: 34100 (of 40000)
## [7] iteration: 35300 (of 40000)
```

```
## [5] iteration: 34700 (of 40000)
## [8] iteration: 34600 (of 40000)
## [4] iteration: 34500 (of 40000)
## [6] iteration: 35200 (of 40000)
## [3] iteration: 34300 (of 40000)
## [1] iteration: 34800 (of 40000)
## [2] iteration: 34200 (of 40000)
## [7] iteration: 35400 (of 40000)
## [5] iteration: 34800 (of 40000)
## [8] iteration: 34700 (of 40000)
## [4] iteration: 34600 (of 40000)
## [6] iteration: 35300 (of 40000)
## [3] iteration: 34400 (of 40000)
## [1] iteration: 34900 (of 40000)
## [7] iteration: 35500 (of 40000)
## [2] iteration: 34300 (of 40000)
## [5] iteration: 34900 (of 40000)
## [8] iteration: 34800 (of 40000)
## [4] iteration: 34700 (of 40000)
## [6] iteration: 35400 (of 40000)
## [3] iteration: 34500 (of 40000)
## [1] iteration: 35000 (of 40000)
## [7] iteration: 35600 (of 40000)
## [2] iteration: 34400 (of 40000)
## [5] iteration: 35000 (of 40000)
## [8] iteration: 34900 (of 40000)
## [4] iteration: 34800 (of 40000)
## [6] iteration: 35500 (of 40000)
## [3] iteration: 34600 (of 40000)
## [1] iteration: 35100 (of 40000)
## [7] iteration: 35700 (of 40000)
## [2] iteration: 34500 (of 40000)
## [5] iteration: 35100 (of 40000)
## [8] iteration: 35000 (of 40000)
## [4] iteration: 34900 (of 40000)
## [6] iteration: 35600 (of 40000)
## [3] iteration: 34700 (of 40000)
## [1] iteration: 35200 (of 40000)
## [7] iteration: 35800 (of 40000)
## [2] iteration: 34600 (of 40000)
## [5] iteration: 35200 (of 40000)
## [8] iteration: 35100 (of 40000)
## [4] iteration: 35000 (of 40000)
## [6] iteration: 35700 (of 40000)
## [3] iteration: 34800 (of 40000)
## [1] iteration: 35300 (of 40000)
## [7] iteration: 35900 (of 40000)
## [2] iteration: 34700 (of 40000)
## [5] iteration: 35300 (of 40000)
## [8] iteration: 35200 (of 40000)
## [4] iteration: 35100 (of 40000)
## [6] iteration: 35800 (of 40000)
## [3] iteration: 34900 (of 40000)
## [1] iteration: 35400 (of 40000)
## [7] iteration: 36000 (of 40000)
## [5] iteration: 35400 (of 40000)
```

```
## [2] iteration: 34800 (of 40000)
## [8] iteration: 35300 (of 40000)
## [4] iteration: 35200 (of 40000)
## [6] iteration: 35900 (of 40000)
## [3] iteration: 35000 (of 40000)
## [1] iteration: 35500 (of 40000)
## [7] iteration: 36100 (of 40000)
## [5] iteration: 35500 (of 40000)
## [2] iteration: 34900 (of 40000)
## [8] iteration: 35400 (of 40000)
## [4] iteration: 35300 (of 40000)
## [6] iteration: 36000 (of 40000)
## [3] iteration: 35100 (of 40000)
## [1] iteration: 35600 (of 40000)
## [7] iteration: 36200 (of 40000)
## [5] iteration: 35600 (of 40000)
## [2] iteration: 35000 (of 40000)
## [8] iteration: 35500 (of 40000)
## [4] iteration: 35400 (of 40000)
## [6] iteration: 36100 (of 40000)
## [3] iteration: 35200 (of 40000)
## [1] iteration: 35700 (of 40000)
## [7] iteration: 36300 (of 40000)
## [5] iteration: 35700 (of 40000)
## [2] iteration: 35100 (of 40000)
## [8] iteration: 35600 (of 40000)
## [4] iteration: 35500 (of 40000)
## [6] iteration: 36200 (of 40000)
## [1] iteration: 35800 (of 40000)
## [7] iteration: 36400 (of 40000)
## [3] iteration: 35300 (of 40000)
## [5] iteration: 35800 (of 40000)
## [2] iteration: 35200 (of 40000)
## [8] iteration: 35700 (of 40000)
## [4] iteration: 35600 (of 40000)
## [6] iteration: 36300 (of 40000)
## [7] iteration: 36500 (of 40000)
## [1] iteration: 35900 (of 40000)
## [3] iteration: 35400 (of 40000)
## [5] iteration: 35900 (of 40000)
## [2] iteration: 35300 (of 40000)
## [8] iteration: 35800 (of 40000)
## [4] iteration: 35700 (of 40000)
## [6] iteration: 36400 (of 40000)
## [7] iteration: 36600 (of 40000)
## [1] iteration: 36000 (of 40000)
## [3] iteration: 35500 (of 40000)
## [5] iteration: 36000 (of 40000)
## [2] iteration: 35400 (of 40000)
## [8] iteration: 35900 (of 40000)
## [4] iteration: 35800 (of 40000)
## [6] iteration: 36500 (of 40000)
## [7] iteration: 36700 (of 40000)
## [1] iteration: 36100 (of 40000)
## [3] iteration: 35600 (of 40000)
## [5] iteration: 36100 (of 40000)
```

```
## [2] iteration: 35500 (of 40000)
## [8] iteration: 36000 (of 40000)
## [4] iteration: 35900 (of 40000)
## [6] iteration: 36600 (of 40000)
## [7] iteration: 36800 (of 40000)
## [1] iteration: 36200 (of 40000)
## [3] iteration: 35700 (of 40000)
## [5] iteration: 36200 (of 40000)
## [2] iteration: 35600 (of 40000)
## [8] iteration: 36100 (of 40000)
## [4] iteration: 36000 (of 40000)
## [6] iteration: 36700 (of 40000)
## [7] iteration: 36900 (of 40000)
## [1] iteration: 36300 (of 40000)
## [3] iteration: 35800 (of 40000)
## [5] iteration: 36300 (of 40000)
## [8] iteration: 36200 (of 40000)
## [2] iteration: 35700 (of 40000)
## [4] iteration: 36100 (of 40000)
## [6] iteration: 36800 (of 40000)
## [7] iteration: 37000 (of 40000)
## [1] iteration: 36400 (of 40000)
## [3] iteration: 35900 (of 40000)
## [5] iteration: 36400 (of 40000)
## [8] iteration: 36300 (of 40000)
## [2] iteration: 35800 (of 40000)
## [4] iteration: 36200 (of 40000)
## [6] iteration: 36900 (of 40000)
## [7] iteration: 37100 (of 40000)
## [1] iteration: 36500 (of 40000)
## [3] iteration: 36000 (of 40000)
## [5] iteration: 36500 (of 40000)
## [8] iteration: 36400 (of 40000)
## [2] iteration: 35900 (of 40000)
## [4] iteration: 36300 (of 40000)
## [6] iteration: 37000 (of 40000)
## [7] iteration: 37200 (of 40000)
## [1] iteration: 36600 (of 40000)
## [3] iteration: 36100 (of 40000)
## [5] iteration: 36600 (of 40000)
## [8] iteration: 36500 (of 40000)
## [2] iteration: 36000 (of 40000)
## [4] iteration: 36400 (of 40000)
## [6] iteration: 37100 (of 40000)
## [7] iteration: 37300 (of 40000)
## [1] iteration: 36700 (of 40000)
## [3] iteration: 36200 (of 40000)
## [5] iteration: 36700 (of 40000)
## [8] iteration: 36600 (of 40000)
## [6] iteration: 37200 (of 40000)
## [4] iteration: 36500 (of 40000)
## [2] iteration: 36100 (of 40000)
## [7] iteration: 37400 (of 40000)
## [1] iteration: 36800 (of 40000)
## [3] iteration: 36300 (of 40000)
## [5] iteration: 36800 (of 40000)
```

```
## [8] iteration: 36700 (of 40000)
## [6] iteration: 37300 (of 40000)
## [4] iteration: 36600 (of 40000)
## [2] iteration: 36200 (of 40000)
## [7] iteration: 37500 (of 40000)
## [1] iteration: 36900 (of 40000)
## [3] iteration: 36400 (of 40000)
## [5] iteration: 36900 (of 40000)
## [8] iteration: 36800 (of 40000)
## [6] iteration: 37400 (of 40000)
## [4] iteration: 36700 (of 40000)
## [2] iteration: 36300 (of 40000)
## [7] iteration: 37600 (of 40000)
## [1] iteration: 37000 (of 40000)
## [3] iteration: 36500 (of 40000)
## [5] iteration: 37000 (of 40000)
## [6] iteration: 37500 (of 40000)
## [8] iteration: 36900 (of 40000)
## [4] iteration: 36800 (of 40000)
## [2] iteration: 36400 (of 40000)
## [7] iteration: 37700 (of 40000)
## [1] iteration: 37100 (of 40000)
## [3] iteration: 36600 (of 40000)
## [5] iteration: 37100 (of 40000)
## [6] iteration: 37600 (of 40000)
## [8] iteration: 37000 (of 40000)
## [4] iteration: 36900 (of 40000)
## [2] iteration: 36500 (of 40000)
## [7] iteration: 37800 (of 40000)
## [1] iteration: 37200 (of 40000)
## [3] iteration: 36700 (of 40000)
## [5] iteration: 37200 (of 40000)
## [6] iteration: 37700 (of 40000)
## [8] iteration: 37100 (of 40000)
## [4] iteration: 37000 (of 40000)
## [2] iteration: 36600 (of 40000)
## [7] iteration: 37900 (of 40000)
## [1] iteration: 37300 (of 40000)
## [3] iteration: 36800 (of 40000)
## [5] iteration: 37300 (of 40000)
## [6] iteration: 37800 (of 40000)
## [8] iteration: 37200 (of 40000)
## [4] iteration: 37100 (of 40000)
## [2] iteration: 36700 (of 40000)
## [7] iteration: 38000 (of 40000)
## [1] iteration: 37400 (of 40000)
## [3] iteration: 36900 (of 40000)
## [5] iteration: 37400 (of 40000)
## [6] iteration: 37900 (of 40000)
## [8] iteration: 37300 (of 40000)
## [4] iteration: 37200 (of 40000)
## [2] iteration: 36800 (of 40000)
## [7] iteration: 38100 (of 40000)
## [1] iteration: 37500 (of 40000)
## [3] iteration: 37000 (of 40000)
## [6] iteration: 38000 (of 40000)
```

```
## [5] iteration: 37500 (of 40000)
## [8] iteration: 37400 (of 40000)
## [4] iteration: 37300 (of 40000)
## [2] iteration: 36900 (of 40000)
## [7] iteration: 38200 (of 40000)
## [1] iteration: 37600 (of 40000)
## [3] iteration: 37100 (of 40000)
## [6] iteration: 38100 (of 40000)
## [5] iteration: 37600 (of 40000)
## [8] iteration: 37500 (of 40000)
## [4] iteration: 37400 (of 40000)
## [7] iteration: 38300 (of 40000)
## [2] iteration: 37000 (of 40000)
## [1] iteration: 37700 (of 40000)
## [6] iteration: 38200 (of 40000)
## [3] iteration: 37200 (of 40000)
## [5] iteration: 37700 (of 40000)
## [8] iteration: 37600 (of 40000)
## [4] iteration: 37500 (of 40000)
## [7] iteration: 38400 (of 40000)
## [2] iteration: 37100 (of 40000)
## [1] iteration: 37800 (of 40000)
## [6] iteration: 38300 (of 40000)
## [3] iteration: 37300 (of 40000)
## [5] iteration: 37800 (of 40000)
## [8] iteration: 37700 (of 40000)
## [4] iteration: 37600 (of 40000)
## [7] iteration: 38500 (of 40000)
## [2] iteration: 37200 (of 40000)
## [1] iteration: 37900 (of 40000)
## [6] iteration: 38400 (of 40000)
## [3] iteration: 37400 (of 40000)
## [5] iteration: 37900 (of 40000)
## [8] iteration: 37800 (of 40000)
## [4] iteration: 37700 (of 40000)
## [7] iteration: 38600 (of 40000)
## [2] iteration: 37300 (of 40000)
## [1] iteration: 38000 (of 40000)
## [6] iteration: 38500 (of 40000)
## [3] iteration: 37500 (of 40000)
## [5] iteration: 38000 (of 40000)
## [8] iteration: 37900 (of 40000)
## [4] iteration: 37800 (of 40000)
## [7] iteration: 38700 (of 40000)
## [2] iteration: 37400 (of 40000)
## [1] iteration: 38100 (of 40000)
## [6] iteration: 38600 (of 40000)
## [3] iteration: 37600 (of 40000)
## [5] iteration: 38100 (of 40000)
## [8] iteration: 38000 (of 40000)
## [4] iteration: 37900 (of 40000)
## [7] iteration: 38800 (of 40000)
## [2] iteration: 37500 (of 40000)
## [1] iteration: 38200 (of 40000)
## [6] iteration: 38700 (of 40000)
## [3] iteration: 37700 (of 40000)
```

```
## [5] iteration: 38200 (of 40000)
## [8] iteration: 38100 (of 40000)
## [4] iteration: 38000 (of 40000)
## [7] iteration: 38900 (of 40000)
## [2] iteration: 37600 (of 40000)
## [1] iteration: 38300 (of 40000)
## [6] iteration: 38800 (of 40000)
## [3] iteration: 37800 (of 40000)
## [5] iteration: 38300 (of 40000)
## [8] iteration: 38200 (of 40000)
## [4] iteration: 38100 (of 40000)
## [7] iteration: 39000 (of 40000)
## [2] iteration: 37700 (of 40000)
## [1] iteration: 38400 (of 40000)
## [6] iteration: 38900 (of 40000)
## [3] iteration: 37900 (of 40000)
## [5] iteration: 38400 (of 40000)
## [8] iteration: 38300 (of 40000)
## [4] iteration: 38200 (of 40000)
## [7] iteration: 39100 (of 40000)
## [2] iteration: 37800 (of 40000)
## [1] iteration: 38500 (of 40000)
## [6] iteration: 39000 (of 40000)
## [3] iteration: 38000 (of 40000)
## [5] iteration: 38500 (of 40000)
## [8] iteration: 38400 (of 40000)
## [7] iteration: 39200 (of 40000)
## [4] iteration: 38300 (of 40000)
## [2] iteration: 37900 (of 40000)
## [1] iteration: 38600 (of 40000)
## [6] iteration: 39100 (of 40000)
## [3] iteration: 38100 (of 40000)
## [5] iteration: 38600 (of 40000)
## [7] iteration: 39300 (of 40000)
## [8] iteration: 38500 (of 40000)
## [4] iteration: 38400 (of 40000)
## [2] iteration: 38000 (of 40000)
## [1] iteration: 38700 (of 40000)
## [6] iteration: 39200 (of 40000)
## [3] iteration: 38200 (of 40000)
## [7] iteration: 39400 (of 40000)
## [5] iteration: 38700 (of 40000)
## [4] iteration: 38500 (of 40000)
## [8] iteration: 38600 (of 40000)
## [2] iteration: 38100 (of 40000)
## [1] iteration: 38800 (of 40000)
## [6] iteration: 39300 (of 40000)
## [3] iteration: 38300 (of 40000)
## [7] iteration: 39500 (of 40000)
## [4] iteration: 38600 (of 40000)
## [5] iteration: 38800 (of 40000)
## [8] iteration: 38700 (of 40000)
## [2] iteration: 38200 (of 40000)
## [1] iteration: 38900 (of 40000)
## [6] iteration: 39400 (of 40000)
## [7] iteration: 39600 (of 40000)
```

```
## [3] iteration: 38400 (of 40000)
## [4] iteration: 38700 (of 40000)
## [5] iteration: 38900 (of 40000)
## [8] iteration: 38800 (of 40000)
## [2] iteration: 38300 (of 40000)
## [1] iteration: 39000 (of 40000)
## [6] iteration: 39500 (of 40000)
## [7] iteration: 39700 (of 40000)
## [3] iteration: 38500 (of 40000)
## [4] iteration: 38800 (of 40000)
## [5] iteration: 39000 (of 40000)
## [8] iteration: 38900 (of 40000)
## [2] iteration: 38400 (of 40000)
## [1] iteration: 39100 (of 40000)
## [6] iteration: 39600 (of 40000)
## [7] iteration: 39800 (of 40000)
## [3] iteration: 38600 (of 40000)
## [4] iteration: 38900 (of 40000)
## [5] iteration: 39100 (of 40000)
## [8] iteration: 39000 (of 40000)
## [1] iteration: 39200 (of 40000)
## [2] iteration: 38500 (of 40000)
## [6] iteration: 39700 (of 40000)
## [7] iteration: 39900 (of 40000)
## [3] iteration: 38700 (of 40000)
## [4] iteration: 39000 (of 40000)
## [5] iteration: 39200 (of 40000)
## [8] iteration: 39100 (of 40000)
## [1] iteration: 39300 (of 40000)
## [6] iteration: 39800 (of 40000)
## [2] iteration: 38600 (of 40000)
## [7] iteration: 40000 (of 40000)
## [3] iteration: 38800 (of 40000)
## [4] iteration: 39100 (of 40000)
## [5] iteration: 39300 (of 40000)
## [8] iteration: 39200 (of 40000)
## [1] iteration: 39400 (of 40000)
## [6] iteration: 39900 (of 40000)
## [2] iteration: 38700 (of 40000)
## [3] iteration: 38900 (of 40000)
## [4] iteration: 39200 (of 40000)
## [5] iteration: 39400 (of 40000)
## [8] iteration: 39300 (of 40000)
## [1] iteration: 39500 (of 40000)
## [6] iteration: 40000 (of 40000)
## [2] iteration: 38800 (of 40000)
## [3] iteration: 39000 (of 40000)
## [4] iteration: 39300 (of 40000)
## [5] iteration: 39500 (of 40000)
## [8] iteration: 39400 (of 40000)
## [1] iteration: 39600 (of 40000)
## [2] iteration: 38900 (of 40000)
## [3] iteration: 39100 (of 40000)
## [4] iteration: 39400 (of 40000)
## [5] iteration: 39600 (of 40000)
## [8] iteration: 39500 (of 40000)
```

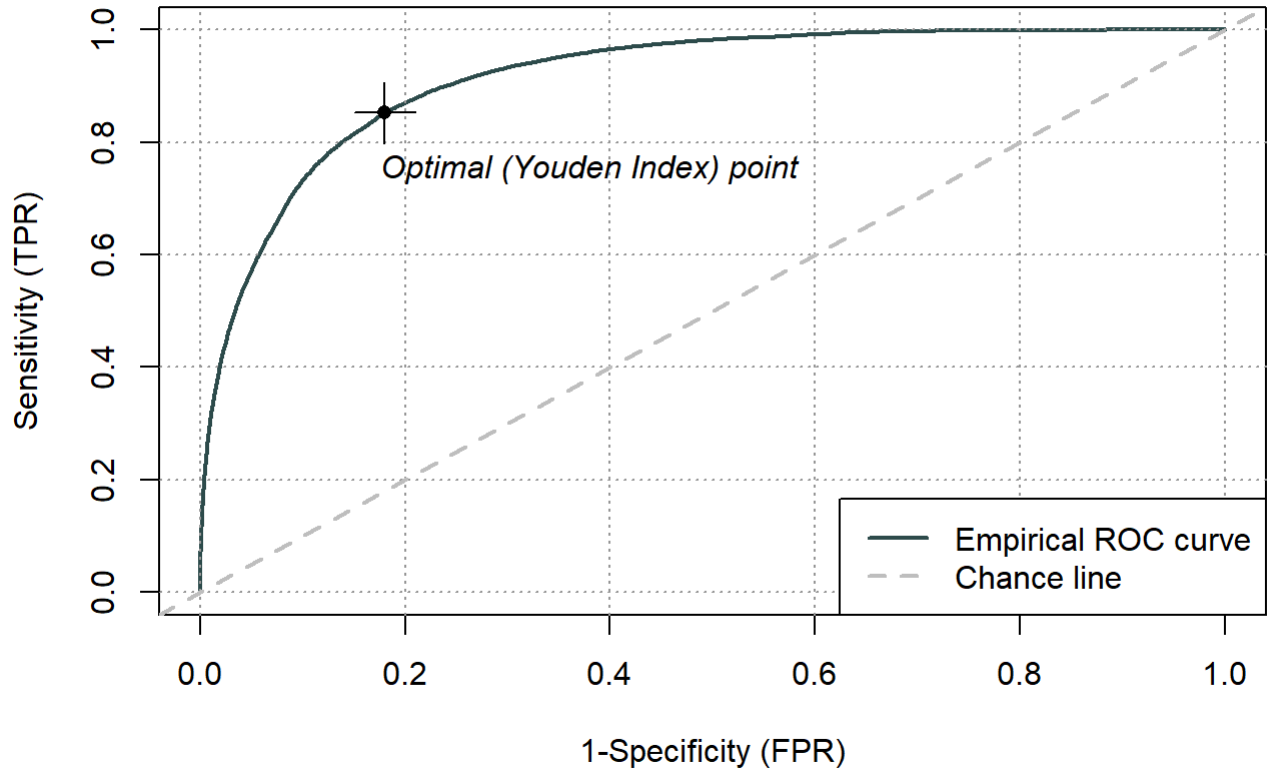
```
## [1] iteration: 39700 (of 40000)
## [2] iteration: 39000 (of 40000)
## [3] iteration: 39200 (of 40000)
## [4] iteration: 39500 (of 40000)
## [5] iteration: 39700 (of 40000)
## [8] iteration: 39600 (of 40000)
## [1] iteration: 39800 (of 40000)
## [2] iteration: 39100 (of 40000)
## [3] iteration: 39300 (of 40000)
## [4] iteration: 39600 (of 40000)
## [5] iteration: 39800 (of 40000)
## [8] iteration: 39700 (of 40000)
## [1] iteration: 39900 (of 40000)
## [2] iteration: 39200 (of 40000)
## [3] iteration: 39400 (of 40000)
## [4] iteration: 39700 (of 40000)
## [5] iteration: 39900 (of 40000)
## [8] iteration: 39800 (of 40000)
## [1] iteration: 40000 (of 40000)
## [2] iteration: 39300 (of 40000)
## [3] iteration: 39500 (of 40000)
## [4] iteration: 39800 (of 40000)
## [5] iteration: 40000 (of 40000)
## [8] iteration: 39900 (of 40000)
## [2] iteration: 39400 (of 40000)
## [4] iteration: 39900 (of 40000)
## [3] iteration: 39600 (of 40000)
## [8] iteration: 40000 (of 40000)
## [2] iteration: 39500 (of 40000)
## [4] iteration: 40000 (of 40000)
## [3] iteration: 39700 (of 40000)
## [2] iteration: 39600 (of 40000)
## [3] iteration: 39800 (of 40000)
## [2] iteration: 39700 (of 40000)
## [3] iteration: 39900 (of 40000)
## [2] iteration: 39800 (of 40000)
## [3] iteration: 40000 (of 40000)
## [2] iteration: 39900 (of 40000)
## [2] iteration: 40000 (of 40000)
## total seconds in loop: 23883.480405
##
## Tree sizes, last iteration:
## [1] 5 5 11 6 7 7 10 5 6 7 10 10 6 10 10 7 8 10
## 12 6 6 3 12 6 9 10 6 9 6 6 10 5 7 6 8 7 4 12
## 10 6 10 4 6 5 6 6 8 5 4 7
## [2] 5 6 4 6 10 4 7 9 10 7 5 7 13 5 8 8 12 6
## 4 4 5 5 5 7 12 9 4 10 3 8 6 12 7 12 4 12 7 10
## 10 9 5 7 12 6 5 8 7 7 5 9
## [3] 10 8 9 7 5 12 5 4 9 5 7 8 9 4 9 12 6 10
## 8 5 8 7 6 10 11 4 7 5 3 9 7 8 9 6 5 10 7 7
## 11 7 7 7 10 7 9 8 8 11 8 7
## [4] 6 10 6 8 9 7 8 9 5 8 9 9 9 8 8 7 8 10
## 10 9 4 10 8 4 8 6 6 7 6 9 6 8 8 9 9 7 11 8
## 4 6 5 9 5 7 11 8 5 9 4 4
## [5] 8 9 2 8 7 7 7 6 13 11 8 8 11 7 10 7 8 3
## 8 7 13 9 6 7 6 6 8 10 7 8 8 11 8 5 7 9 5 7
```

```
## 7 11 4 8 10 6 4 5 7 11 5 7
## [6] 8 5 10 8 12 6 7 6 7 6 7 9 6 3 9 9 8 9
## 5 7 8 8 8 11 8 5 10 7 8 5 6 9 3 7 6 7 11 10
## 5 9 11 6 10 7 8 12 10 9 5 8
## [7] 7 7 6 6 7 9 13 5 7 5 11 12 7 9 6 10 7 7
## 11 15 7 8 6 6 6 10 8 10 7 9 6 6 8 8 9 8 7 6
## 6 5 6 8 6 9 8 6 7 6 4 12
## [8] 10 4 7 10 8 9 10 9 5 6 12 11 6 12 9 5 7 5
## 10 10 10 6 6 5 9 6 13 5 5 7 7 6 6 3 5 7 8 9
## 7 9 7 10 4 10 6 9 6 7 5 5
##
## Variable Usage, last iteration (var:count):
## (1: 73) (2: 75) (3: 56) (4: 52) (5: 77)
## (6: 51) (7: 58) (8: 56) (9: 73) (10: 47)
## (11: 54) (12: 57) (13: 51) (14: 48) (15: 52)
## (16: 78) (17: 47) (18: 59) (19: 75) (20: 104)
## (21: 137) (22: 54) (23: 86) (24: 109) (25: 101)
## (26: 114) (27: 55) (28: 12) (29: 29) (30: 109)
## (31: 45) (32: 67) (33: 64) (34: 103) (35: 111)
## (36: 36) (37: 75) (38: 62)
## DONE BART
```

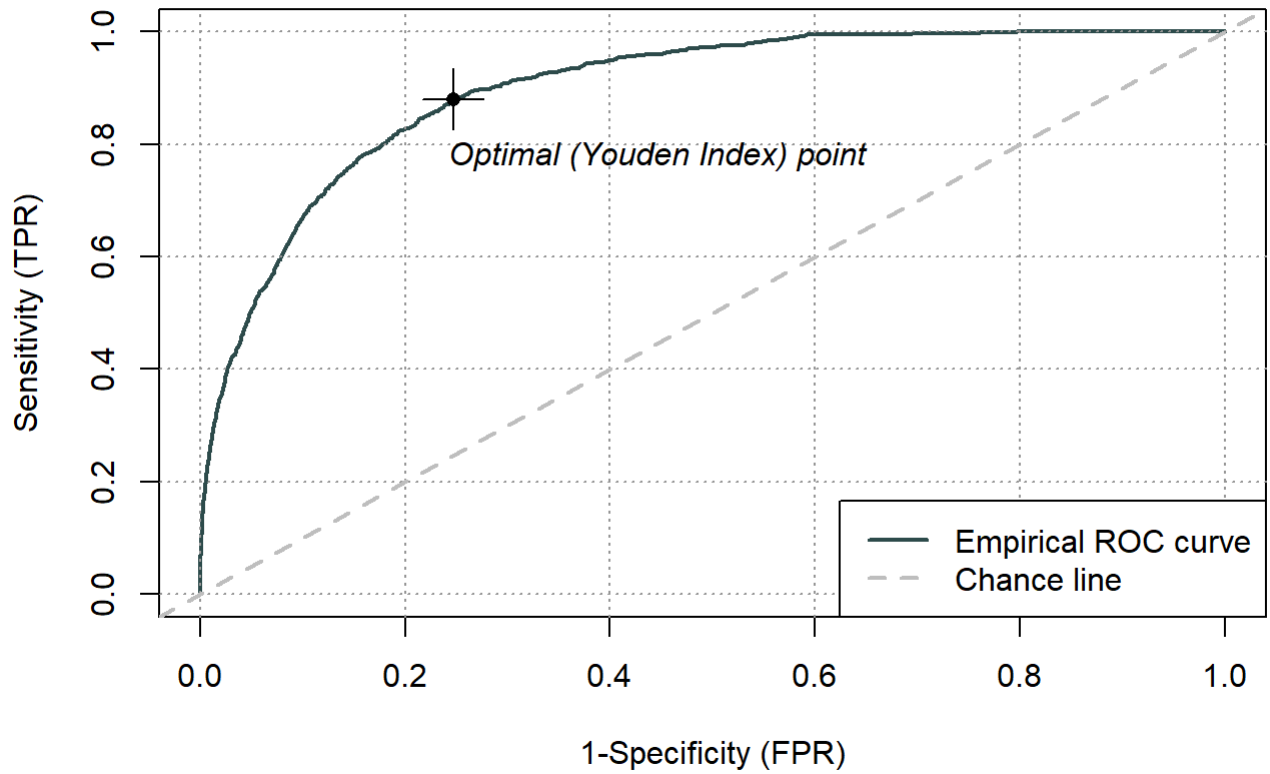
```
predictedTrain <- predict(model, X) |> apply(2,mean)
predictedTest <- predict(model, Test %>% select(T4EXINF,T4COLES,T4SELF,T4SEMUL,T4GROMST,T4SES
EN,T4COMSEL,T4SEDRT,T4BLFDRT,T4ADVDRT,T4CLAIN,T4COGAC,T4CLASM,T4PDGENP,T4EMPS,T4CLSDIS,T4ADL
E,T4PBLE,T4AUTCH,T4FULFIL,T4OPPART,T4SESDEV,T4TLEAD,T4STUD,T4CLIBUL,T4STUHAR,T4CHFAT,T4MOPU,T
4MOSU,T4WELS,T4WLOADT,T4STBEH,T4JSENVT,T4JSPROT,T4VALP,T4JOYTCH,UseDigRes,Leadership)   ) |>
apply(2,mean)
save(model,predictedTrain, predictedTest, file="MainModel.rData")
```

get data for ROC curves

```
# load(file="MainModel.rData")
ROCit_objTr <- rocit(score=model$yhat.train |> pnorm() |> apply(2,mean),class=BTGINTT4$Nordi
c,negref =0)
ROCit_obj   <- rocit(score=model$yhat.test  |> pnorm() |> apply(2,mean),class=Test$Nordic
,negref =0)
ROCIT_NordicTr <- plot(ROCit_objTr)
```




```
ROCIT_Nordic   <- plot(ROCit_obj  )
```



```
save(ROCIT_NordicTr,ROCit_objTr,
     ROCIT_Nordic  ,ROCit_obj  ,
     file="ROCIT_Nordic.rData")
```

get data for trees

```
tree_data <- extractTrees.bart.fixed(model = model, data = BTGINTT4 %>% select(T4EXINF,T4COLE
S,T4SELF,T4SEMUL,T4GROMST,T4SESEN,T4COMSEL,T4SEDRT,T4BLFDRT,T4ADVDRT,T4CLAIN,T4COGAC,T4CLASM,
T4PDGENP,T4EMPS,T4CLSDIS,T4ADLE,T4PBLE,T4AUTCH,T4FULFIL,T4OPPART,T4SESDEV,T4TLEAD,T4STUD,T4CL
IBUL,T4STUHAR,T4CHFAT,T4MOPU,T4MOSU,T4WELS,T4WLOADT,T4STBEH,T4JSENVT,T4JSPROT,T4VALP,T4JOYTC
H,UseDigRes,Leadership) )
```

```
## Extracting Observation Data...
```

```
hideHelper1 <- function(df) {
        class(df) <- c("hideHelper1", class(df))
        df }

tree_data <- hideHelper1(tree_data)
```

create shap values

```
pfun <- function (object, newdata) {
  predict(object , newdata)
}

shap_results <- Explain(model,
          X = as.data.frame(X),
          newdata = Test %>% select(T4EXINF,T4COLES,T4SELF,T4SEMUL,T4GROMST,T4SESEN,T4COMSEL,
T4SEDRT,T4BLFDRT,T4ADVDRT,T4CLAIN,T4COGAC,T4CLASM,T4PDGENP,T4EMPS,T4CLSDIS,T4ADLE,T4PBLE,T4AU
TCH,T4FULFIL,T4OPPART,T4SESDEV,T4TLEAD,T4STUD,T4CLIBUL,T4STUHAR,T4CHFAT,T4MOPU,T4MOSU,T4WELS,
T4WLOADT,T4STBEH,T4JSENVT,T4JSPROT,T4VALP,T4JOYTCH,UseDigRes,Leadership) %>% as.data.frame,
          pred_wrapper = pfun)
```

Check model convergence and calculate ESS

```
macro_predictions <- split( pnorm(model$yhat.train) %>% apply(c(1), mean) %>% t ,
                            rep(1:model$call$nchain , each=model$call$ndpost/model$call$keepe
very) )
mcmc_chains <- lapply(1:length(macro_predictions), function(i) {
  mcmc(macro_predictions[[i]]) }) %>%  as.mcmc.list

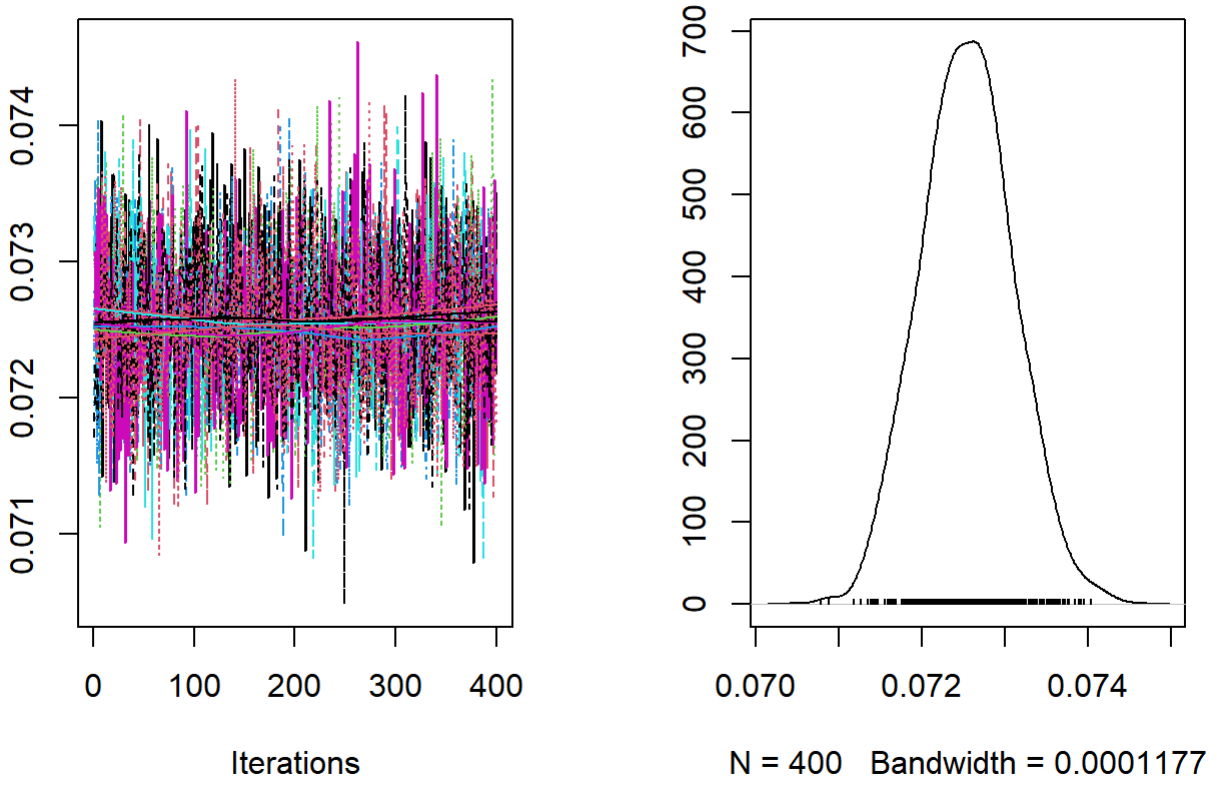
gelman.diag(mcmc_chains)
```

```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## [1,]          1       1.01
```

```
print(coda::effectiveSize(mcmc_chains))
```

```
##     var1
## 3044.548
```

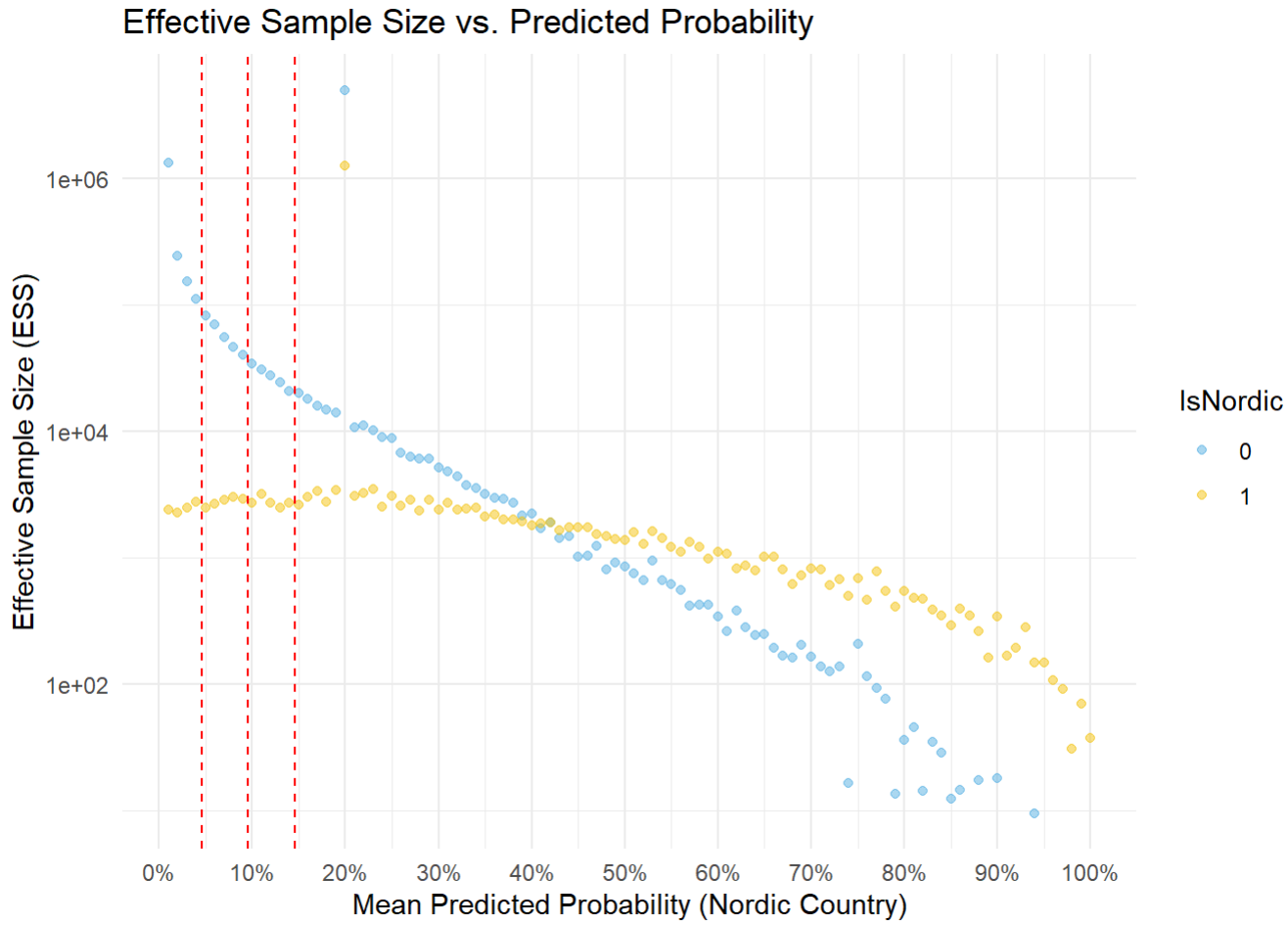
```
plot(mcmc_chains)
```



```
pred_prob_matrix <- model$yhat.train |> pnorm()
# Calculate ESS for each column (teacher) by looking across samples for each case
ess_values <- apply(pred_prob_matrix, 2, effectiveSize)
# Calculate the mean probability for each teacher
mean_probs <- colMeans(pred_prob_matrix)
# Create a data frame combining Teacher ID, Mean Probability, and ESS
teacher_diagnostics <- data.frame(
  IsNordic=model$fit$data@y,
  teacher_id = 1:length(mean_probs),
  mean_prob = mean_probs,
  ess = ess_values  )

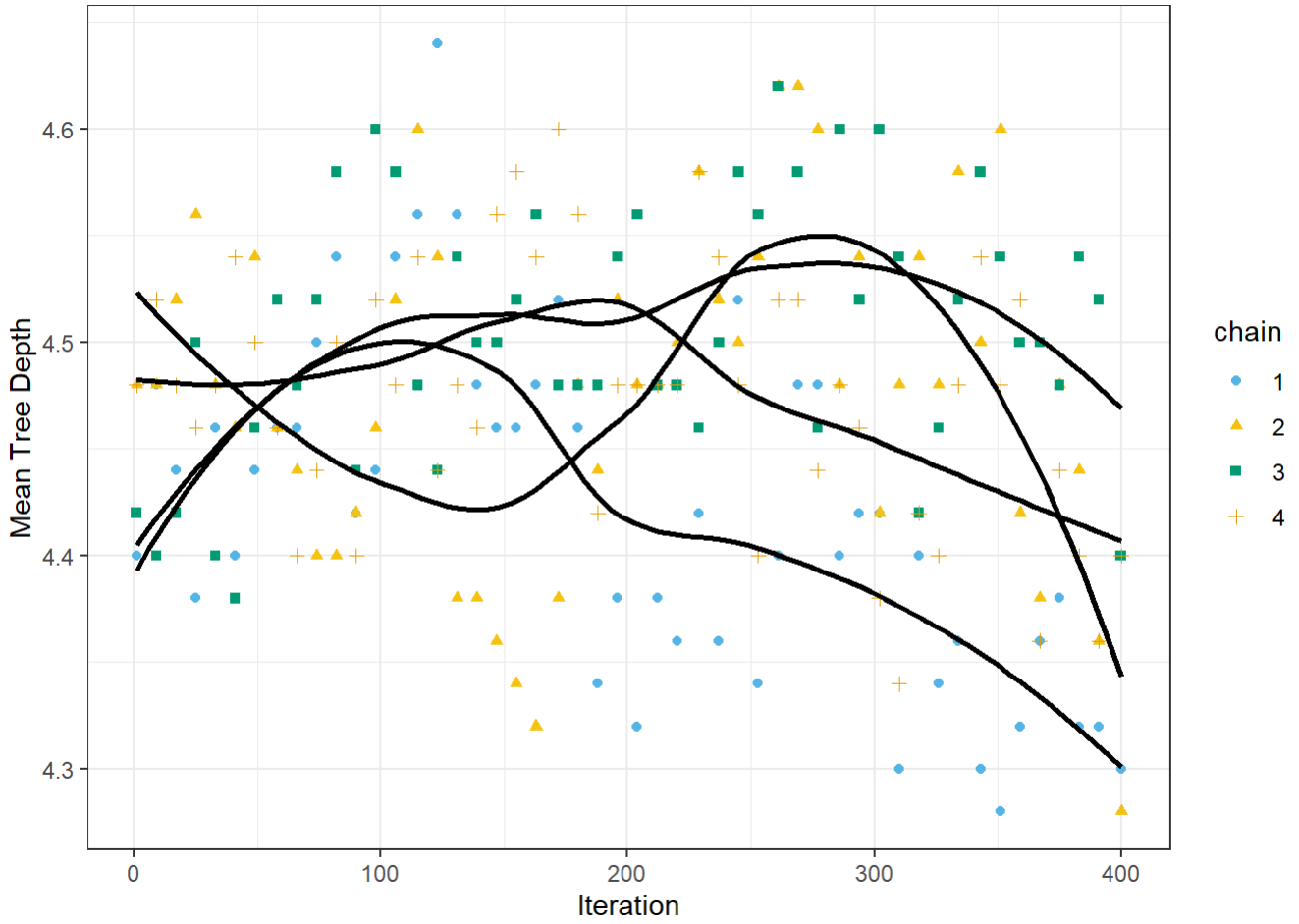
cutoff_zone <- teacher_diagnostics[teacher_diagnostics$mean_prob >= ROCIT_NordicTr$optimal[4]
-0.05 &
                                   teacher_diagnostics$mean_prob <= ROCIT_NordicTr$optimal[4]
+0.05, ]

teacher_diagnostics |>
  mutate(mean_prob=cut(mean_prob,seq(0,1,0.01),include.lowest=T) |> factor() |> as.integer()
,
         mean_prob=mean_prob/100,
         IsNordic=factor(IsNordic) ) |>
  group_by(mean_prob,IsNordic) |>
  summarize(ess=sum(ess,na.rm=T) , .groups="drop") |>
  ggplot(aes(x = mean_prob, y = ess,color=IsNordic)) +
  geom_point(alpha = 0.5) +
  geom_vline(xintercept = c(ROCIT_NordicTr$optimal[4]-0.05,
                            ROCIT_NordicTr$optimal[4],
                            ROCIT_NordicTr$optimal[4]+0.05),
             linetype = "dashed", color = "red") +
  labs(title = "Effective Sample Size vs. Predicted Probability",
       x = "Mean Predicted Probability (Nordic Country)",
       y = "Effective Sample Size (ESS)") +
  theme_minimal()+
  scale_x_continuous(breaks=seq(0,1,0.1),label=scales::percent)+
  scale_y_log10()
```

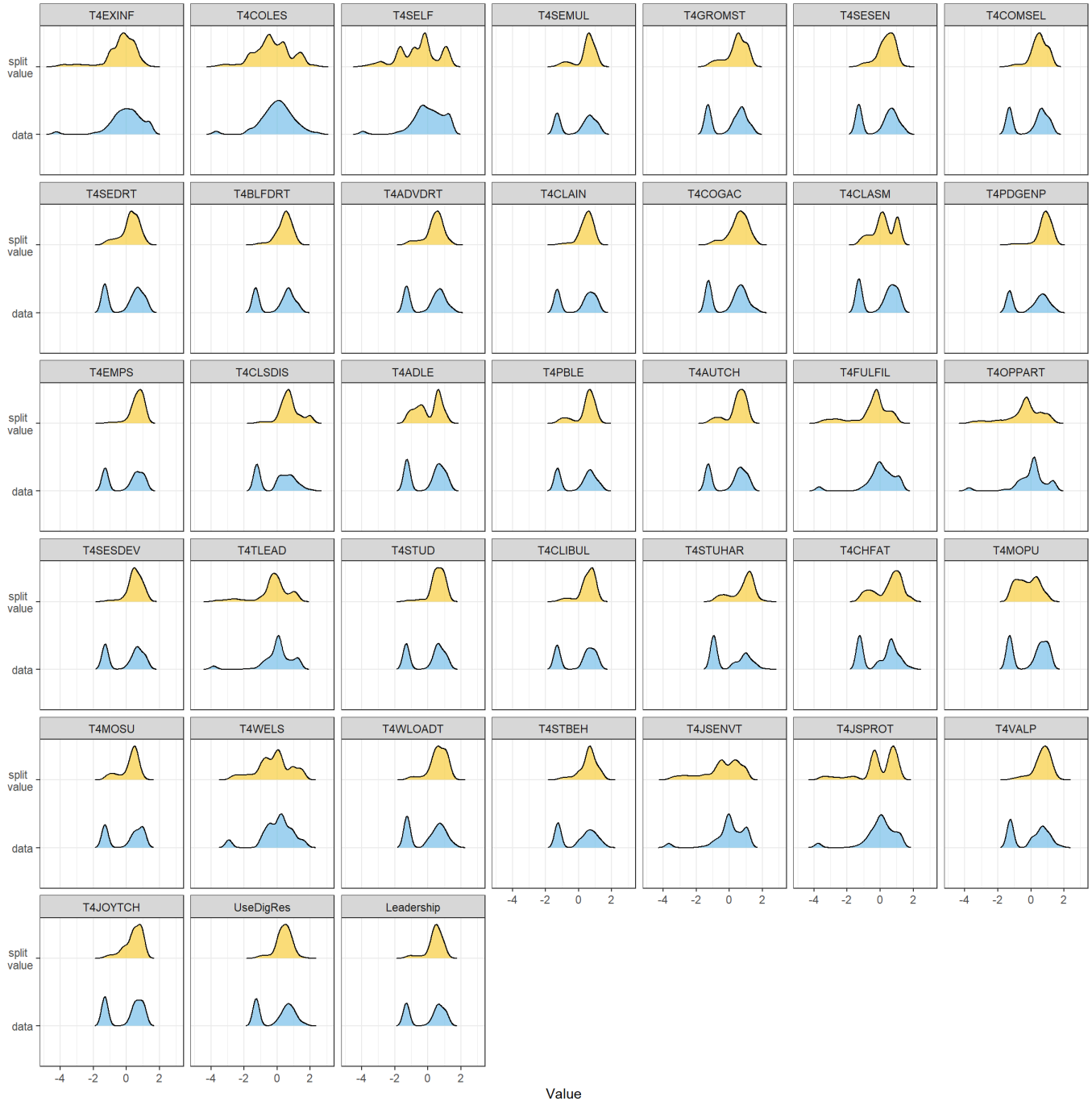


tree information

```
treeDepthf <- function (trees) {
    maxIter <- max(trees$structure$iteration)
    newTrees <- trees$structure %>%
      count(iteration, treeNum, chain, depthMax) %>%
      group_by(iteration,chain) %>%
      summarize(avgDepth = mean(depthMax),
                .groups="drop") %>%
      mutate(chain=factor(chain))
    ylimMax <- max(newTrees$avgDepth)
    p <- ggplot(newTrees, aes(iteration, avgDepth,color=chain,shape=chain)) +
      geom_point() +
      geom_smooth(formula = y ~ x, method = "loess", colour = "black", se = F) +
      theme_bw() +
      xlab("Iteration") +
      ylab("Mean Tree Depth")
    return(p)
}
treeDepthf( trees = tree_data)
```




```
splitDensity(trees = tree_data, data = BTGINTT4, display = 'dataSplit' )
```

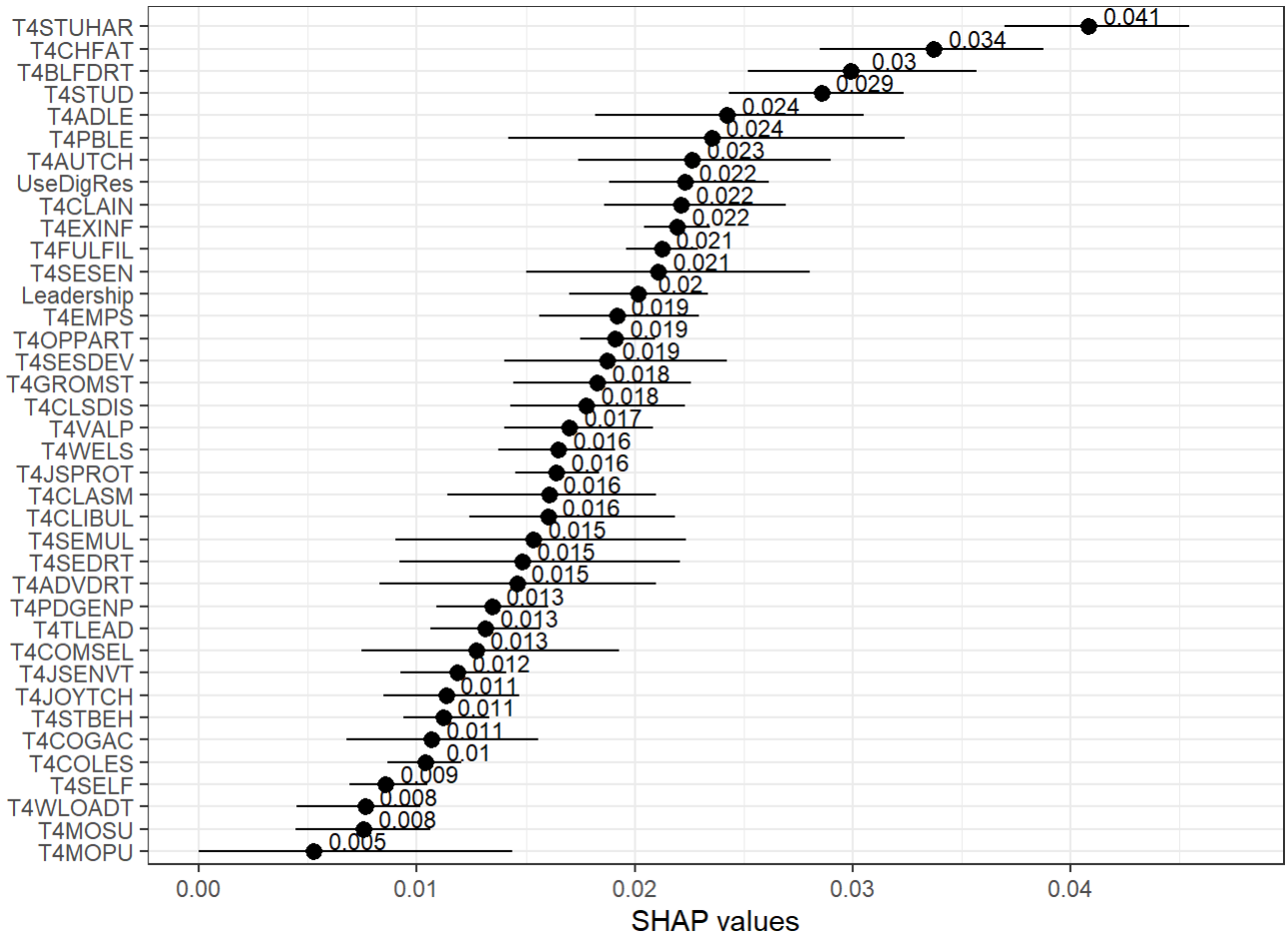

T4EXINF
T4COLES
T4SELF
T4SEMUL
T4GROMST
T4SESEN
T4COMSEL
T4SEDRT
T4BLFDRT
T4ADVDRT
T4CLAIN
T4COGAC
T4CLASM
T4PDGENP
T4EMPS
T4CLSDIS
T4ADLE
T4PBLE
T4AUTCH
T4FULFIL
T4OPPART
T4SESDEV
T4TLEAD
T4STUD
T4CLIBUL
T4STUHAR
T4CHFAT
T4MOPU
T4MOSU
T4WELS
T4WLOADT
T4STBEH
T4JSENVT
T4JSPROT
T4VALP
T4JOYTCH
UseDigRes
Leadership
split
value
data
-4
-2
0
2
Value

Results

Variable selection and variable importance

```
shap_results$phis %>%
  lapply( function(x) colMeans(abs(x)) ) %>%
  lapply( function(x) data.frame(m   =mean(x),
                                 q10=quantile(x,0.10) ,
                                 q90=quantile(x,0.90) ) ) %>%
  bind_rows(.id="Var") -> phi_CI

phi_CI %>% arrange(m) %>%
  mutate(Var=factor(Var,unique(Var)) ) %>%
  ggplot(aes(x=Var,y=m,ymin=q10,ymax=q90)) +
    geom_point()+
    geom_text(aes(label=round(m,3)),position=position_nudge(0.48,0.002),size=3 )+
    geom_pointrange()+
    theme_bw() +
    coord_flip(xlim=c(1,38.3))+
    labs(x="",y="SHAP values")
```



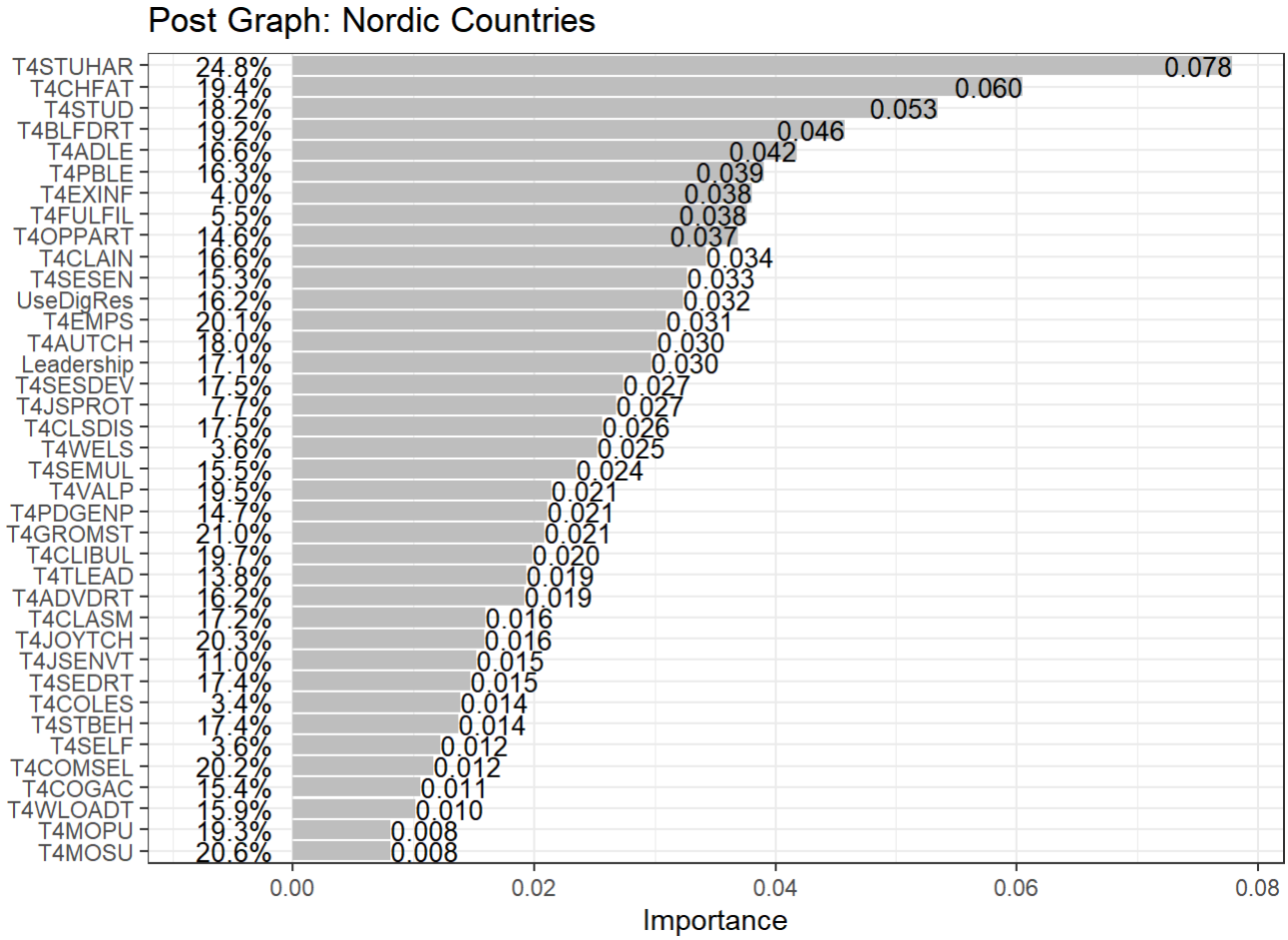

```
ggsave(file="customshapresults.png",height=14,width=14,units="cm")
```

Plot separate SHAP plots across the nordics and non-nordics.

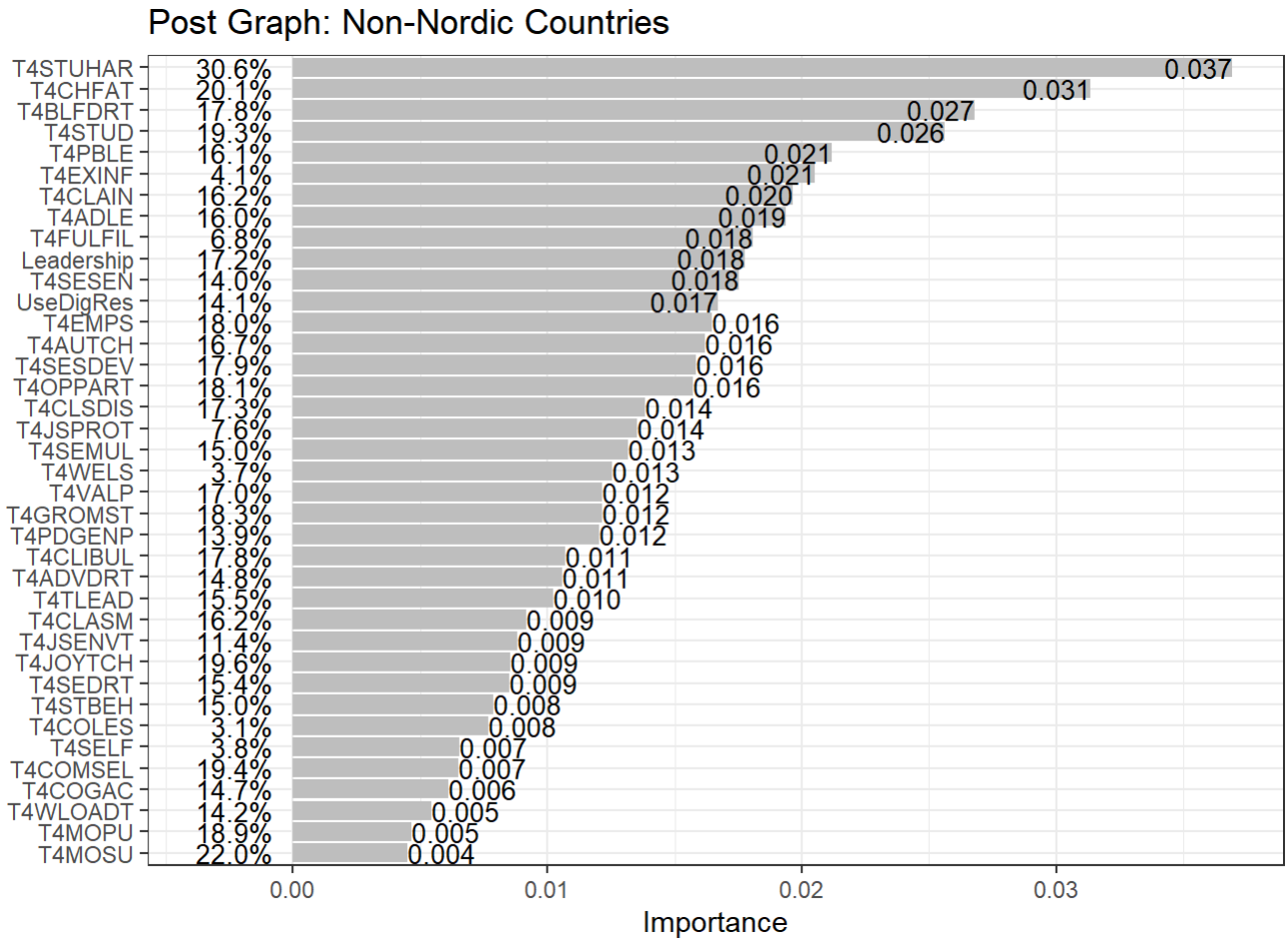
```
# 1. Duplicate your main SHAP object for the TRUE group
shap_nordic_true <- shap_results
# Filter the SHAP values (phis) and the row predictions (fx)
shap_nordic_true$phis <- lapply(shap_results$phis, function(x) x[Test$Nordic==1,, drop = FALS
E])
shap_nordic_true$newdata <- shap_nordic_true$newdata[Test$Nordic==1,, drop = FALSE]
shap_nordic_true$fx   <- shap_results$fx[Test$Nordic == 1]

# 2. Duplicate your main SHAP object for the FALSE group
shap_nordic_false <- shap_results
shap_nordic_false$phis <- lapply(shap_results$phis, function(x) x[Test$Nordic==0,, drop = FAL
SE])
shap_nordic_false$newdata <- shap_nordic_false$newdata[Test$Nordic==0,, drop = FALSE]
shap_nordic_false$fx   <- shap_results$fx[Test$Nordic == 0]

plot(shap_nordic_true,  average = "post", title= "Post Graph: Nordic Countries")
```

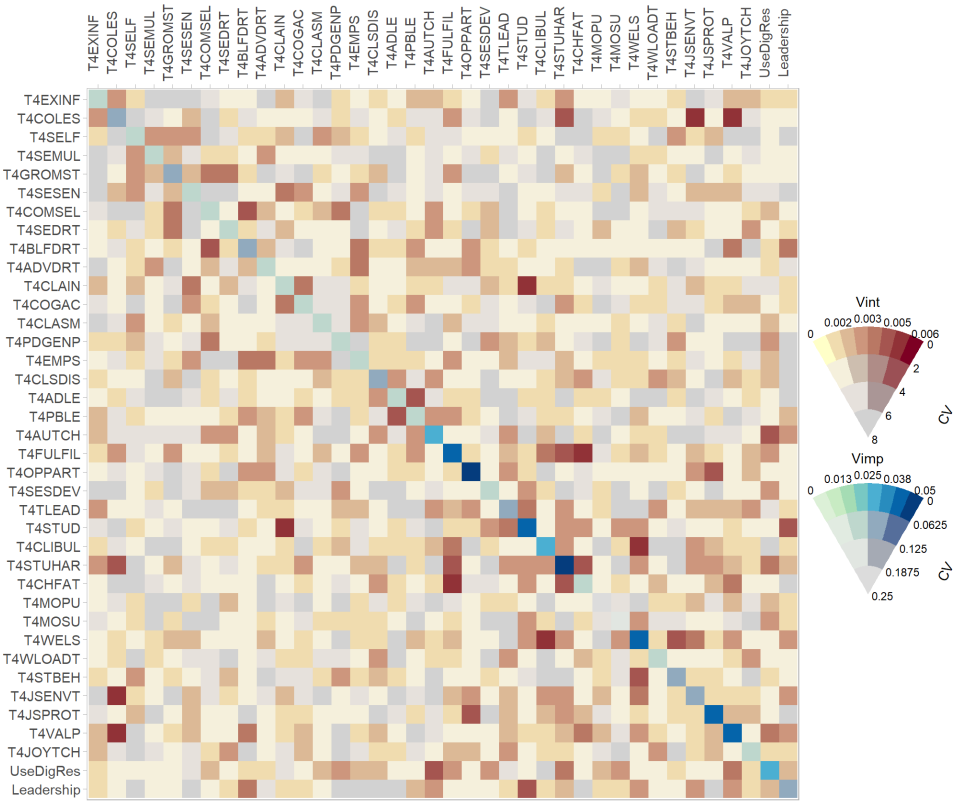



```
plot(shap_nordic_false, average = "post", title= "Post Graph: Non-Nordic Countries")
```



Interaction investigator

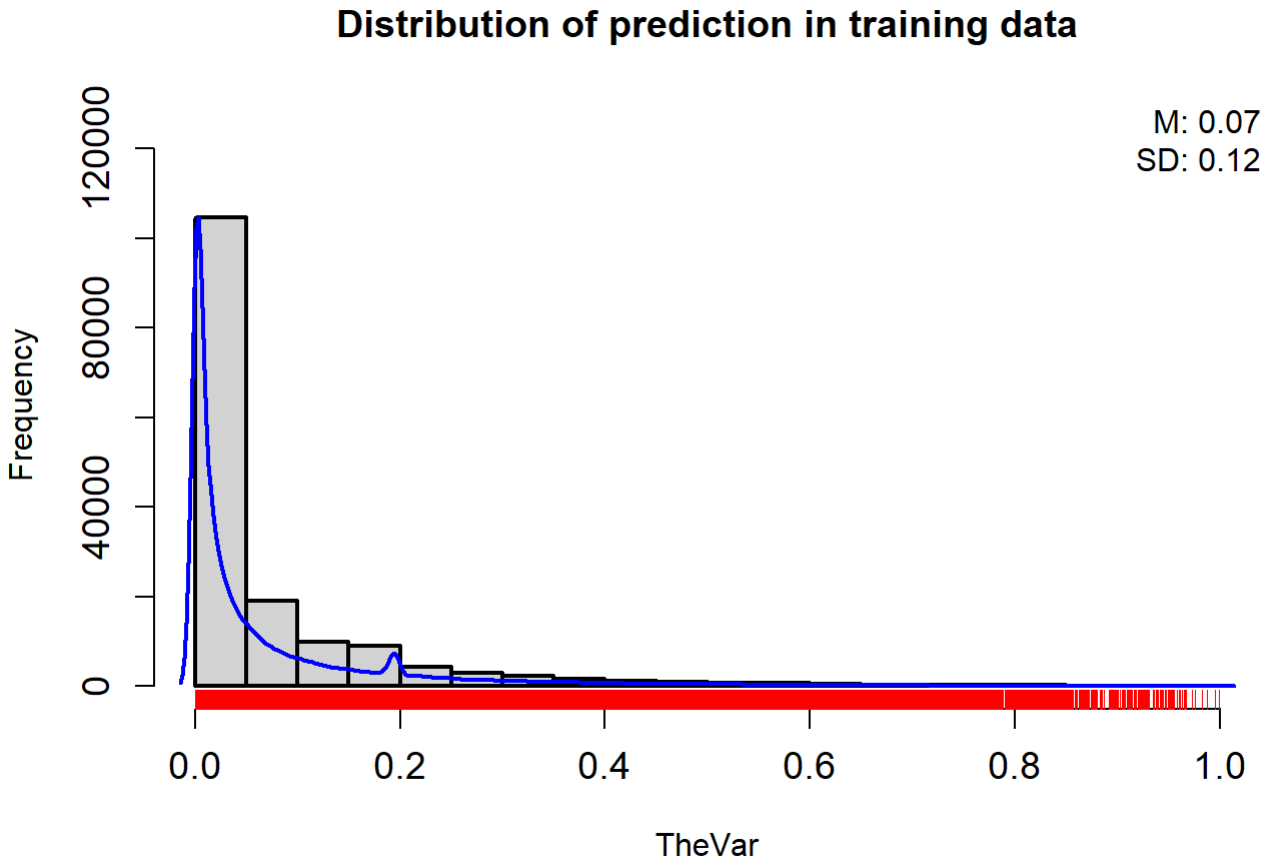
```
vsupMat <- viviBartMatrix(tree_data,
                          type = 'vsup',
                          metric = 'propMean',
                          metricError = "CV")
viviBartPlot(vsupMat,
             max_desat = 1,
             pow_desat = 0.6,
             max_light = 0.6,
             pow_light = 1,angle=90,
             label = 'CV')
```



distribution of predicted scores

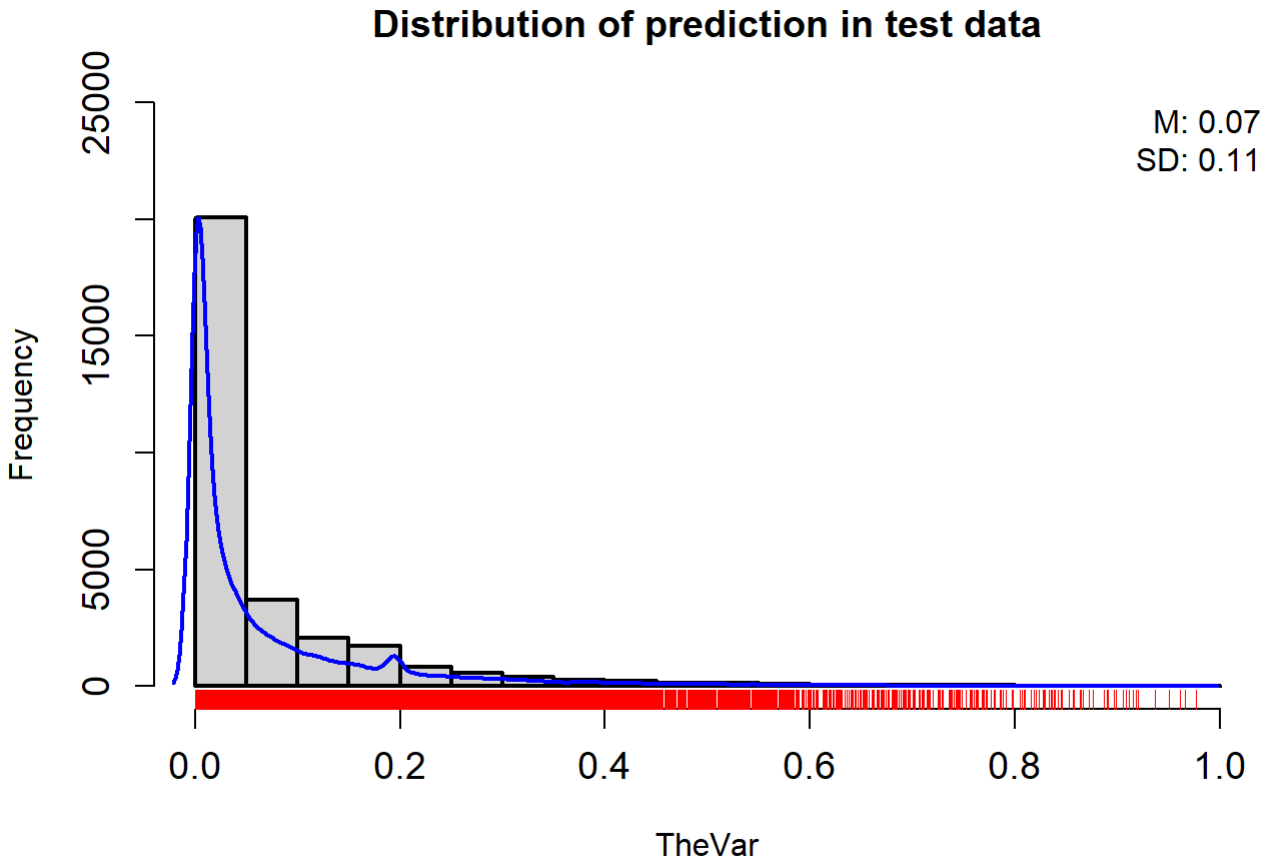
```
hist.rug(model$yhat.train |> pnorm() |> apply(2,mean),main="Distribution of prediction in tra
ining data")
```

```
##    vars      n mean   sd median trimmed  mad min max range skew kurtosis se
## X1    1 158831 0.07 0.12   0.02    0.04 0.03   0   1     1 2.95    10.56  0
```




```
hist.rug(model$yhat.test  |> pnorm() |> apply(2,mean),main="Distribution of prediction in tes
t data")
```

```
##    vars     n mean   sd median trimmed  mad min  max range skew kurtosis se
## X1    1 30515 0.07 0.11   0.02    0.04 0.03   0 0.98  0.98 2.94       11  0
```



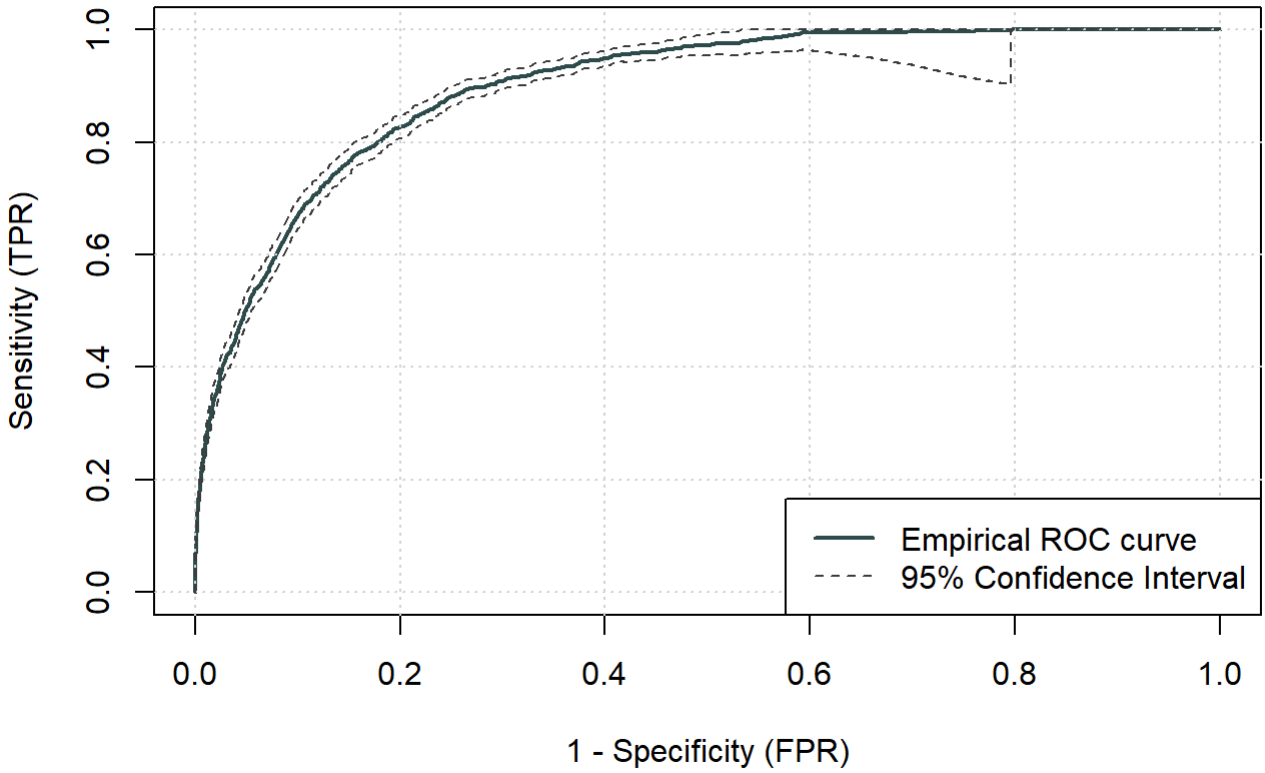
Print Prediction Tables using test data

```
str( ROCIT_Nordic   )
```

```
## List of 6
##  $ method                   : chr "empirical"
##  $ AUC                      : num 0.898
##  $ Cutoff                   : num [1:30078] Inf 0.976 0.966 0.966 0.961 ...
##  $ TPR/Sensitivity/Recall   : num [1:30078] 0 0.000759 0.001519 0.002278 0.003037 ...
##  $ FPR/1-Specificity        : num [1:30078] 0 0 0 0 0 0 0 0 0 0 ...
##  $ optimal Youden Index point: Named num [1:4] 0.6332 0.2468 0.88 0.0737
##   ..- attr(*, "names")= chr [1:4] "value" "FPR" "TPR" "cutoff"
```

```
ciROC(ROCit_obj, method = "binomial") %>% plot(legend=TRUE)
```

```
## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm): collapsing to uniq
ue 'x' values
```




```
ciAUC(ROCit_obj, delong = F)
```

```
##
##    estimated AUC : 0.897927175715377
##    AUC estimation method : empirical
##
##    CI of AUC
##    confidence level = 95%
##    lower = 0.886631104581589     upper = 0.909223246849166
```

Tables for classification

```
BTGINTT4 %>% transmute(CNTRY,predicted=model$yhat.train |> pnorm() |> apply(2,mean)>=ROCIT_No
rdicTr$optimal[4]) %>% table
```

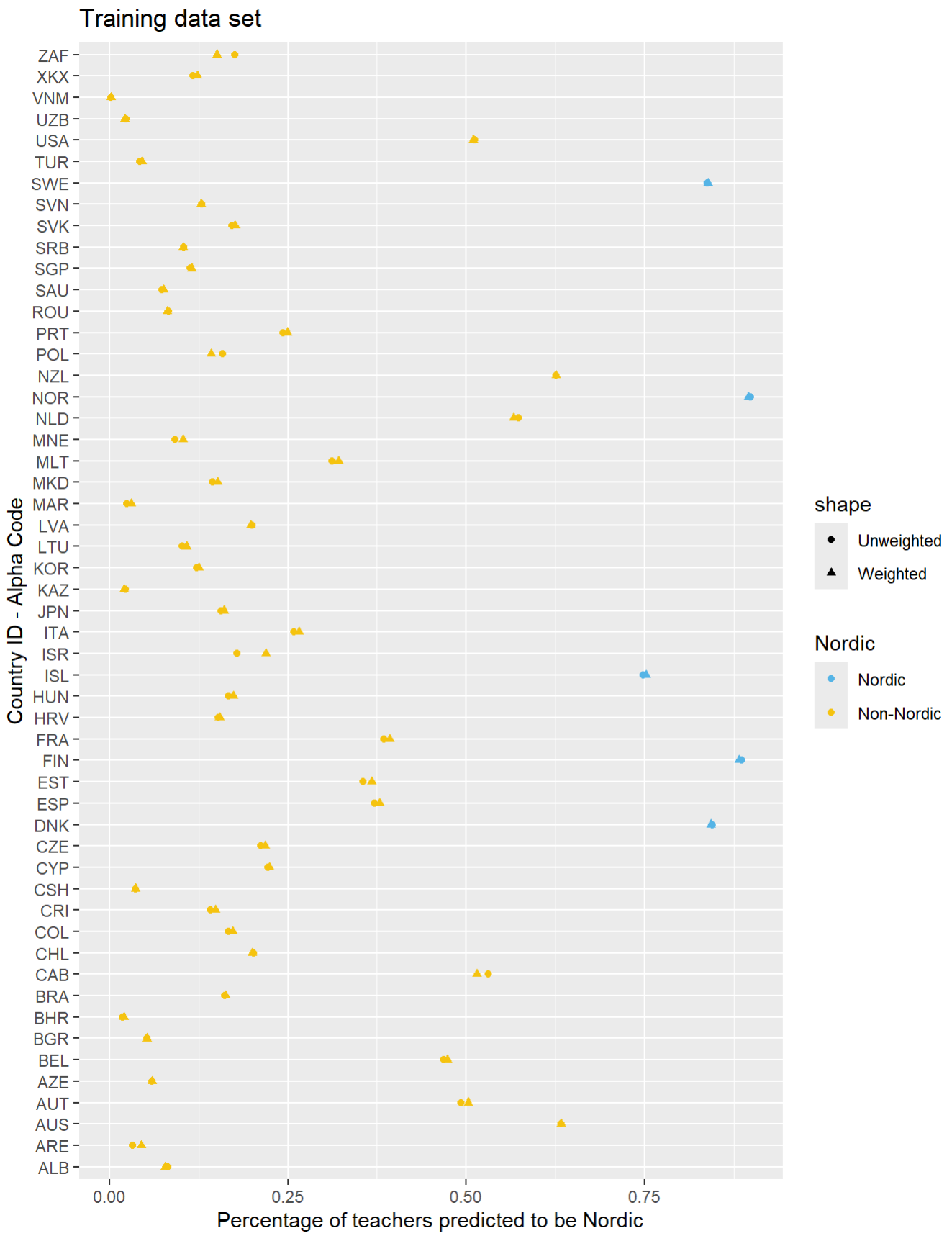
```
##      predicted
## CNTRY FALSE TRUE
##   ALB  3330  295
##   ARE  3804  126
##   AUS   999 1721
##   AUT  1970 1914
##   AZE  3640  232
##   BEL  2120 1871
##   BGR  3075  169
##   BHR  2739   52
##   BRA  2263  436
##   CAB   765  865
##   CHL  1663  421
##   COL  2244  450
##   CRI  2334  384
##   CSH  3532  133
##   CYP  1249  357
##   CZE  3143  848
##   DNK   346 1886
##   ESP  2509 1484
##   EST  1933 1064
##   FIN   437 3414
##   FRA  1590  995
##   HRV  2414  433
##   HUN  2905  580
##   ISL   322  954
##   ISR  2376  517
##   ITA  2583  900
##   JPN  2694  498
##   KAZ  3904   89
##   KOR  2445  339
##   LTU  3404  386
##   LVA  2943  735
##   MAR  3899   97
##   MKD  3071  517
##   MLT  1382  627
##   MNE  1101  112
##   NLD   486  652
##   NOR   139 1234
##   NZL   402  672
##   POL  2865  538
##   PRT  2378  764
##   ROU  3190  288
##   SAU  2753  219
##   SGP  2674  341
##   SRB  3220  372
##   SVK  3116  647
##   SVN  2217  330
##   SWE   447 2312
##   TUR  3808  170
##   USA   873  915
##   UZB  3880   92
##   VNM  3890    8
##   XKX  2606  346
##   ZAF  2439  519
```

```
Test     %>% transmute(CNTRY,predicted=model$yhat.test  |> pnorm() |> apply(2,mean)>=ROCIT_No
rdicTr$optimal[4]) %>% table
```

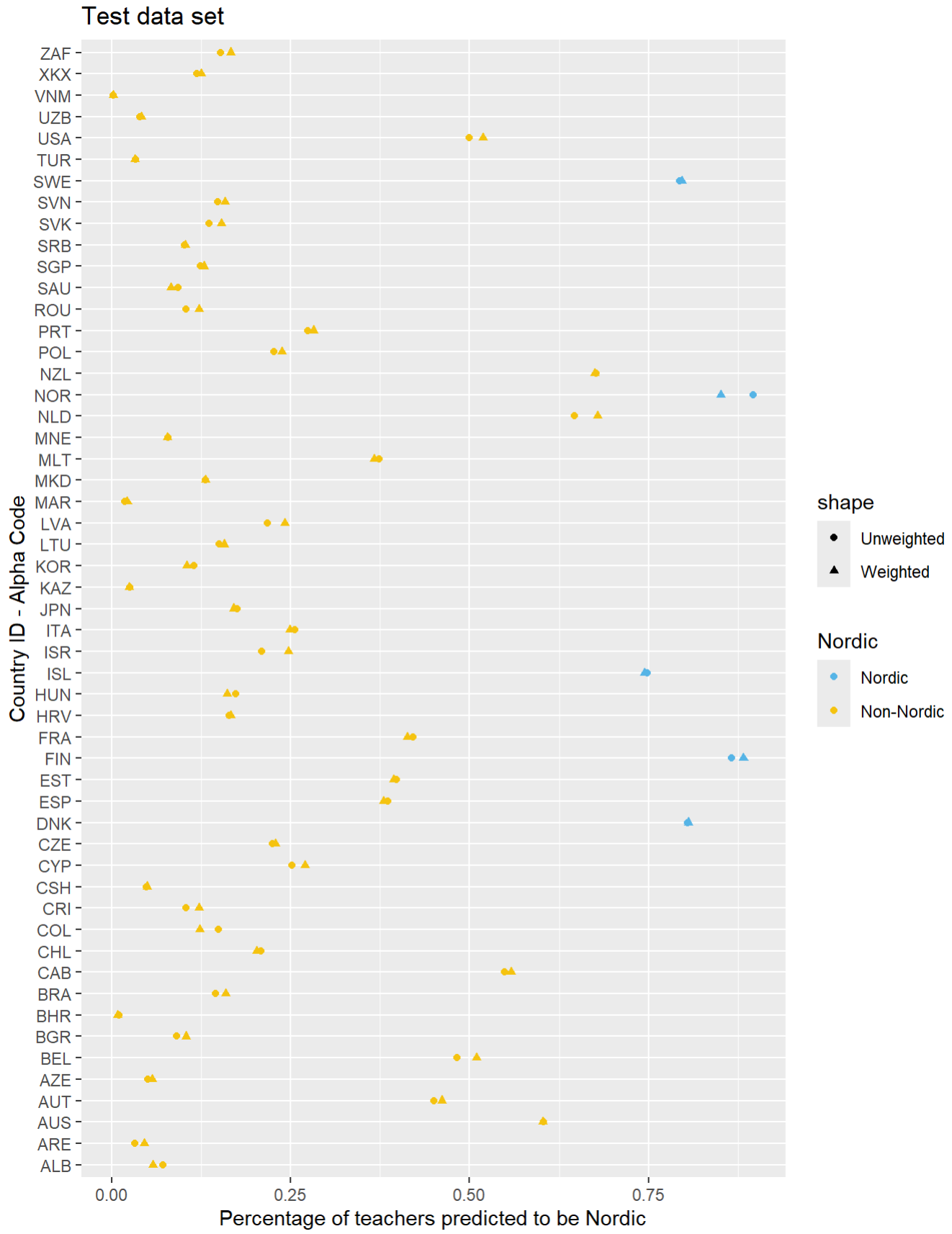
```
##      predicted
## CNTRY FALSE TRUE
##   ALB   375   29
##   ARE   425   14
##   AUS   125  190
##   AUT   248  203
##   AZE   413   22
##   BEL   613  571
##   BGR   350   35
##   BHR   310    3
##   BRA   265   45
##   CAB    87  106
##   CHL   186   49
##   COL   269   47
##   CRI   276   32
##   CSH   393   20
##   CYP   140   47
##   CZE  1843  535
##   DNK    50  206
##   ESP  3134 1971
##   EST   206  136
##   FIN    60  389
##   FRA   169  123
##   HRV   264   52
##   HUN   335   70
##   ISL    38  113
##   ISR   264   70
##   ITA   291  100
##   JPN   302   64
##   KAZ  3282   85
##   KOR   278   36
##   LTU   368   65
##   LVA   321   89
##   MAR  2078   38
##   MKD   351   53
##   MLT   159   95
##   MNE   130   11
##   NLD    47   86
##   NOR    16  139
##   NZL    42   88
##   POL   294   86
##   PRT   268  101
##   ROU   354   41
##   SAU   303   31
##   SGP   298   42
##   SRB   373   42
##   SVK   369   58
##   SVN   242   42
##   SWE    63  243
##   TUR   922   32
##   USA   100  100
##   UZB   441   18
##   VNM   449    1
##   XKX   290   39
##   ZAF   291   52
```

Comparison of weighted and unweighted results

```
BTGINTT4 %>%
  mutate(predicted = model$yhat.train |> pnorm() |> apply(2,mean),
         Nordic=factor(Nordic,1:0,c("Nordic","Non-Nordic")) ) %>%
  group_by(CNTRY,Nordic) %>%
  summarize(Unwg_pred=mean(predicted          >=ROCIT_NordicTr$optimal[4]),
            Wgt_pred =weighted.mean(predicted >=ROCIT_NordicTr$optimal[4],unlabelled(TCHWGT))
,
            .groups="drop") %>%
  ggplot(aes(y=CNTRY,color=Nordic))+
    geom_point(aes(x=Unwg_pred,shape="Unweighted"))+
    geom_point(aes(x=Wgt_pred,shape="Weighted"))+
    labs(x="Percentage of teachers predicted to be Nordic",title="Training data set")
```


Training data set
Country ID - Alpha Code
ZAF
XKX
VNM
UZB
USA
TUR
SWE
SVN
SVK
SRB
SGP
SAU
ROU
PRT
POL
NZL
NOR
NLD
MNE
MLT
MKD
MAR
LVA
LTU
KOR
KAZ
JPN
ITA
ISR
ISL
HUN
HRV
FRA
FIN
EST
ESP
DNK
CZE
CYP
CSH
CRI
COL
CHL
CAB
BRA
BHR
BGR
BEL
AZE
AUT
AUS
ARE
ALB
0.00
0.25
0.50
0.75
Percentage of teachers predicted to be Nordic
shape
Unweighted
Weighted
Nordic
Nordic
Non-Nordic

```
Test %>%
  mutate(predicted =model$yhat.test |> pnorm() |> apply(2,mean),
         Nordic=factor(Nordic,1:0,c("Nordic","Non-Nordic")) ) %>%
  group_by(CNTRY,Nordic) %>%
  summarize(Unwg_pred=mean(predicted>=ROCIT_NordicTr$optimal[4]),
            Wgt_pred =weighted.mean(predicted>=ROCIT_NordicTr$optimal[4],unlabelled(TCHWGT))
,
            .groups="drop") %>%
  ggplot(aes(y=CNTRY,color=Nordic))+
    geom_point(aes(x=Unwg_pred,shape="Unweighted"))+
    geom_point(aes(x=Wgt_pred,shape="Weighted"))+
    labs(x="Percentage of teachers predicted to be Nordic",title="Test data set")
```


Test data set
Country ID - Alpha Code
ZAF
XKX
VNM
UZB
USA
TUR
SWE
SVN
SVK
SRB
SGP
SAU
ROU
PRT
POL
NZL
NOR
NLD
MNE
MLT
MKD
MAR
LVA
LTU
KOR
KAZ
JPN
ITA
ISR
ISL
HUN
HRV
FRA
FIN
EST
ESP
DNK
CZE
CYP
CSH
CRI
COL
CHL
CAB
BRA
BHR
BGR
BEL
AZE
AUT
AUS
ARE
ALB
0.00
0.25
0.50
0.75
Percentage of teachers predicted to be Nordic
shape
Unweighted
Weighted
Nordic
Nordic
Non-Nordic

```
Test %>%
  mutate(predicted =model$yhat.test |> pnorm() |> apply(2,mean),
         Nordic=factor(Nordic,1:0,c("Nordic","Non-Nordic")) ) %>%
  group_by(Nordic) %>%
  summarize(Unwg_pred=mean(predicted>=ROCIT_NordicTr$optimal[4]),
            Wgt_pred =weighted.mean(predicted>=ROCIT_NordicTr$optimal[4],unlabelled(TCHWGT))
,
            .groups="drop") |>
  knitr::kable(caption="Comparison of weighted versus non-weighted, aggregate results")
```

Comparison of weighted versus non-weighted, aggregate results

Nordic	Unwg_pred	Wgt_pred
Nordic	0.8276386	0.8269853
Non-Nordic	0.2008699	0.2356533

Accuracy in Test data

```
Test %>% transmute(CNTRY,predicted=ifelse(model$yhat.test |> pnorm() |> apply(2,mean) >ROCIT_
NordicTr$optimal[4],"Nordic","Non") ) %>%
  table %>% as.data.frame %>%
  mutate(CNTRY=ifelse(CNTRY %in% c("DNK","NOR","SWE","FIN","ISL"),"Nordic","Non") ) %>%
  group_by(predicted,CNTRY) %>%
  summarize(Freq=sum(Freq) , .groups="drop") %>%
  with(
    data.frame(TruePos =Freq[predicted=="Nordic" & CNTRY=="Nordic"]/sum(Freq[CNTRY=="Nordi
c"]),
               FalsePos=Freq[predicted!="Nordic" & CNTRY=="Nordic"]/sum(Freq[CNTRY=="Nordi
c"]),
               TrueNeg =Freq[predicted!="Nordic" & CNTRY!="Nordic"]/sum(Freq[CNTRY!="Nordi
c"]),
               FalseNeg=Freq[predicted=="Nordic" & CNTRY!="Nordic"]/sum(Freq[CNTRY!="Nordi
c"])
               )
  )
```

```
##     TruePos  FalsePos   TrueNeg  FalseNeg
## 1 0.8276386 0.1723614 0.7991301 0.2008699
```

Session info

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows Server 2022 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] C
##
## time zone: Europe/Oslo
## tzcode source: internal
##
## attached base packages:
## [1] splines   grid      stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] future.apply_1.20.2 furrr_0.4.0         future_1.75.0       data.table_1.18.4   bartX
Viz_1.0.11
##  [6] SuperLearner_2.0-42 nnls_1.6            gam_1.22-7          foreach_1.5.2       label
led_2.16.0
## [11] tinytex_0.60        arm_1.15-3          lme4_2.0-6          MASS_7.3-66         haven
_2.5.5
## [16] posterior_1.7.0     ROCit_2.1.2         coda_0.19-4.1       survey_4.5          survi
val_3.8-9
## [21] Matrix_1.7-5        lubridate_1.9.5     forcats_1.0.1       stringr_1.6.0       dplyr
_1.2.1
## [26] purrr_1.2.2         readr_2.2.0         tidyr_1.3.2         tibble_3.3.1        ggplo
t2_4.0.3
## [31] tidyverse_2.0.0     bartMan_0.2.1       dbarts_0.9-33
##
## loaded via a namespace (and not attached):
##   [1] bitops_1.1-0            polyclip_1.10-7         gggenes_0.7.0           lifecycle_1.
0.5
##   [5] Rdpack_2.6.6            rstatix_1.1.0           rrapply_1.2.8           globals_0.1
9.1
##   [9] lattice_0.22-9          backports_1.5.1         magrittr_2.0.5          sass_0.4.10
##  [13] rmarkdown_2.31          jquerylib_0.1.4         yaml_2.3.12             otel_0.2.0
##  [17] doRNG_1.8.6.3           BART_2.9.10             cowplot_1.2.0           DBI_1.3.0
##  [21] minqa_1.2.8             RColorBrewer_1.1-3      DendSer_1.0.3           abind_1.4-8
##  [25] ggraph_2.2.2            itertools_0.1-3         tensorA_0.36.2.1        tweenr_2.0.3
##  [29] gdtools_0.5.1           seriation_1.5.8         ggrepel_0.9.8           listenv_1.0.
0
##  [33] missForest_1.6.1        parallelly_1.48.0       codetools_0.2-20        ggforce_0.5.
0
##  [37] tidyselect_1.2.1        farver_2.1.2            viridis_0.6.5           TSP_1.2.7
##  [41] jsonlite_2.0.0          tidygraph_1.3.1         Formula_1.2-6           ggridges_0.
5.7
##  [45] iterators_1.0.14        systemfonts_1.3.2       tools_4.6.1             ggnewscale_
0.5.2
##  [49] ragg_1.5.2              Rcpp_1.1.2              glue_1.8.1              mnormt_2.1.2
##  [53] gridExtra_2.3.1         mgcv_1.9-4              xfun_0.60               ranger_0.18.
0
##  [57] distributional_0.8.1    ca_0.71.1               withr_3.0.3             fastmap_1.2.
0
```

```
##  [61] mitools_2.4            boot_1.3-32            caTools_1.18.4         digest_0.6.3
9
##  [65] gclus_1.3.3            timechange_0.4.0       R6_2.6.1               textshaping_
1.0.5
##  [69] colorspace_2.1-3       gtools_3.9.5           see_0.14.1             generics_0.
1.4
##  [73] fontLiberation_0.1.0   graphlayouts_1.2.5     htmlwidgets_1.6.4      pkgconfig_2.
0.3
##  [77] rJava_1.0-18           gtable_0.3.6           registry_0.5-1         S7_0.2.2
##  [81] htmltools_0.5.9        bartMachineJARs_1.2.2  fontBitstreamVera_0.1.1 carData_3.0-
6
##  [85] scales_1.4.0           reformulas_0.4.4       knitr_1.51             rstudioapi_
0.19.0
##  [89] tzdb_0.5.0             reshape2_1.4.5         checkmate_2.3.4        nlme_3.1-169
##  [93] nloptr_2.2.1           cachem_1.1.0           KernSmooth_2.23-26     parallel_4.
6.1
##  [97] pillar_1.11.1          vctrs_0.7.3            gplots_3.3.0           randomForest
_4.7-1.2
## [101] ggpubr_1.0.0           ggfittext_0.10.4       car_3.1-5              cluster_2.1.
8.2
## [105] evaluate_1.0.5         cli_3.6.6              compiler_4.6.1         rlang_1.3.0
## [109] rngtools_1.5.2         ggsignif_0.6.4         labeling_0.4.3         plyr_1.8.9
## [113] ggiraph_0.9.6          stringi_1.8.9          psych_2.6.5            viridisLite_
0.4.3
## [117] fontquiver_0.2.1       hms_1.1.4              patchwork_1.3.2        rbibutils_2.
4.1
## [121] igraph_2.3.3           broom_1.0.13           memoise_2.0.1          bslib_0.12.0
## [125] bartMachine_1.4.2
```